Differential Geometry

Wang Yuyao

Contents

I Classical Differential Geometry **1**

0 Introduction **2**

1 Curve Theory **3**

1.1 Regular Curve . . . 3

1.2 Local Theory of Curves . . . 4

1.2.1 Intuition and Basic Concepts . . . 4

1.2.2 Reparametrization . . . 5

1.2.3 Arc-length . . . 5

1.2.4 Frenet Frame and Curvature of plane curve . . . 6

1.2.5 Fundamental Theorem for Plane Curve . . . 8

1.2.6 Space Curve . . . 11

1.3 Fundamental Theorem for Space Curve . . . 13

1.4 Global Theory of Plane Curves . . . 15

1.4.1 Closed Curve and Simple Curve . . . 15

1.4.2 Hopf Theorem . . . 16

1.4.3 Isoperimetric Inequality . . . 18

2 Surface Theory **20**

2.1 Local Theory of Surfaces . . . 20

2.1.1 Regular Surface . . . 20

2.1.2 Change of Coordinates . . . 23

2.1.3 Tangent Plane . . . 25

2.1.4 First Fundamental Form (f.f.f.) . . . 27

2.1.5 Arc-length and Area . . . 29

2.2 Gauss Map . . . 31

2.2.1 Orientability . . . 31

2.2.2 Differential . . . 31

2.2.3 Classification of Points of Surface . . . 32

2.2.4 Second Fundamental Form . . . 33

2.2.5 Weingarten Equation . . . 36

2.2.6 Umbilical Points . . . 39
2.3 Third Fundamental Form . . . 39
2.4 Minimal Surface . . . 40
2.4.1 Definition . . . 40
2.4.2 Isothermal Parametrization . . . 41
2.4.3 Perspective of Complex Analysis . . . 43
2.5 Gaussian Curvature . . . 46

II Modern Differential Geometry 47

3 Smooth Manifolds 48
3.1 Topological Manifolds . . . 48
3.1.1 Definitions and Examples . . . 48
3.1.2 Topological Properties . . . 51
3.2 Smooth Manifolds . . . 54
3.2.1 Intuition and Construction . . . 54
3.2.2 Local Coorodinate Representations . . . 57
3.2.3 Examples of Smooth Manifolds . . . 57
3.3 Manifolds with Boundary . . . 62
3.3.1 Topological Manifolds with Boundary . . . 62
3.3.2 Smooth Structure on Topological Manifolds with Boundary . . . 65

4 Smooth Maps between Manifolds 68
4.1 Smooth Functions on Manifolds . . . 68
4.2 Smooth Maps between Manifolds . . . 69
4.3 Diffeomorphisms between Manifolds . . . 74

5 Partition of Unity 76
5.1 Cutoff Functions and Smooth Bump Functions . . . 76
5.2 Definition and Existence of Partition of Unity . . . 77
5.3 Applications . . . 79
5.3.1 Extension of Bump Functions . . . 79
5.3.2 Extension of Smooth Functions from Closed Subsets . . . 80
5.3.3 Exaustion Functions . . . 81
5.3.4 Level Sets . . . 81

6 Tangent Vectors 82
6.1 Geometric Tangent Vectors . . . 82
6.2 Tangent Vectors on Manifolds . . . 85
6.3 Differential of Smooth Maps . . . 85

6.4 More Properties of Tangent Vectors 86
6.4.1 Locality 86
6.4.2 Identification 87
6.4.3 Dimension 88
6.5 Computations in Coordinates 89
6.5.1 Coordinate Vectors and Basis 89
6.5.2 Differentials in Coordinates 90
6.5.3 Change of Coordinates 91
6.6 Tangent Bundles 92
6.7 Velocity of Curves in Manifolds 94

A Preliminary Theorems from Analysis **97**

Part I

Classical Differential Geometry

Chapter 0

Introduction

This part is devoted to studying curves and surfaces in $\mathbb{R}^3$. More precisely, this part is going to address the following questions:

Curves

- **How to describe a curve?**
  e.g.: $\{x = y, z = 0\}$ or equivalently $\{\begin{pmatrix} t \\ t \\ 0 \end{pmatrix} \mid t \in \mathbb{R}\}$
- **Properties of curves?**
  e.g.: length, curvature.
- **If we know the curvature of a curve at every point, can we recover the whole curve?**
- **If we suppose that we have a closed curve in $\mathbb{R}^2$ of a given length, what is the largest possible area bounded by the curve?** (Isoperimetric Inequality)

Surfaces

- **How to describe a surface? More precisely, what kind of surfaces shall we study?**
  e.g.: We'd not learn surfaces with corners, self-intersection, and cusps ("infinite tangent").
  Remark: These kinds of surfaces will be discussed in the discrete Gauss–Bonnet Theorem.
- **For the "smooth" surface, how can we study the "area" and "curvature" of them?**
- **How to compute the "shortest" distance between two points of a surface?** (Geodesics)

Chapter 1

Curve Theory

1.1 Regular Curve

Definition 1.1.1. Regular Curve
A curve $\alpha : [c, d] \to \mathbb{R}^3$ is said to be *regular* if the derivative does not vanish in the sense of norm, i.e.: on $[c, d]$

$$|\alpha'| \neq 0$$

(α is taken to be differentiable).

Proposition 1.1.2. *Riemann Sum and Arc-length of Regular Curve*
Let $\alpha : [c, d] \to \mathbb{R}^3$ be a regular curve with partition $\mathcal{P} = \{c = t_0 < t_1 < \cdots < t_n = d\}$ and let

$$l(\alpha, \mathcal{P}) := \sum_{i=0}^{n-1} |\alpha(t_i) - \alpha(t_{i+1})|$$

Then the arc-length of α is equal to

$$\int_c^d \left| \frac{d\alpha}{dt} \right| dt = \sup \{ l(\alpha, \mathcal{P}) \mid \mathcal{P} \text{ any partition of } [c, d] \}$$

Proposition 1.1.3. *Well-defined-ness of Regular Curve*
Let $\alpha : [a, b] \to \mathbb{R}^3$ be a regular curve and let $g : [c, d] \to [a, b]$ be a reparametrization with $\beta = \alpha \circ g$. Then:

$$\int_a^b \left| \frac{d\alpha}{dt} \right| dt = \int_c^d \left| \frac{d\beta}{ds} \right| ds$$

Proof. It immediately follows from the Chain Rule. □

1.2 Local Theory of Curves

1.2.1 Intuition and Basic Concepts

Intuition: There are many ways to parametrize the same curve, but we want to pick one such that the "tangent" is moving at *unit* speed, i.e.: $\forall a \leq s \leq b$, $|\frac{d\beta}{ds}| = 1$, or equivalently, $\forall a \leq s \leq b$,

$$\int_a^b \left|\frac{d\beta}{ds}\right| ds = b - a$$

Definition 1.2.1. Plane Curve
The *trace* of a curve α is defined to be $\text{im}(\alpha)$ ($\subseteq \mathbb{R}^3$). If $\text{im}(\alpha)$ lies on a plane $P \subseteq \mathbb{R}^3$, then α is called a *plane curve*.
Remark: By default, we express our vector in $\mathbb{R}^3$ by the usual basis

$$\mathcal{A} := \left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

i.e.: $\alpha = (x(t), y(t), z(t))_{\mathcal{A}} = \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix}_{\mathcal{B}}$. However, our results also hold by using coordinate vectors of another orthonormal basis $\mathcal{B} = \{u_1, u_2, u_3\}$.
If α is a plane curve, i.e.: $\alpha(t) \in P$ with $P = \{z_0 + pu_1 + gu_2 \mid p, g \in \mathbb{R}\}$, then

$$\alpha(t) = (z_0)_{\mathcal{B}} + (a(t), b(t), 0)_{\mathcal{B}} \in \mathbb{R}^3$$

To simplify further, one can just study the curve $\beta : I \to \mathbb{R}^2$ given by $\beta(t) := (a(t), b(t))$, and then all our results on plane curves can be obtained from $\alpha : I \xrightarrow{\beta} \mathbb{R}^2 \xrightarrow{\gamma} \mathbb{R}^3$ where $\gamma(a, b) = (a, b, 0)_{\mathcal{B}} = au_1 + bu_2$.

Definition 1.2.2. Tangent Vector
Let the curve $\alpha : I \to \mathbb{R}^3$ be regular. The *velocity vector* of α at $t \in I$ is given by $\alpha'(t)$ and the *speed* of α at $t \in I$ is given by $|\alpha'(t)|$. Then we define the *tangent vector* $T(t)$ of α by

$$T(t) := \frac{\alpha'(t)}{|\alpha'(t)|}$$

Furthermore, the tangent line at $t = t_0$ is: $l(t_0) := \alpha(t_0) + \lambda T(t_0)$.
Remark: The cusp is not a regular curve since $|\alpha'(0)| = 0$, but we allow self-intersection, i.e.: $\exists t_1 \neq t_2$ s.t.: $\alpha(t_1) = \alpha(t_2)$. Moreover, they are also allowed to have different tangent vectors $T(t_1) \neq T(t_2)$.

1.2.2 Reparametrization

Definition 1.2.3. Reparametrization
The *reparametrization* of a curve $\alpha : [a, b] \to \mathbb{R}^3$ is a bijective function $g : [c, d] \to [a, b]$ such that g is $\mathcal{C}^\infty$ and g^{-1} is $\mathcal{C}^1$ (differentiable with continuous derivative).

Proposition 1.2.4. *Invariance of Regularity under Reparametrization*
If α is a regular curve and g is a reparametrization, then $\beta := \alpha \circ g$ is regular.

Proof. Note that $g \circ g^{-1} = \text{id}$. Hence, $g'(g^{-1}(t)) \cdot (g^{-1})'(t) = 1, \forall t$. Therefore, by the bijectivity of g, $g'(s) \neq 0$, $\forall s$. Thus,

$$|\beta'(s)| = |\alpha'(g(s))| \cdot |g'(s)| \neq 0.$$

□

Proposition 1.2.5. *"Invariance" of Tangent Vector under Reparametrization*
Let α be a regular curve and let g be a reparametrization with $\beta = \alpha \circ g$. If T is the tangent vector of α at $t = g(s_0)$ and if S is the tangent vector of β at $s = s_0$, then

$$S = \text{sgn}\left(\left.\frac{dg}{ds}\right|_{s_0}\right) \cdot T$$

Proof. Note that $g'(s_0)/|g'(s_0)| = \text{sgn}(g'(s_0))$. Hence, by chain rule, we're done. □

1.2.3 Arc-length

Proposition 1.2.6. *Basic Formula for computing Arc-length*

$$\int_\alpha ds = \int_c^d \left|\frac{d\alpha}{dt}\right| dt = \int_c^d \sqrt{(x')^2 + (y')^2 + (z')^2}\, dt$$

Remark: $\sqrt{(x')^2 + (y')^2 + (z')^2} = ds$

Proposition 1.2.7. *Invariance of Arc-length under Reparametrization*
Let α be a regular curve and let g be a reparametrization with $\beta = \alpha \circ g$. Then

$$\int_\alpha dt = \int_\beta ds$$

Lemma 1.2.8. *Preparation for defining Arc-length Parametrization*
Let $t_0 \in [a, b]$. Consider $h : [a, b] \to \mathbb{R}$ given by

$$h(t) := \int_{t_0}^t \left|\frac{d\alpha}{ds}\right| ds$$

Then:

- *$h : [a, b] \to [h(a), h(b)]$ is $\mathcal{C}^\infty$*

- h^{-1} *exists and is also* $\mathcal{C}^\infty$

Proof. The second property is immediately from IVT. □

Definition 1.2.9. Arc-length Parametrization
Let $\alpha : [a, b] \to \mathbb{R}^3$ be regular and $t_0 \in I$ with $g = h^{-1}$ a reparametrization, where h is given in the Preparation for defining Arc-length Parametrization lemma. Then $\beta = \alpha \circ g : [h(a), h(b)] \to \mathbb{R}^3$ is *parametrization by arc-length* (*p.a.l.*).

Proposition 1.2.10. *Advantage of P.A.L.*
Let β *be a p.a.l. Then* $|d\beta/ds| = 1$.

Proof. $|d\beta/ds| = |\beta'(g(s))| \cdot |g'(s)| = |h'| \cdot \frac{1}{|h'|} = 1$. □

Corollary 1.2.11. *of Advantage of P.A.L. Proposition*
If β *is a p.a.l., then* $\beta' = T$.

1.2.4 Frenet Frame and Curvature of plane curve

In this section, we assume $\alpha : I \to \mathbb{R}^2$ where α is a p.a.l.

Definition 1.2.12. Normal Vector
The *normal vector* $N(s)$ of α at $t = s$ is the vector

$$N(s) := J(T(s))$$

where J is the 90° anticlockwise rotation $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, i.e.: $J : \begin{pmatrix} a \\ b \end{pmatrix} \mapsto \begin{pmatrix} -b \\ a \end{pmatrix}$.

Definition 1.2.13. Frenet Frame
The set $\{T(s), N(s)\}$ forms an ordered, oriented basis of $\mathbb{R}^2$ which is orthonormal. It's called the *Frenet frame* of α.

Definition 1.2.14. Curvature
Since $\langle T(s), T(s)\rangle = 1$, then $2\langle T'(s), T(s)\rangle = \frac{d}{ds}\langle T(s), T(s)\rangle = 0$, so there must be some scalar $K(s)$, s.t.: $T'(s) = K(s)N(s)$. The *curvature* of α at s is the value $K(s)$.

Proposition 1.2.15. $N'(s) = -K(s)T(s)$.

Proof. **Method 1**: Since $\langle N(s), N(s)\rangle = 1$, then $2\langle N'(s), N(s)\rangle = 0$, so there must be some scalar $c(s)$, s.t.: $N'(s) = c(s)T(s)$. Since $\langle T(s), N(s)\rangle = 0$, then

$$0 = \frac{d}{ds}\langle T(s), N(s)\rangle = \langle T'(s), N(s)\rangle + \langle T(s), N'(s)\rangle$$

Hence,

$$0 = \langle K(s)N(s), N(s)\rangle + \langle T(s), c(s)T(s)\rangle = K(s) + c(s).$$

Method 2: Note that

$$J^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I.$$

Hence,

$$N'(s) = \frac{d}{ds}J(T(s)) = J(T'(s)) = J(K(s)N(s)) = K(s)J(N(s)) = K(s)J(J(T(s))) = K(s)\cdot(-T(s)).$$

□

Remark: $K(s)$ records "the change of $T(s)$":

- $K(s) \equiv 0 \iff T'(s) \equiv 0 \iff T(s) = \underline{v}$ for some constant vector $\underline{v}$.
- $K(s) > 0 \Rightarrow T(s)$ is turning towards $N(s)$, say, T' along the direction of N.
- $K(s) < 0 \Rightarrow T(s)$ is turning away $N(s)$, say, T' against the direction of N.

Example: Let $\alpha(s) := \underline{u} + r\left(\cos\frac{s}{r}, \sin\frac{s}{r}\right)$ be a circle centered at $\underline{u}$ of radius r. Then

$$T_\alpha(s) := \alpha'(s) = \left(-\sin\frac{s}{r}, \cos\frac{s}{r}\right) \Rightarrow N_\alpha(s) := J(T_\alpha(s)) = \left(-\cos\frac{s}{r}, -\sin\frac{s}{r}\right)$$

and $T'_\alpha(s) = \frac{1}{r}\left(-\cos\frac{s}{r}, -\sin\frac{s}{r}\right)$. Hence, by the def of curvature, $K_\alpha(s) = \frac{1}{r}$.
By contrast, consider $\beta(s) := \underline{u} + r\left(\cos(-\frac{s}{r}), \sin(-\frac{s}{r})\right)$, which is the opposite orientation of the above circle α. Then after a simple computation, we have $K_\beta(s) = -\frac{1}{r}$.

Proposition 1.2.16. *Curvature Formula for Regular Plane Curve*
Let $\alpha : I \to \mathbb{R}^2$ be a regular curve with p.a.l. $\beta = \alpha \circ g$. Then the curvature of α is:

$$K_\alpha(s) := \frac{\det(\alpha'(t), \alpha''(t))}{|\alpha'(t)|^3}$$

where $t = g(s)$.

Proof. By the def of p.a.l., $\frac{ds}{dt} = \frac{d}{dt}(g^{-1}(t)) = |\alpha'(t)|$. Hence,

$$\frac{dt}{ds} = \frac{1}{|\alpha'(t)|}$$

Then by the Chain Rule,

$$\beta'(s) = \alpha'(t)\frac{dt}{ds} = \frac{\alpha'(t)}{|\alpha'(t)|}$$

and then

$$\beta''(s) = \frac{d}{ds}\frac{\alpha'(t)}{|\alpha'(t)|} = \frac{dt}{ds}\frac{d}{dt}\frac{\alpha'(t)}{|\alpha'(t)|} = \frac{1}{|\alpha'(t)|}\cdot\frac{\alpha''(t)|\alpha'(t)| - \alpha'(t)\cdot\frac{d}{dt}|\alpha'(t)|}{|\alpha'(t)|^2}$$

Since $\{T_\beta, N_\beta\}$ is orthonormal, then $\det(T_\beta(s), N_\beta(s)) = 1$, $\forall s$. Hence,

$$K_\beta(s) = K_\beta(s)\det(T_\beta(s), N_\beta(s)) = \det(T_\beta(s), K_\beta(s)N_\beta(s)) := \det(T_\beta(s), T'_\beta(s))$$

$$:= \det(\beta'(s), \beta''(s)) = \det\left(\frac{\alpha'(t)}{|\alpha'(t)|}, \frac{\alpha''(t)|\alpha'(t)| - \alpha'(t)\cdot\frac{d}{dt}|\alpha'(t)|}{|\alpha'(t)|^3}\right)$$

$$= \det\left(\frac{\alpha'(t)}{|\alpha'(t)|}, \frac{\alpha''(t)}{|\alpha'(t)|^2}\right) - \det\left(\frac{\alpha'(t)}{|\alpha'(t)|}, \frac{\alpha'(t)\cdot\frac{d}{dt}|\alpha'(t)|}{|\alpha'(t)|^3}\right)$$

$$= \det\left(\frac{\alpha'(t)}{|\alpha'(t)|}, \frac{\alpha''(t)}{|\alpha'(t)|^2}\right) - \frac{\frac{d}{dt}|\alpha'(t)|}{|\alpha'(t)|^4}\cdot\det(\alpha'(t), \alpha'(t))$$

$$= \frac{1}{|\alpha'(t)|^3}\det(\alpha'(t), \alpha''(t)) - 0$$

□

1.2.5 Fundamental Theorem for Plane Curve

Definition 1.2.17. Orthogonal Group
We define the *orthogonal group* and *special orthogonal group* by

$$O(n) := \{A \in M_{n\times n}(\mathbb{R}) \mid A^TA = I\}$$

respectively

$$SO(n) := \{A \in M_{n\times n}(\mathbb{R}) \mid A^TA = I, \det(A) = 1\}$$

Example: $n = 2$.

$$SO(2) = \left\{\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}\right\}$$

consists of rotation matrices, and

$$O(2) = \left\{\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}, \begin{pmatrix}\cos\theta & \sin\theta\\ \sin\theta & -\cos\theta\end{pmatrix}\right\} = \text{“rotation” + “reflection”}.$$

Proposition 1.2.18. *Orthonormal Basis preserving Orthogonality*
If $\{\underline{u}_1, \cdots, \underline{u}_n\}$ is an orthonormal basis of $\mathbb{R}^n$ under usual dot product, and if $A \in O(n)$, then so is $\{A\underline{u}_1, \cdots, A\underline{u}_n\}$.

Proof. $\forall i, j$,

$$\langle A\underline{u}_i, A\underline{u}_j\rangle = (A\underline{u}_i)^TA\underline{u}_j = \underline{u}_i^TA^TA\underline{u}_j = \underline{u}_i^T\underline{u}_j = \langle\underline{u}_i, \underline{u}_j\rangle = \begin{cases}1, & \text{if } i = j\\ 0, & \text{if } i \neq j\end{cases}$$

□

Definition 1.2.19. Rigid Motion
We say a map $\mathbb{R}^n \to \mathbb{R}^n$ is a *rigid motion* if it is of the form:

$$\underline{x} \mapsto A\underline{x} + \underline{b}$$

where $A \in O(n)$ and $\underline{b} \in \mathbb{R}^n$. Furthermore, if $A \in SO(n)$, then we call the rigid motion *orientation preserving.*

Theorem 1.2.20. *Fundamental Theorem for Plane Curve*

1. *(Existence). For any $\mathcal{C}^\infty$ function $f : I \to \mathbb{R}$, $\exists$ p.a.l. plane curve $\alpha : I \to \mathbb{R}^2$, s.t.:*

$$K_\alpha(s) = f(s) \quad \forall s \in I, \text{ with } \alpha(0) = x_0$$

2. *(Uniqueness). Suppose that there are two curves α, α^* p.a.l., s.t.: $K_\alpha = K_{\alpha^*}$, $\forall s \in I$, then they only differ by a rigid motion, i.e.: $\alpha^* = \Phi \circ \alpha$ for some $\Phi : \mathbb{R}^2 \to \mathbb{R}^2$ rigid motion.*

Proof. 1. Consider a system of ODE

$$\begin{cases} t'(s) = f(s)n(s) \\ n'(s) = -f(s)t(s) \end{cases} \iff \begin{pmatrix} t \\ n \end{pmatrix}' = \begin{pmatrix} 0 & f \\ -f & 0 \end{pmatrix} \begin{pmatrix} t \\ n \end{pmatrix}$$

with $t(0) = p \neq 0$ unit length and $n(0) = q = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} p$. Then by the existence and uniqueness of ODE solutions give a unique solution for t and n.

Since

$$\frac{d}{ds}\langle t, n\rangle = \langle t', n\rangle + \langle t, n'\rangle = \langle fn, n\rangle + \langle t, -ft\rangle = f(\langle n, n\rangle - \langle t, t\rangle)$$

and

$$\frac{d}{ds}\langle t, t\rangle = \langle t', t\rangle + \langle t, t'\rangle = 2f\langle t, n\rangle \quad \text{and} \quad \frac{d}{ds}\langle n, n\rangle = -2f\langle t, n\rangle$$

then if we denote $a(s) := \langle t(s), t(s)\rangle$, $b(s) := \langle t(s), n(s)\rangle$, and $c(s) := \langle n(s), n(s)\rangle$, then we obtain another ODE system

$$\begin{cases} a' = 2fb \\ b' = f(c - a) \\ c' = -2fb \end{cases}$$

with initial value $(a(0), b(0), c(0)) = (1, 0, 1)$. Note that $(a, b, c) \equiv (1, 0, 1)$ satisfies this system, so by the existence and uniqueness of ODE solutions, $\langle t(s), t(s)\rangle \equiv 1$, $\langle t(s), n(s)\rangle \equiv 0$, and $\langle n(s), n(s)\rangle \equiv 1$.

Then we define $\alpha(s) := x_0 + \int_0^s t(\sigma)\, d\sigma$. We can check:

- $\alpha(0) = x_0$

- $\alpha'(s) = t(s)$.
- Since $|\alpha'(s)| = |t(s)| = \langle t, t\rangle \equiv 1$, then α is a p.a.l.
- $T'_\alpha(s) := \alpha''(s) = t'(s) = f(s)n(s)$

Hence, by def, it remains to show that $n(s) = N_\alpha(s)$: Since $\langle t, n\rangle \equiv 0$, then $n(s) = J(t(s))$ or $n(s) = -J(t(s))$, where $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Note that when $s = 0$, $n(0) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} t(0) = q$, i.e.: the first case holds at $s = 0$. Moreover, note that the function $\|n(s) - \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} t(s)\|$ is continuous, so we must have

$$n(s) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} t(s)$$

Since $T_\alpha(s) := \alpha'(s) = t(s)$, then by def, $n(s) = N_\alpha(s)$.

2. Denote $K(s) := K_\alpha(s) = K_{\alpha^*}(s)$, $\forall s \in I$. Let $A \in SO(2)$ be such that

$$\begin{pmatrix} T_{\alpha^*}(0) \\ N_{\alpha^*}(0) \end{pmatrix} = A \begin{pmatrix} T_\alpha(0) \\ N_\alpha(0) \end{pmatrix}$$

and let $b := \alpha^*(0) - A\alpha(0)$. Consider $\Phi(x) := Ax + b$ and $\tilde{\alpha} := \Phi \circ \alpha$. Then

$$\|\tilde{\alpha}'(s)\| = \left\| \frac{d}{ds}(A\alpha(s) + b) \right\| = \|A\alpha'(s)\| = 1$$

since $A \in SO(2)$ and α is p.a.l. Hence, $\tilde{\alpha}$ is a p.a.l.
$\tilde{\alpha}' = A\alpha' \Rightarrow T_{\tilde{\alpha}} = AT_\alpha \Rightarrow K_{\tilde{\alpha}} N_{\tilde{\alpha}} = T'_{\tilde{\alpha}} = AT'_\alpha = AKN_\alpha$

$$\Rightarrow K_{\tilde{\alpha}} N_{\tilde{\alpha}} = KA \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} T_\alpha = K \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} AT_\alpha = K \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} T_{\tilde{\alpha}} = KN_{\tilde{\alpha}}$$

$$\Rightarrow K_{\tilde{\alpha}} = K \qquad (*)$$

Now consider $f(s) := \langle T_{\tilde{\alpha}}, T_{\alpha^*}\rangle + \langle N_{\tilde{\alpha}}, N_{\alpha^*}\rangle$. Then

$$f'(s) = \frac{d}{ds}\left(\langle T_{\tilde{\alpha}}, T_{\alpha^*}\rangle + \langle N_{\tilde{\alpha}}, N_{\alpha^*}\rangle\right) = \langle T'_{\tilde{\alpha}}, T_{\alpha^*}\rangle + \langle T_{\tilde{\alpha}}, T'_{\alpha^*}\rangle + \langle N'_{\tilde{\alpha}}, N_{\alpha^*}\rangle + \langle N_{\tilde{\alpha}}, N'_{\alpha^*}\rangle$$

$$= \langle K_{\tilde{\alpha}} N_{\tilde{\alpha}}, T_{\alpha^*}\rangle + \langle T_{\tilde{\alpha}}, KN_{\alpha^*}\rangle + \langle -K_{\tilde{\alpha}} T_{\tilde{\alpha}}, N_{\alpha^*}\rangle + \langle N_{\tilde{\alpha}}, -KT_{\alpha^*}\rangle$$

$$\overset{(*)}{=} K\left(\langle N_{\tilde{\alpha}}, T_{\alpha^*}\rangle + \langle T_{\tilde{\alpha}}, N_{\alpha^*}\rangle - \langle T_{\tilde{\alpha}}, N_{\alpha^*}\rangle - \langle N_{\tilde{\alpha}}, T_{\alpha^*}\rangle\right) = 0.$$

Hence, $f(s) =$ constant. By the def of A, $T_{\tilde{\alpha}}(0) = AT_\alpha(0) = T_{\alpha^*}(0)$ and $N_{\tilde{\alpha}}(0) = AN_\alpha(0) = N_{\alpha^*}(0)$. Thus

$$f(s) \equiv f(0) = \langle T_{\alpha^*}(0), T_{\alpha^*}(0)\rangle + \langle N_{\alpha^*}(0), N_{\alpha^*}(0)\rangle = 1 + 1 = 2.$$

Since $T_{\tilde{\alpha}}$, T_{α^*}, $N_{\tilde{\alpha}}$ and N_{α^*} are unit vectors, then $T_{\tilde{\alpha}}(s) = T_{\alpha^*}(s)$ and $N_{\tilde{\alpha}}(s) = N_{\alpha^*}(s)$, $\forall s \in I$. Integrate $T_{\tilde{\alpha}}(s) = T_{\alpha^*}(s)$ and substitute the initial condition $\tilde{\alpha}(0) = A\alpha(0) + b = \alpha^*(0)$. Then we get $\tilde{\alpha}(s) = \alpha^*(s)$, $\forall s \in I$, that is, $\alpha^* = \Phi \circ \alpha$.

□

1.2.6 Space Curve

Definition 1.2.21. Frenet-Serret Frame
Let $\alpha : I \to \mathbb{R}^3$ be a regular p.a.l. Define the *tangent vector* of α by $T(s) := \alpha'(s)$. Furthermore, if $T'(s) \neq 0$, then we define

- *normal vector*: $N(s) := T'(s)/\|T'(s)\|$.
- *binormal vector*: $B(s) := T(s) \times N(s)$.

The *Frenet-Serret frame* of α is the orthonormal basis $\{T, N, B\}$.
Rmk: Unlike the case of plane curve, the normal vector N is simply along the same direction of T', so the curvature is always positive.

Definition 1.2.22. Curvature
The *curvature* of a regular p.a.l. is defined by $K(s) := \|T'(s)\|$.

Definition 1.2.23. Torsion
Since $\langle B, B\rangle = 1$, then $0 = 2\langle B', B\rangle$, so $B' \perp B$. On the other hand, since

$$B' = T' \times N + T \times N' = KN \times N + T \times N' = 0 + T \times N' \perp T$$

then $B'(s) = \tau(s)N(s)$ for some scalar $\tau(s)$. The scalar is defined to be the *torsion* of α.
Rmk: τ tells how far α away from being a plane curve.

Theorem 1.2.24. *Frenet-Serret Equations*

$$\begin{pmatrix} T \\ N \\ B \end{pmatrix}' = \begin{pmatrix} 0 & K & 0 \\ -K & 0 & -\tau \\ 0 & \tau & 0 \end{pmatrix} \begin{pmatrix} T \\ N \\ B \end{pmatrix}$$

Proof. By the def of curvature and torsion, it suffices to show that

$$N' = -KT - \tau B$$

Since $0 = \langle N, N\rangle' = 2\langle N', N\rangle$, then $N'(s) = a(s)T(s) + b(s)B(s)$ for some scalar $a(s), b(s)$. Hence,

$$\begin{cases} \langle N', T\rangle = a\langle T, T\rangle + b\langle B, T\rangle = a \Rightarrow -K = a \\ \langle N', B\rangle = a\langle T, B\rangle + b\langle B, B\rangle = b \Rightarrow -\tau = b \end{cases}$$

□

Rmk: If α is any regular curve, then the Frenet-Serret frame of α is obtained by that of $\beta = \alpha \circ s$ p.a.l.

Theorem 1.2.25. *Plane Curve Criterion*
Let α be a regular p.a.l. curve with $K \neq 0$. Then the followings are equivalent:

1. *α is a plane curve.*
2. *B is a constant vector.*
3. *$\tau(s) \equiv 0$, $\forall s \in I$.*

Proof. $(2 \Leftrightarrow 3)$: It's immediately from one of the Frenet-Serret equations

$$B' = \tau N$$

$(1 \Rightarrow 2)$: Let $\alpha = (f, g, 0)_{\mathcal{B}}$ for some orthonormal basis $\mathcal{B}$. Then

$$T(s) := \alpha'(s) = (f'(s), g'(s), 0)_{\mathcal{B}}$$

$$\Rightarrow T'(s) = (f''(s), g''(s), 0)_{\mathcal{B}}$$

By the def of $N = T'/\|T'\|$, we know that $N(s) = (n_1(s), n_2(s), 0)_{\mathcal{B}}$ for some $n_1(s), n_2(s)$. Then by def, we can compute $B(s)$:

$$B(s) := \begin{vmatrix} b_1 & b_2 & b_3 \\ f'(s) & g'(s) & 0 \\ n_1(s) & n_2(s) & 0 \end{vmatrix} = (0, 0, f'n_2 - g'n_1)_{\mathcal{B}}$$

where $\{b_1, b_2, b_3\} = \mathcal{B}$ (ordered). Since B is a unit vector, then it must be $(0, 0, 1)_{\mathcal{B}}$ or $(0, 0, -1)_{\mathcal{B}}$. Hence, B is a constant.
$(2 \Rightarrow 1)$: Since B is a constant, then $B' = 0$. Hence,

$$\langle \alpha(s) - \alpha(0), B \rangle' = \langle \alpha', B \rangle + \langle \alpha(s) - \alpha(0), B' \rangle = \langle T, B \rangle + 0 = 0 + 0 = 0.$$

Thus, $\langle \alpha(s) - \alpha(0), B \rangle =$ constant. Since $\langle \alpha(0) - \alpha(0), B \rangle = 0$, then $\forall s \in I$, $\langle \alpha(s) - \alpha(0), B \rangle \equiv 0$, which is a plane equation. Therefore, $\alpha(s)$ lies on a plane. □

Proposition 1.2.26. *Straight Line Characterization*
α is a straight line iff $\exists x_0 \in \mathbb{R}^3$, s.t.: every tangent line of α passes through x_0.

Proof. $(\Leftarrow)$: $\forall s \in I$, the tangent line is parallel to $T(s)$ and passes through two points x_0 and $\alpha(s)$. Hence,

$$\alpha(s) + \lambda(s)T(s) = x_0$$

for some scalar $\lambda(s)$. Then we have:

$$0 = \alpha'(s) + \lambda'(s)T(s) + \lambda(s)T'(s) = (1 + \lambda'(s))T(s) + (\lambda(s)K(s))N(s)$$

Since $T \perp N$, then $1 + \lambda'(s) = \lambda(s)K(s) = 0$, so it's easy to see that $K(s) \equiv 0\ \forall s \in I$. Hence, $T(s) =$ constant. Therefore, α is a straight line.
($\Rightarrow$): Trivial. □

Proposition 1.2.27. *Normal Plane and Sphere Criterion*
Let the normal plane of $\alpha(s)$ be the plane perpendicular to $T(s)$. Suppose α is regular and p.a.l. such that every normal plane of $\alpha(s)$ passes through a fixed point x_0. Then $\alpha(s)$ lies on a sphere.

Proof. The normal plane of $\alpha(s)$ has the equation:

$$\langle T(s), x - \alpha(s)\rangle = 0$$

Since x_0 lies on the plane, then $\langle T(s), x_0 - \alpha(s)\rangle = 0$, $\forall s \in I$. Hence,

$$\frac{d}{ds}\langle x_0 - \alpha(s), x_0 - \alpha(s)\rangle = -2\langle T(s), x_0 - \alpha(s)\rangle = 0$$

Thus, $\langle x_0 - \alpha(s), x_0 - \alpha(s)\rangle =$ constant, i.e.: $\alpha(s)$ lies on a sphere of radius $\sqrt{\text{constant}}$ centered at x_0. □

1.3 Fundamental Theorem for Space Curve

Theorem 1.3.1. *Fundamental Theorem for Space Curve*

1. *(Existence): Let $k, t : I \to \mathbb{R}$ be smooth functions with $k(s) > 0$, let $x_0 \in \mathbb{R}^3$ be a fixed point, and let $\{D, E, F\}$ be an orthonormal basis of $\mathbb{R}^3$. Then there exists a p.a.l. $\alpha : I \to \mathbb{R}^3$, s.t.:*

$$\begin{pmatrix} T(0) \\ N(0) \\ B(0) \end{pmatrix} = \begin{pmatrix} D \\ E \\ F \end{pmatrix}, \quad \begin{cases} k(s) = K_\alpha(s) \\ t(s) = \tau_\alpha(s) \end{cases} \qquad \text{and} \quad \alpha(0) = x_0.$$

2. *(Uniqueness): Suppose $\alpha, \alpha^* : I \to \mathbb{R}^3$ are two curves with $K_\alpha(s) = K_{\alpha^*}(s)$ and $\tau_\alpha(s) = \tau_{\alpha^*}(s)$, $\forall s \in I$. Then α, α^* differ by an orientation preserving rigid motion.*

Proof. **1.** Construct an ODE system, based on the Frenet-Serret Equations,

$$\begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}' = \begin{pmatrix} 0 & k(s) & 0 \\ -k(s) & 0 & t(s) \\ 0 & -t(s) & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}$$

with the initial condition

$$\begin{pmatrix} u_1(0) \\ u_2(0) \\ u_3(0) \end{pmatrix} = \begin{pmatrix} D \\ E \\ F \end{pmatrix}.$$

Then by the existence and uniqueness of ODE sols, $\exists!$ sol. of the above system $\{u_1(s), u_2(s), u_3(s)\}$.
Claim 1: $\{u_1(s), u_2(s), u_3(s)\}$ is an orthonormal basis.
Proof of claim 1: Denote, to simplify,

$$U(s) := \begin{pmatrix} u_1(s) \\ u_2(s) \\ u_3(s) \end{pmatrix} \quad \text{and} \quad M(s) := \begin{pmatrix} 0 & k(s) & 0 \\ -k(s) & 0 & t(s) \\ 0 & -t(s) & 0 \end{pmatrix}.$$

Consider $P(s) := U(s)U^T(s)$. Then $P_{ij}(s) = \langle u_i(s), u_j(s) \rangle$. Since

$$P'(s) = U'(s)U^T(s) + U(s)\left(U^T(s)\right)' = M(s)U(s)U^T(s) + U(s)U^T(s)M^T(s) = M(s)P(s) - P(s)M(s).$$

and $P(s) \equiv I$ satisfies this equation, then by the existence and uniqueness of ODE sols, $P(s) \equiv I$ is the only possible sol. Hence, claim 1 holds. □
Define $\alpha(s) := x_0 + \int_0^s u_1(\sigma)\, d\sigma$. Check:

- $\alpha'(s) = u_1(s)$ with $\|u_1(s)\| = 1$, so α is regular and p.a.l.
- $T'_\alpha(s) := \alpha''(s) = u_1'(s) = k(s)u_2(s)$, so

$$K_\alpha(s) := \|T'_\alpha(s)\| = k(s)\|u_2(s)\| = k(s) \quad \text{and then} \quad N_\alpha(s) = u_2(s).$$

- $u_3(s) = u_1(s) \times u_2(s) = T_\alpha(s) \times N_\alpha(s) := B_\alpha(s)$.
- $B'_\alpha = u_3' \Rightarrow -\tau_\alpha(s)N_\alpha(s) = -t(s)u_2(s) \Rightarrow \tau_\alpha = t$, since N_α and u_2 are unit vectors.

2. Let $K(s) := K_\alpha(s) = K_{\alpha^*}(s)$ and $\tau(s) := \tau_\alpha(s) = \tau_{\alpha^*}(s)$, and let $A \in SO(3)$ be such that

$$\begin{pmatrix} T^*(0) \\ N^*(0) \\ B^*(0) \end{pmatrix} = A \begin{pmatrix} T(0) \\ N(0) \\ B(0) \end{pmatrix}$$

and let $b_0 := \alpha^*(0) - A\alpha(0)$. Consider $\Phi(x) := Ax + b_0$ and $\tilde{\alpha} := \Phi \circ \alpha$. Then $\tilde{\alpha}$ is p.a.l. and it's easy to check that:

$$\tilde{\alpha}(0) = \alpha^*(0), \quad \begin{pmatrix} T_{\tilde{\alpha}}(s) \\ N_{\tilde{\alpha}}(s) \\ B_{\tilde{\alpha}}(s) \end{pmatrix} = \begin{pmatrix} AT_\alpha(s) \\ AN_\alpha(s) \\ AB_\alpha(s) \end{pmatrix}, \quad K_{\tilde{\alpha}} = K, \text{ and } \tau_{\tilde{\alpha}} = \tau.$$

Hence, $T_{\tilde{\alpha}}(0) = T_{\alpha^*}(0)$, $N_{\tilde{\alpha}}(0) = N_{\alpha^*}(0)$ and $B_{\tilde{\alpha}}(0) = B_{\alpha^*}(0)$ $(*)$.

Now consider

$$f(s) := \langle T_{\tilde{\alpha}}(s), T_{\alpha^*}(s)\rangle + \langle N_{\tilde{\alpha}}(s), N_{\alpha^*}(s)\rangle + \langle B_{\tilde{\alpha}}(s), B_{\alpha^*}(s)\rangle.$$

Then

$$\begin{aligned} f'(s) &= \langle T'_{\tilde{\alpha}}, T_{\alpha^*}\rangle + \langle T_{\tilde{\alpha}}, T'_{\alpha^*}\rangle + \langle N'_{\tilde{\alpha}}, N_{\alpha^*}\rangle + \langle N_{\tilde{\alpha}}, N'_{\alpha^*}\rangle + \langle B'_{\tilde{\alpha}}, B_{\alpha^*}\rangle + \langle B_{\tilde{\alpha}}, B'_{\alpha^*}\rangle \\ &= \langle K_{\tilde{\alpha}} N_{\tilde{\alpha}}, T_{\alpha^*}\rangle + \langle T_{\tilde{\alpha}}, K N_{\alpha^*}\rangle + \langle -K_{\tilde{\alpha}} T_{\tilde{\alpha}} + \tau_{\tilde{\alpha}} B_{\tilde{\alpha}}, N_{\alpha^*}\rangle \\ &+ \langle N_{\tilde{\alpha}}, -K T_{\alpha^*} + \tau B_{\alpha^*}\rangle + \langle -\tau_{\tilde{\alpha}} N_{\tilde{\alpha}}, B_{\alpha^*}\rangle + \langle B_{\tilde{\alpha}}, -\tau N_{\alpha^*}\rangle \\ &= K\left(\langle N_{\tilde{\alpha}}, T_{\alpha^*}\rangle + \langle T_{\tilde{\alpha}}, N_{\alpha^*}\rangle - \langle T_{\tilde{\alpha}}, N_{\alpha^*}\rangle - \langle N_{\tilde{\alpha}}, T_{\alpha^*}\rangle\right) \\ &+ \tau\left(\langle B_{\tilde{\alpha}}, N_{\alpha^*}\rangle + \langle N_{\tilde{\alpha}}, B_{\alpha^*}\rangle - \langle N_{\tilde{\alpha}}, B_{\alpha^*}\rangle - \langle B_{\tilde{\alpha}}, N_{\alpha^*}\rangle\right) \\ &= 0. \end{aligned}$$

Hence, $f(s) =$ constant, so by $(*)$,

$$f(s) \equiv f(0) = \langle T_{\alpha^*}(0), T_{\alpha^*}(0)\rangle + \langle N_{\alpha^*}(0), N_{\alpha^*}(0)\rangle + \langle B_{\alpha^*}(0), B_{\alpha^*}(0)\rangle = 3.$$

Since $T_{\tilde{\alpha}}$, T_{α^*}, $N_{\tilde{\alpha}}$, N_{α^*}, $B_{\tilde{\alpha}}$ and B_{α^*} are unit vectors, then $T_{\tilde{\alpha}}(s) = T_{\alpha^*}(s)$, $N_{\tilde{\alpha}}(s) = N_{\alpha^*}(s)$, and $B_{\tilde{\alpha}}(s) = B_{\alpha^*}(s)$, $\forall s \in I$. Integrate $T_{\tilde{\alpha}}(s) = T_{\alpha^*}(s)$ and substitute the initial condition $\tilde{\alpha}(0) = A\alpha(0) + b_0 = \alpha^*(0)$. Then we get $\tilde{\alpha}(s) = \alpha^*(s)$, $\forall s \in I$, that is $\alpha^* = \Phi \circ \alpha$ □

1.4 Global Theory of Plane Curves

1.4.1 Closed Curve and Simple Curve

Definition 1.4.1. Closed Curve and Simple Curve
A regular curve β is *closed* if $\beta(t) = \beta(t + a)$, for some $a > 0$, $\forall t$. The *period* of β is the lowest possible value of $a > 0$.

Furthermore, β is a *simple* curve iff β is injective, i.e.: β has no self-intersection.

Remark: For any simple closed curve $\alpha : [0, a] \to \mathbb{R}^2$ with $\alpha(s) = (x(s), y(s))$, $y([0, a])$ is closed and bounded in $\mathbb{R}$. Hence, by the Weierstrass Theorem, $\exists m \in [0, a]$ s.t. $y(m)$ is a global minimum in $y([0, a])$. Therefore, we can study $\tilde{\alpha}(s) := \alpha(m + s)$, which begins with the smallest possible y-value, and hence whose initial tangent vector $T_{\tilde{\alpha}}(0)$ is horizontal, i.e.: $\tilde{\alpha}'(0) = (x'(m), 0)$, and whose image lies on or above $\tilde{\alpha}(0)$.
Since $|T| = 1$, then we can define $0 \le \theta(s) \le 2\pi$ to be such that $T(s) = (\cos(\theta(s)), \sin(\theta(s)))$. Then

$$T'(s) = (-\theta'(s)\sin(\theta(s)), \theta'(s)\cos(\theta(s))) = \theta'(s)\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} T(s) := \theta'(s) N(s)$$

Hence, by def, $\theta'(s) = K(s)$.

1.4.2 Hopf Theorem

Definition 1.4.2. Rotation Index
The *rotation index* of a simple closed curve $\alpha : I \to \mathbb{R}^2$ with period a is given by

$$i_\alpha := \frac{1}{2\pi}(\theta(a) - \theta(0))$$

where θ is defined to be such that $T(s) = (\cos(\theta(s)), \sin(\theta(s)))$.

Theorem 1.4.3. *Hopf Let α be a simple closed curve. Then its rotation index must be 1 or -1.*

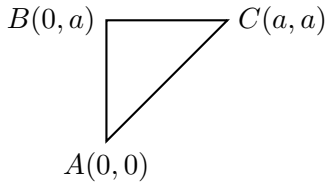
Proof. Consider $\Delta := \{(u, v) \mid 0 \leq u \leq v \leq a\} \subseteq \mathbb{R}^2$. Then we define the secant function $\sigma : \Delta \to \mathbb{R}^2$ of α by

$$\sigma(u, v) := \begin{cases} \dfrac{\alpha(v) - \alpha(u)}{\|\alpha(v) - \alpha(u)\|}, & \text{if } u < v \text{ and } (u, v) \neq (0, a) \\ T(u), & \text{if } u = v \\ -T(0), & \text{if } (u, v) = (0, a) \end{cases}$$

which is continuous and differentiable, in fact a differentiable vector field. Let $\phi : \Delta \to \mathbb{R}$ be a continuous function that is the angle of $\sigma(u, v)$, i.e.:

$$\sigma(u, v) = \Big(\cos(\phi(u, v)), \sin(\phi(u, v))\Big)$$

Then by the above remark, it's clear that $\phi(u, u) = \theta(u)$. Hence, if we denote ϕ as



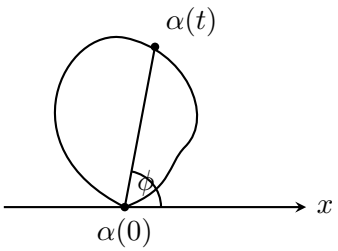
then

$$\int_{\vec{AC}} d\phi = \int_{[0,a]} d\theta = \int_0^a \theta'(s)\, ds = \theta(a) - \theta(0) := 2\pi i_\alpha$$

Since by Green's Theorem,

$$\oint_{\partial\Delta} d\phi = \oint_{\partial\Delta} \frac{\partial\phi}{\partial u}\, du + \frac{\partial\phi}{\partial v}\, dv \overset{\text{Green's}}{=\!=\!=} \iint_\Delta \left(\frac{\partial^2\phi}{\partial v \partial u} - \frac{\partial^2\phi}{\partial u \partial v}\right) du\, dv = 0$$

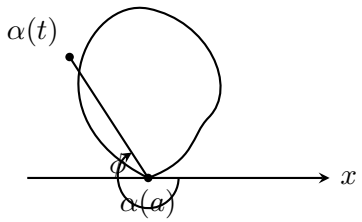
then for computing i_α, we can compute $\int_{\vec{AB}} d\phi$ and $\int_{\vec{BC}} d\phi$ separately.
On $\vec{AB}$, since by the above remark, we WLOG let



then

$$\int_{\vec{AB}} d\phi = \begin{cases} \pi - 0, & \text{counterclockwise} \\ 0 - \pi, & \text{clockwise} \end{cases}$$

Similarly, on $\vec{BC}$,



we have

$$\int_{\vec{BC}} d\phi = \begin{cases} 2\pi - \pi, & \text{counterclockwise} \\ \pi - 2\pi, & \text{clockwise} \end{cases}$$

Hence, $i_\alpha = 1$, counterclockwise or $i_\alpha = -1$, clockwise. □

Corollary 1.4.4.

$$\int_\alpha K(s)\, ds = 2\pi \quad or \quad -2\pi$$

Proof. It's immediately from the Hopf's Theorem and the computation

$$i_\alpha := \frac{1}{2\pi}(\theta(a) - \theta(0)) = \frac{1}{2\pi}\int_0^a \theta'(s)\, ds = \frac{1}{2\pi}\int_0^a K(s)\, ds = \frac{1}{2\pi}\int_\alpha K(s)\, ds.$$ □

Remark:

- In fact, for any closed spatial curve α, the total curvature satisfies that

$$\int_\alpha K(s)ds \geq 2\pi \text{ or } \leq -2\pi$$

  where the equality holds iff α is a plane curve.

- If a simple closed curve α has strictly positive curvature, then α is strictly convex.

1.4.3 Isoperimetric Inequality

Lemma 1.4.5. *Area Formula*
Let α be a simple, closed plane curve bounding a region R. Anti-clockwise, the area A of R is equal to

$$A = \int_\alpha x\,dy = -\int_\alpha y\,dx$$

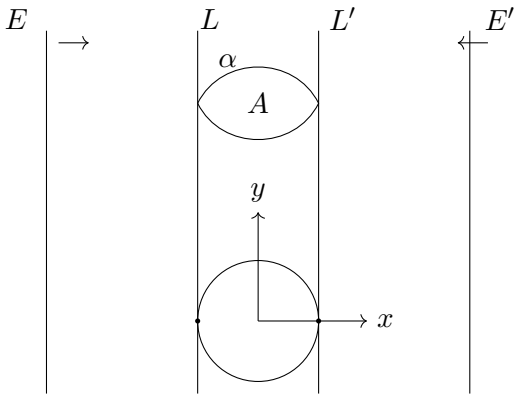
Proof. The second equality is directly by Green's Thm. □ □

Theorem 1.4.6. *Isoperimetric Inequality*
Let α be a simple, closed plane curve of length ℓ and let A be the area bounded by α. Then

$$\ell^2 \geq 4\pi A$$

where the equality holds iff α is a circle.

Proof. WLOG, let $\alpha(s) = (x(s), y(s))$ be a p.a.l., so $\alpha(\ell) = \alpha(0)$. Let E and E' be two parallel lines which do not meet α and lie on either side of α. Then move them together until they first meet α. We hence obtain two parallel tangent lines L and L' to α, so that α is entirely between L and L'. Consider a circle which is tangent to both L and L' and does not meet α. Then take a coordinate system with origin at the center of the circle and x-axis perpendicular to L and L'.



Denote $\alpha \cap L =: \{\alpha(s_0)\}$ for some $0 < s_0 < \ell$ and $\alpha \cap L' =: \{\alpha(0)\}$ (assuming). Then parametrize the circle by $\beta(s) = (z(s), w(s))$, $0 \leq s \leq \ell$, where

$$z(s) := x(s) \quad \text{and} \quad w(s) := \begin{cases} \sqrt{r^2 - x^2(s)}\,, & \text{if } 0 \leq s \leq s_0 \\ -\sqrt{r^2 - x^2(s)}\,, & \text{if } s_0 \leq s \leq \ell \end{cases}$$

(where r denotes the radius of the circle). Then by lemma above, we have:

$$A = \int_\alpha x\,dy = \int_0^\ell x(s) \cdot \frac{dy}{ds}(s)\,ds = \int_0^\ell xy'\,ds$$

and

$$\pi r^2 = -\int_\beta w\,dz = -\int_0^\ell wz'\,ds$$

Hence, compute:

$$A + \pi r^2 = \int_0^\ell (xy' - wz')\,ds = \int_0^\ell (xy' - wx')\,ds = \int_0^\ell \langle (x', y'), (-w, x)\rangle\,ds$$

$$\overset{(*)}{\leq} \int_0^\ell \left|\langle (x', y'), (w, x)\rangle\right| ds \overset{(**)}{\leq} \int_0^\ell |(x', y')| \cdot |(-w, x)|\,ds$$

where $(**)$ holds by the Cauchy-Schwarz inequality. Since α is p.a.l., then $|(x', y')| = 1$. Since $z(s) = x(s)$ and $(z, w) \in$ circle of radius r, then $|(-w, x)| = r$. Thus, $A + \pi r^2 = \int_0^\ell 1 \cdot r\,ds = r \cdot \ell$. Therefore,

$$r \cdot \ell \geq A + \pi r^2 \overset{(***)}{\geq} 2\sqrt{A\pi r^2} \Rightarrow \ell^2 \geq 4\pi A.$$

For the condition of equality, it suffices to show that the inequalities $(*)$, $(**)$, and $(***)$ all hold exactly when α is a circle.

($\Leftarrow$): We suppose first that α is a circle. Then $A = \pi r^2$, $\ell = 2\pi r$, so we're done.

($\Rightarrow$): Suppose that $(*)$, $(**)$, and $(***)$ are all equalities. Then

$$(*)\ \checkmark \Rightarrow \langle (x', y'), (-w, x)\rangle \geq 0$$

$$(**)\ \checkmark \Rightarrow (x', y') = \lambda(-w, x) \text{ with } |\lambda| = \frac{1}{r}$$

$$\Rightarrow \lambda = \frac{1}{r} \overset{\text{ODE}}{\underset{x^2+w^2=r^2}{\Longrightarrow}} x^2 + (y - c)^2 = r^2 \quad \text{for some constant } c.$$

□

Chapter 2

Surface Theory

2.1 Local Theory of Surfaces

2.1.1 Regular Surface

Definition 2.1.1. Parametrized Surface
A *parametrized surface* is a function $X : U \subseteq \mathbb{R}^2 \to \mathbb{R}^3$ which is differentiable. Furthermore, the image of X is called the *trace* of X.
Example: $X : [0, 2\pi) \times [0, \pi] \to \mathbb{R}^3$ given by

$$(\theta, \varphi) \longmapsto \begin{pmatrix} r\cos\theta \cdot \sin\varphi \\ r\sin\theta \cdot \sin\varphi \\ r\cos\varphi \end{pmatrix}$$

gives a sphere S^2. But this raises problems:

- $\forall \theta \in [0, 2\pi)$, $X(\theta, 0) = (0, 0, r)$, which means that when $\varphi = 0$, X is not bijective.
- Since $\left.\dfrac{\partial X}{\partial \theta}\right|_{(0,0)} = (0, 0, 0)$, then $\dfrac{\partial X}{\partial \theta}$ and $\dfrac{\partial X}{\partial \varphi}$ are linearly dependent at $(0, 0)$, so the Jacobian matrix has rank 1. This means that at $(0, 0)$, we can not compute the "normal vector".

To fix these problems, we use semi circles to cover S^2, since each semi circle can be parametrized by a (unit) open ball in $\mathbb{R}^2$.

Definition 2.1.2. Regular Surface
A subset $S \subseteq \mathbb{R}^3$ is called a *regular surface* if $\forall p \in S$, $\exists$ a neighborhood $V \subseteq \mathbb{R}^3$ of p (V can be seen as an open ball centered at p and radius of some $\varepsilon > 0$), and a map $X : U \to V \cap S$ for some (open connected) $U \subseteq \mathbb{R}^2$, such that the followings hold:

1. X is differentiable, i.e.: $\forall (u, v) \in U$, $X(u, v) = (x(u, v), y(u, v), z(u, v))$ has partial derivatives of all orders.
2. X is a homeomorphism from U onto $V \cap S$.

3. X is *regular*, i.e.: $\forall q \in U$,
$$dX_q = \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} \end{pmatrix}(q)$$
is of rank 2.

Example:

- Let $S^2 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$. We use 6 coordinate patches to cover S^2: Let $U := \{(u, v) \in \mathbb{R}^2 \mid u^2 + v^2 < 1\}$. Consider a map $X_1 : U \to V \cap S^2$ given by
$$(u, v) \longmapsto \left(u, v, \sqrt{1 - u^2 - v^2}\right)$$
Then <u>check</u>:
  1. u, v, $\sqrt{1 - u^2 - v^2}$ have infinite partial derivatives.
  2. X_1 is a bijection, with $X_1^{-1} : (x, y, z) \mapsto (x, y)$ continuous.
  3. $\forall (u, v) \in U$, compute
$$(dX_1)_q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ * & \# \end{pmatrix}(q)$$
  where $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is of rank 2, so $(dX_1)_q$ is of rank 2.

  Similarly, we can define
$$\begin{pmatrix} u \\ v \end{pmatrix} \xmapsto{X_2} \begin{pmatrix} u \\ v \\ -\sqrt{1 - u^2 - v^2} \end{pmatrix}, \quad \begin{pmatrix} u \\ v \end{pmatrix} \xmapsto{X_3} \begin{pmatrix} u \\ \sqrt{1 - u^2 - v^2} \\ v \end{pmatrix}, \quad \begin{pmatrix} u \\ v \end{pmatrix} \xmapsto{X_4} \begin{pmatrix} u \\ -\sqrt{1 - u^2 - v^2} \\ v \end{pmatrix}$$
$$\begin{pmatrix} u \\ v \end{pmatrix} \xmapsto{X_5} \begin{pmatrix} \sqrt{1 - u^2 - v^2} \\ u \\ v \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} u \\ v \end{pmatrix} \xmapsto{X_6} \begin{pmatrix} -\sqrt{1 - u^2 - v^2} \\ u \\ v \end{pmatrix}$$
  Then $\forall p \in S^2$, p must lie in (at least) one $V_i \cap S^2$, so we can take $\{X_i : U \to V_i \cap S^2 \mid i = 1, 2, \cdots, 6\}$ to conclude that S^2 is a regular surface.

- Let $f : U \subseteq \mathbb{R}^2 \to \mathbb{R}$ be a smooth function. Consider the *graph* of f
$$S := \text{graph}(f) := \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \;\middle|\; \begin{pmatrix} x \\ y \end{pmatrix} \in U,\ z = f(x, y) \right\}$$

To prove that S is a regular surface, in fact, we only need one coordinate patch $X : U \to S$ given by $(u, v) \mapsto (u, v, f(u, v))$. Check:

1. u, v, $f(u, v)$ are all differentiable (smooth).
2. X is bijective, with $X^{-1} : (x, y, z) \mapsto (x, y)$ continuous.
3. $dX_q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ * & \# \end{pmatrix} (q)$ is of rank 2, $\forall q \in U$.

Therefore, $S = \text{graph}(f)$ is a regular surface. Furthermore, we may ask, does the converse hold?

Proposition 2.1.3. *Regular Surfaces are Locally Graphs*
Any regular surface S is locally a graph, or equivalently, in combination with the above example, we conclude that a surface $S \subseteq \mathbb{R}^3$ is regular iff S is locally a graph of some smooth function.

Proof. It suffices to prove the first statement: $\forall p \in S$, let $X : U \xrightarrow{\cong} V \cap S$ satisfy the three conditions, with $p = X(q)$. By the regularity of X,

$$dX_q = \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} \end{pmatrix} (q)$$

is of rank 2, so at least one of the three matrices below has nonzero determinant:

$$\begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} (q), \quad \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} \end{pmatrix} (q), \quad \text{and} \quad \begin{pmatrix} \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} \end{pmatrix} (q)$$

WLOG, assume the first one has nonzero determinant, so by the Inverse Function Thm, the function $f(u, v) := (x(u, v), y(u, v))$, which is smooth since X and $(x, y, z) \mapsto (x, y)$ are smooth, has a differentiable inverse $g = f^{-1} : (x, y) \mapsto (u(x, y), v(x, y))$ onto some neighborhood of q. That is,

$$X \circ g : (x, y) \longmapsto \big(x, y, z\big(u(x, y), v(x, y)\big)\big)$$

where $z : \mathbb{R}^2 \to \mathbb{R}$ given by $(x, y) \mapsto z\big(u(x, y), v(x, y)\big)$ is the desired graph function, which is smooth, on a neighborhood of $p = X(q)$. □

- Let $f : \mathbb{R}^3 \to \mathbb{R}$ be a smooth map and let $a \in \mathbb{R}$ be such that $\forall p \in f^{-1}(a)$ with $df_p \neq 0$, i.e.: a is a *regular value* of f. Then $f^{-1}(a) \subseteq \mathbb{R}^3$ is a regular surface.

Proof. $\forall p \in f^{-1}(a)$, assume WLOG that $\dfrac{\partial f}{\partial z}(p) \neq 0$. Consider $F := f - c_a$ where c_a is the constant map. Then $\forall p \in f^{-1}(a)$, $F(p) = f(p) - a = 0$ and $\dfrac{\partial F}{\partial z}(p) = \dfrac{\partial f}{\partial z}(p) \neq 0$. Hence, we can apply the Implicit Function Thm, $\exists\ g : U \subseteq \mathbb{R}^2 \to \mathbb{R}$ smooth, s.t.: $F(u, v, g(u, v)) = 0$. Consider

$$W := \{(u, v, g(u, v)) \mid (u, v) \in U\}$$

containing p clearly. Since $F(W) = \{0\}$, then $p \in W \subseteq F^{-1}(0)$. Therefore, $\exists$ a neighborhood V of p, s.t.:

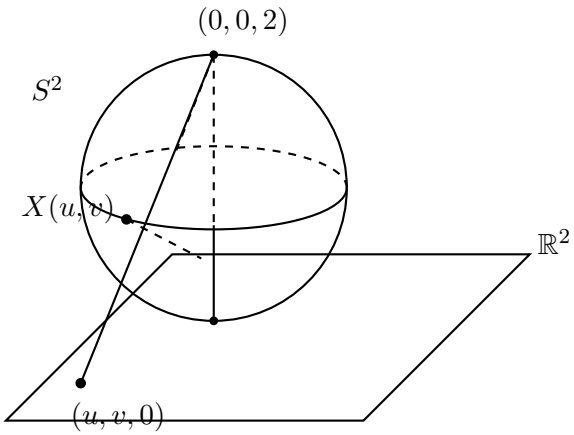
$$V \cap f^{-1}(a) = V \cap F^{-1}(0) = W = \{(x, y, z) \mid (x, y) \in U,\ z = g(x, y)\}$$

i.e.: $f^{-1}(a)$ is locally a graph of smooth function. □

- Consider $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$. Let $F = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1$, which is clearly differentiable. $\forall w = (x, y, z) \in F^{-1}(0)$, since WLOG, $x \neq 0$, then $dF_w \neq 0$, so 0 is a regular value of F. Hence, by above, $F^{-1}(0) = \{(x, y, z) \mid \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1\} \subseteq \mathbb{R}^3$ is a regular surface.

2.1.2 Change of Coordinates

Motivation: Consider the stereographic projection of S^2:



That is, define $X : \mathbb{R}^2 \to S^2 \subseteq \mathbb{R}^3$ by $X(u, v) :=$ the intersection of the straight line joining $(u, v, 0)$ and $(0, 0, 2)$. This gives a coordinate patch of $S^2 \setminus \{(0, 0, 2)\}$. This gives a way to cover S^1, different from the first example of regular surface. Hence, there are many ways to get coordinate patches for a fixed surface.

Proposition 2.1.4. *Change of Coordinates*
Let $p \in S$ be in the image of two parametrizations $X : U \to S$, $Y : V \to S$, i.e.: $p \in W = X(U) \cap Y(V)$. Then

$$\psi := X^{-1} \circ Y : Y^{-1}(W) \longrightarrow X^{-1}(W)$$

is a diffeomorphism.

Proof. Since X is regular at p, then WLOG, we can conclude that the map

$$(u,v) \xmapsto{X} \big(x(u,v), y(u,v), z(u,v)\big) \xmapsto{\pi} \big(x(u,v), y(u,v)\big)$$

is invertible on some neighborhood of $X^{-1}(p)$. Hence,

$$\psi = X^{-1} \circ Y = \underbrace{(\pi \circ X)^{-1}}_{\text{differentiable}} \circ \underbrace{(\pi \circ Y)}_{\text{differentiable}}$$

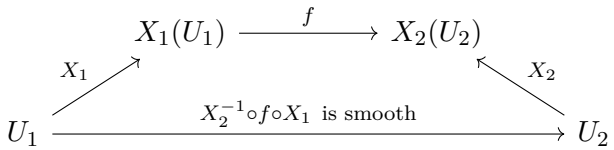
is differentiable. Similarly, ψ^{-1} is differentiable. Therefore, by the bijectivity of X, Y, ψ is a differentiable. □

Definition 2.1.5. Smooth Function on Regular Surface
Let S be a regular surface and let $f : S \to \mathbb{R}$ be a function. We say f is *smooth* at $p \in S$ if $\exists$ coordinate patch $X : U \to V \cap S$ with $p \in X(U)$, s.t.: $f \circ X$ is smooth at $q = X^{-1}(p) \in U$.
Remark:

- In fact, for any parametrization $Y : U' \to V' \cap S$ with $p \in Y(U')$,

$$f \circ Y = (f \circ X) \circ (X^{-1} \circ Y)$$

  is smooth on $Y^{-1}\big(X(U) \cap Y(U')\big)$.
- One can extend this definition:
  - to the function $f : S \to \mathbb{R}^n$, and
  - to $f : S_1 \to S_2$ to be *smooth* if $\exists$ X_1, X_2 coordinates patches of S_1 respectively S_2, such that



Example: If $f : \mathbb{R}^3 \to \mathbb{R}$ is smooth, then $f|_S : S \to \mathbb{R}$ is smooth, since $\forall X : U \to S \cap V$, $f \circ X : (u,v) \mapsto f\big(x(u,v), y(u,v), z(u,v)\big)$ is a composition of smooth functions.

Proposition 2.1.6. *Double Cone is Not Regular*
Consider $F : \mathbb{R}^3 \to \mathbb{R}$ given by $(x,y,z) \mapsto x^2 + y^2 - z^2$. Then $F^{-1}(0)$ is not a regular surface.

Proof. WARNING: We can not just use $dF_p = (2x, 2y, 2z) = 0$ at $p = (0,0,0)$, since perhaps the surface $F^{-1}(0)$ can be described as a regular surface by other parametrizations.
Suppose on the contrary that $F^{-1}(0)$ is a regular surface. Then a neighborhood of $(0,0,0)$ can be described as one of the three functions:

$$(x,y) \mapsto (x,y,f(x,y)), \quad (x,z) \mapsto (x,g(x,z),z), \quad \text{and} \quad (y,z) \mapsto (h(y,z),y,z)$$

WLOG, let the first one describe locally $F^{-1}(0)$ at $(0,0,0)$. $\forall$ open ball $V = B(0,\varepsilon)$,

$$(x_0, y_0, z_0) \in V \cap S \iff (x_0, y_0, -z_0) \in V \cap S.$$

Hence, f is not a function, contradiction. □

2.1.3 Tangent Plane

Definition 2.1.7. Total Derivative
Let $p \in S$ be in the image of two parametrizations $X_1 : U \to V \cap S$, $X_2 : \tilde{U} \to \tilde{V} \cap S$. Then the *total derivative* at p is defined as:

$$d\psi := \begin{pmatrix} \frac{\partial \tilde{u}}{\partial u} & \frac{\partial \tilde{u}}{\partial v} \\ \frac{\partial \bar{v}}{\partial u} & \frac{\partial \bar{v}}{\partial v} \end{pmatrix} = \frac{\partial(\tilde{u}, \bar{v})}{\partial(u, v)}$$

which is the Jacobian matrix of the transition map $\psi = X_2^{-1} \circ X_1 : X_1^{-1}(W) \to X_2(W)$ where $W := X_1(U) \cap X_2(\tilde{U}) \ni p$.
Rmk: By the change of coordinates Prop., ψ is a diffeomorphism, so $(d\psi)_p$ is invertible.

Proposition 2.1.8.

$$\frac{\partial X_1}{\partial u} = \frac{\partial \tilde{u}}{\partial u}\frac{\partial X_2}{\partial \tilde{u}} + \frac{\partial \bar{v}}{\partial u}\frac{\partial X_2}{\partial \bar{v}} \quad \textit{and} \quad \frac{\partial X_1}{\partial v} = \frac{\partial \tilde{u}}{\partial v}\frac{\partial X_2}{\partial \tilde{u}} + \frac{\partial \bar{v}}{\partial v}\frac{\partial X_2}{\partial \bar{v}}.$$

Proof. It's immediately from the fact that $\forall (u,v) \in X_1^{-1}(W)$, $X_1(u,v) = X_2\big(\tilde{u}(u,v), \bar{v}(u,v)\big)$. □

Corollary 2.1.9.

$$\operatorname{Span}\left\{\frac{\partial X_1}{\partial u}, \frac{\partial X_1}{\partial v}\right\} = \operatorname{Span}\left\{\frac{\partial X_2}{\partial \tilde{u}}, \frac{\partial X_2}{\partial \bar{v}}\right\}.$$

Rmk: This coro. gives the well-defined-ness of tangent plane (we'll introduce it later), i.e.: the tangent plane is not affected by using different coordinate patches.

Definition 2.1.10. Tangent Plane
Let $X : U \to V \cap S$ be a coordinate patch of a regular surface at a point $X(q) = p \in S$. Then the *tangent plane* at $p \in S$ is defined as:

$$T_pS := \operatorname{Im}(dX_p) = \operatorname{Span}\left\{\frac{\partial X}{\partial u}(q), \frac{\partial X}{\partial v}(q)\right\}.$$

Example: S^2. A coordinate patch is given by $X(u,v) = (u, \sqrt{1-u^2-v^2}, v)$. Then for $p = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) = X\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$, one has

$$dX_p = \begin{pmatrix} 1 & 0 \\ -\frac{u}{\sqrt{1-u^2-v^2}} & -\frac{v}{\sqrt{1-u^2-v^2}} \\ 0 & 1 \end{pmatrix} \Rightarrow T_p(S^2) = \operatorname{Span}\left\{\begin{pmatrix} 1 \\ -\frac{1}{\sqrt{3}} \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -\frac{1}{\sqrt{3}} \\ 1 \end{pmatrix}\right\} = \operatorname{Span}\left\{\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}\right\}.$$

- For a (local) graph, $X(u,v) = (u, v, f(u,v))$, $p \in S$ with $p = X(u_0, v_0)$,

$$T_pS = \text{Span}\left\{ \begin{pmatrix} 1 \\ 0 \\ \frac{\partial f}{\partial u}(u_0, v_0) \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \frac{\partial f}{\partial v}(u_0, v_0) \end{pmatrix} \right\}$$

or $\langle x - p, n \rangle = 0$, where n is normal to T_pS.

Definition 2.1.11. Normal Vector
The *normal vector* of a point p on a coordinate patch $X : U \to V \cap S$ is defined as:

$$N(p) := \frac{\frac{\partial X}{\partial u} \times \frac{\partial X}{\partial v}}{\| \frac{\partial X}{\partial u} \times \frac{\partial X}{\partial v} \|} (q)$$

where $q := X^{-1}(p)$.
Example: S^2.

$$N\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) = \frac{\begin{pmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix} \times \begin{pmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix}}{\| \begin{pmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix} \times \begin{pmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix} \|} = \left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

- Local graph:

$$N(p) = \frac{\left(-\frac{\partial f}{\partial u}, -\frac{\partial f}{\partial v}, 1\right)}{\| \left(-\frac{\partial f}{\partial u}, -\frac{\partial f}{\partial v}, 1\right) \|}$$

Remark: Observe that $N(p)$ can be "+" or "−" along the same direction, so it may be affected by X up to ± 1. This problem will be fixed by the "orientability".

Definition 2.1.12. Tangent Vector
A *tangent vector* at $p \in S$ is the tangent vector $\alpha'(0)$ of a regular curve $\alpha : (-\varepsilon, \varepsilon) \to S \subseteq \mathbb{R}^3$ with $\alpha(0) = p$.

Proposition 2.1.13. *The set of tangent vectors at $p \in S$ is precisely equal to T_pS.*

Proof. ($\subseteq$): Let $\alpha : (-\varepsilon, \varepsilon) \to S$ be a regular curve with $\alpha(0) = p$. Since regular surfaces are locally graphs, then $\exists\ Y : U \to V \cap S$, s.t.:

$$Y(u,v) = (u, v, f(u,v)) \quad \text{or} \quad (u, g(u,v), v) \quad \text{or} \quad (h(u,v), u, v).$$

WLOG, let $Y(u,v) = (u, v, f(u,v))$. By taking ε to be small enough, we may assume $\alpha\big((-\varepsilon, \varepsilon)\big) \subseteq V \cap S$. Now define

$$\beta = \left((-\varepsilon, \varepsilon) \xrightarrow{\alpha} V \cap S \xrightarrow{Y^{-1}} U \subseteq \mathbb{R}^2 \right)$$

where Y^{-1} is differentiable. Hence, $\alpha = Y \circ \beta = X \circ (X^{-1} \circ Y \circ \beta) = X \circ (\psi \circ \beta)$ where α, X, ψ, and β are all differentiable. Therefore,

$$\alpha'(0) = d\alpha_0 = dX_q \circ d(\psi \circ \beta)_0 = \left(\frac{\partial X}{\partial u}, \frac{\partial X}{\partial v}\right)(q) \cdot \begin{pmatrix} a \\ b \end{pmatrix} \in \text{Span}\left\{\frac{\partial X}{\partial u}(q), \frac{\partial X}{\partial v}(q)\right\} = T_pS.$$

($\supseteq$): $\forall w \in T_pS$, let $w = \left(\frac{\partial X}{\partial u}(q), \frac{\partial X}{\partial v}(q)\right)\begin{pmatrix} a \\ b \end{pmatrix}$ by the def of tangent plane, for some $a, b \in \mathbb{R}$. Then define $\gamma : (-\varepsilon, \varepsilon) \to \mathbb{R}^2$ by $t \mapsto q + t\begin{pmatrix} a \\ b \end{pmatrix}$ and then define $\alpha := X \circ \gamma$. Then we have:

$$\alpha(0) := X(\gamma(0)) = X(q) = p \quad \text{and} \quad \alpha'(0) = dX_q \circ d\gamma_0 := \left(\frac{\partial X}{\partial u}(q), \frac{\partial X}{\partial v}(q)\right) \cdot \begin{pmatrix} a \\ b \end{pmatrix}\Bigg|_{t=0} = w.$$

□

Example: Suppose $f : \mathbb{R}^3 \to \mathbb{R}$ is a smooth function and $S = f^{-1}(a)$ where a is a regular value of f. Then S is a regular surface. Let $\alpha : (-\varepsilon, \varepsilon) \to S$ be any regular curve on S, with $\alpha(0) = p$. Then

$$f(\alpha(t)) \in f(f^{-1}(a)) = \{a\} \Rightarrow f(\alpha(t)) = a, \quad \forall t \in (-\varepsilon, \varepsilon).$$

Hence, $df_p \circ d\alpha_0 = 0$, i.e.: $df_p \cdot \alpha'(0) = 0$. Therefore, df_p is normal to T_pS.

2.1.4 First Fundamental Form (f.f.f.)

Definition 2.1.14. First Fundamental Form (f.f.f.)
The *f.f.f.* of S is given by the quadratic form $I_p(v) := \langle v, v \rangle$ for $v \in T_pS$.
Remark: Let $X : U \to V \cap S$ be a coordinate patch with $X(q) = p$. Then

$$\mathcal{B} = \left\{X_u := \frac{\partial X}{\partial u}(q), X_v := \frac{\partial X}{\partial v}(q)\right\}$$

is a basis of T_pS. If we write $E_p := \langle X_u, X_u \rangle_p$, $F_p := \langle X_u, X_v \rangle_p$ and $G_p := \langle X_v, X_v \rangle_p$ (for brevity, we often drop "p"), then $\forall v, w \in T_pS$, $\exists\ a, b, c, d \in \mathbb{R}$, s.t.:

$$\langle v, w \rangle_p = \langle aX_u + bX_v, cX_u + dX_v \rangle_p$$

$$= ac\,\langle X_u, X_u \rangle_p + (bc + ad)\,\langle X_u, X_v \rangle_p + bd\,\langle X_v, X_v \rangle_p$$

$$= ac\,E + (bc + ad)\,F + bd\,G$$

$$= \begin{pmatrix} a & b \end{pmatrix}\begin{pmatrix} E & F \\ F & G \end{pmatrix}\begin{pmatrix} c \\ d \end{pmatrix}.$$

Hence, $\forall v \in T_pS$, $I_p(v) = \begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}$ where $v = aX_u + bX_v$. Therefore, we can retrieve I_p from the matrix

$$\begin{pmatrix} E & F \\ F & G \end{pmatrix} := \begin{pmatrix} \langle X_u, X_u \rangle_p & \langle X_u, X_v \rangle_p \\ \langle X_v, X_u \rangle_p & \langle X_v, X_v \rangle_p \end{pmatrix}.$$

We then will often call this matrix the f.f.f. as well.

Example:

- *Cylinder.* $X(u, v) = (\cos u, \sin u, v)$, $0 < u < 2\pi$, $v \in \mathbb{R}$.

$$\Rightarrow X_u = (-\sin u, \cos u, 0) \quad \text{and} \quad X_v = (0, 0, 1).$$

$$\Rightarrow E = \langle X_u, X_u \rangle = \sin^2 u + \cos^2 u = 1, \quad F = \langle X_u, X_v \rangle = 0, \quad \text{and} \quad G = \langle X_v, X_v \rangle = 1.$$

$$\Rightarrow I_p = \begin{pmatrix} E & F \\ F & G \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

- *Plane.* $X(u, v) = x_0 + uW_1 + vW_2 \Rightarrow I_p = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
- *Sphere.* $X(u, v) = (u, v, \sqrt{1 - u^2 - v^2})$

$$\Rightarrow X_u = \left(1, 0, \frac{-u}{\sqrt{1 - u^2 - v^2}}\right) \quad \text{and} \quad X_v = \left(0, 1, \frac{-v}{\sqrt{1 - u^2 - v^2}}\right)$$

$$\Rightarrow E = 1 + \frac{u^2}{1 - u^2 - v^2} = \frac{1 - v^2}{1 - u^2 - v^2}, \quad F = \frac{uv}{1 - u^2 - v^2}, \quad \text{and} \quad G = \frac{1 - u^2}{1 - u^2 - v^2}$$

$$\Rightarrow I_p = \begin{pmatrix} \frac{1 - v^2}{1 - u^2 - v^2} & \frac{uv}{1 - u^2 - v^2} \\ \frac{uv}{1 - u^2 - v^2} & \frac{1 - u^2}{1 - u^2 - v^2} \end{pmatrix}$$

On the other hand, consider another coordinate patch $\tilde{X}(\theta, \phi) := (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$.

$$\Rightarrow \tilde{X}_\theta = (\cos\theta\cos\phi, \cos\theta\sin\phi, -\sin\theta) \quad \text{and} \quad \tilde{X}_\phi = (-\sin\theta\sin\phi, \sin\theta\cos\phi, 0).$$

$$\Rightarrow \tilde{E} = 1, \quad \tilde{F} = 0, \quad \text{and} \quad \tilde{G} = \sin^2\theta \Rightarrow I_p = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2\theta \end{pmatrix}$$

Therefore, it's natural to ask: if we have two parametrizations $X : U \to V \cap S$ and $\tilde{X} : \tilde{U} \to \tilde{V} \cap S$, then how are their f.f.f.s related?

Proposition 2.1.15. *Let $\psi = X^{-1} \circ \tilde{X}$ be the transition map. Then*

$$\begin{pmatrix} \tilde{E} & \tilde{F} \\ \tilde{F} & \tilde{G} \end{pmatrix} = (d\psi)^t \begin{pmatrix} E & F \\ F & G \end{pmatrix} (d\psi)$$

i.e.: the related f.f.f.s are conjugated by the Jacobian matrix of ψ.

Proof.

$$\begin{pmatrix} \tilde{E} & \tilde{F} \\ \tilde{F} & \tilde{G} \end{pmatrix} = \left(\frac{\partial \tilde{X}}{\partial \tilde{u}}, \frac{\partial \tilde{X}}{\partial \bar{v}}\right)^t \left(\frac{\partial \tilde{X}}{\partial \tilde{u}}, \frac{\partial \tilde{X}}{\partial \bar{v}}\right) = (d\tilde{X})^t (d\tilde{X}) = (dX \circ d\psi)^t (dX \circ d\psi)$$
$$= (d\psi)^t \Big((dX)^t (dX)\Big)(d\psi) = (d\psi)^t \begin{pmatrix} E & F \\ F & G \end{pmatrix} (d\psi).$$

□

Corollary 2.1.16.

$$\sqrt{\tilde{E}\tilde{G} - \tilde{F}^2} = \sqrt{EG - F^2} \cdot \left|\det\left(\frac{\partial(u,v)}{\partial(\tilde{u},\bar{v})}\right)\right|$$

Proof. It's immediately from the fact that $\psi : (\tilde{u}, \bar{v}) \mapsto (u(\tilde{u}, \bar{v}), v(\tilde{u}, \bar{v}))$ gives the Jacobian matrix

$$d\psi = \frac{\partial(u,v)}{\partial(\tilde{u},\bar{v})} \Rightarrow \det(d\psi) = \det\left((d\psi)^t\right) = \det\left(\frac{\partial(u,v)}{\partial(\tilde{u},\bar{v})}\right).$$

□

2.1.5 Arc-length and Area

Arc-length of Regular Curve on Regular Surface

Proposition 2.1.17. *Let $\alpha : I \to U \subseteq \mathbb{R}^2$ be a regular curve on U, let $S \subseteq \mathbb{R}^3$ be a regular surface, and let $X : U \to V \cap S$ be a coordinate patch. Then the curve $\beta = X \circ \alpha : I \to S$ has length*

$$\int_I |\beta'(t)|\, dt = \int_I \sqrt{\left\langle \frac{d(X\circ\alpha)}{dt}, \frac{d(X\circ\alpha)}{dt} \right\rangle}\, dt = \int_I \sqrt{(dX \cdot d\alpha)^t (dX \cdot d\alpha)}\, dt$$
$$= \int_I \sqrt{(x'(t)\ y'(t))(dX)^t dX \begin{pmatrix} x'(t) \\ y'(t) \end{pmatrix}}\, dt = \int_I \sqrt{(x'(t)\ y'(t))\, I_p \begin{pmatrix} x'(t) \\ y'(t) \end{pmatrix}}\, dt$$
$$= \int_I \sqrt{E(x'(t))^2 + G(y'(t))^2 + 2F\, x'(t)\, y'(t)}\, dt$$

Area of Regular Surface

Definition 2.1.18. Area
Let $X : U \to V \cap S$ be a coordinate patch of S and let $R \subseteq S$ be such that $X(Q) = R$ for some $Q \subseteq U$. Then the *area* of R is given by:

$$\int_Q d\sigma = \int_Q |X_u \times X_v|\, du\, dv$$

Remark: Note that

$$|X_u \times X_v|^2 = |X_u|^2|X_v|^2 \sin^2\theta = |X_u|^2|X_v|^2(1-\cos^2\theta) = |X_u|^2|X_v|^2 - \langle X_u, X_v\rangle^2$$

Hence, $|X_u \times X_v| = \sqrt{EG-F^2}$. If $\tilde{X} : \tilde{U} \to \tilde{V} \cap S$ is another coordinate patch of S, and $\psi := X^{-1} \circ \tilde{X} : \tilde{Q} \to Q$ is a transition map, then

$$\int_{\tilde{Q}} \sqrt{\tilde{E}\tilde{G}-\tilde{F}^2}\, d\tilde{u}\, d\tilde{v} = \int_{\tilde{Q}} \sqrt{EG-F^2} \cdot \left|\det\left(\frac{\partial(u,v)}{\partial(\tilde{u},\tilde{v})}\right)\right| d\tilde{u}\, d\tilde{v} = \int_Q \sqrt{EG-F^2}\, du\, dv$$

Therefore, the area of R is independent of the choice of coordinate patches.
Example:

- *Cylinder.* $X(u,v) = (\cos u, \sin u, v)$, $0 < u < 2\pi$, $v \in (a,b)$. Then $I_p = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow EG - F^2 = 1$. Hence,
$$\text{the area of } X(u,v) = \int_a^b \int_0^{2\pi} 1\, du\, dv = 2\pi(b-a).$$

- *Sphere.* $X(\theta,\phi) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$ with $0 < \theta \le \pi$, $0 < \phi < 2\pi$ covers the sphere except for poles. Hence,
$$\text{the area of } S^2 = \int_0^{\pi} \int_0^{2\pi} \sqrt{\sin^2\theta}\, d\phi\, d\theta = 4\pi.$$

- *Torus.* $X(u,v) = \big((R+r\cos v)\cos u, (R+r\cos v)\sin u, r\sin v\big)$, with $0 < u < 2\pi$, $0 < v < 2\pi$. Then

  $X_u = \big(-(R+r\cos v)\sin u, (R+r\cos v)\cos u, 0\big), \quad X_v = \big(-r\sin v\cos u, -r\sin v\sin u, r\cos v\big)$

  Hence,

  $$\begin{cases} E = \langle X_u, X_u\rangle = (R+r\cos v)^2\sin^2 u + (R+r\cos v)^2\cos^2 u = (R+r\cos v)^2 \\ F = \langle X_u, X_v\rangle = (R+r\cos v)\sin v\cos u\sin u - (R+r\cos v)\sin v\cos u\sin u + 0 = 0 \\ G = \langle X_v, X_v\rangle = r^2\sin^2 v\cos^2 u + r^2\sin^2 v\sin^2 u + r^2\cos^2 v = r^2 \end{cases}$$

  Therefore,
  $$I_p = \begin{pmatrix} (R+r\cos v)^2 & 0 \\ 0 & r^2 \end{pmatrix}$$
  so $\sqrt{EG-F^2} = \sqrt{(R+r\cos v)^2 r^2} = r(R+r\cos v) \Rightarrow$ Area of $T^2 = \int_0^{2\pi}\int_0^{2\pi} r(R+r\cos v)\, du\, dv = 4\pi^2 Rr$.

2.2 Gauss Map

2.2.1 Orientability

Goal: We want to define the curvature of a regular surface. Recall that in the case of curve, in particular, of plane curve,

$$T' = KN \quad \text{and} \quad N' = -KT$$

Note that we need a global concept, normal vector. We deal with this problem by definition:

Definition 2.2.1. Orientable Surface
A surface S is called *orientable* if there is a smooth map $N : S \to S^2$, s.t.: $N(p) \perp T_pS$, $\forall p \in S$.
Remark:

- The codomain of the smooth map is S^2 instead of $\mathbb{R}^3$, which is because the normal vector is unit.
- This map N is called the *Gauss map* of S, which is well-defined up to a sign.

Example:

- If S is covered by exactly one coordinate patch $X : U \to S$, then

$$N(p) = \frac{X_u \times X_v}{\|X_u \times X_v\|}\,(q) \quad \text{where } q = X^{-1}(p)$$

  e.g.: every graph is orientable.

- Let $f : \mathbb{R}^3 \to \mathbb{R}$ be a smooth function, and let $a \in \mathbb{R}$ be a regular value. Then consider $F := f - a$, so $S = F^{-1}(0)$ is a regular surface. Hence,

$$N(p) := dF_p = \left(\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z}\right)(p)$$

2.2.2 Differential

Definition 2.2.2. Differential
Let $f : S_1 \to S_2$ be a smooth map. The *differential* of f is defined as follows: $\forall w \in T_pS_1$, let $\alpha : (-\varepsilon, \varepsilon) \to S_1$ be a curve on S_1 such that $\alpha(0) = p$ and $\alpha'(0) = w$. Then $f \circ \alpha : (-\varepsilon, \varepsilon) \to S_2$ is such that $(f \circ \alpha)(0) = f(p)$. Hence, we can define well-definedly:

$$df_p(w) := (f \circ \alpha)'(0) \in T_{f(p)}S_2.$$

Proposition 2.2.3. *The image of $dN_p : T_pS \to T_{N(p)}S^2$ is precisely in T_pS.*

Proof. Let $X_{(u,v)}$ be a coordinate patch of S. Consider $\beta : (-\varepsilon, \varepsilon) \to U^\circ$ given by $t \mapsto (t, 0)$ and $\alpha := X \circ \beta$. Then $\alpha(0) = p$ and $\alpha'(0) = X_u$. Then

$$dN_p(X_u) := (N \circ \alpha)'(0) := (N \circ X \circ \beta)'(0) = \frac{d}{dt}(N \circ X)(t_0) := (N \circ X)_u$$

Similarly, $dN_p(X_v) = \frac{d}{dt}(N \circ X)(t_0) := (N \circ X)_v$. Since $dN_p : T_pS \to T_{N(p)}S^2$ is a linear transformation and $\{X_u, X_v\}$ forms a basis for T_pS, then we can get dN_p from $dN_p(X_u) = (N \circ X)_u$ and $dN_p(X_v) = (N \circ X)_v$. Since every vector on the unit sphere S^2 is perpendicular to the tangent space at that point, then $T_{N(p)}S^2 \perp N(p)$, so

$$\langle dN_p, N_{f(p)} \rangle = 0 \Rightarrow \langle (N \circ X)_u, N \rangle = \langle (N \circ X)_v, N \rangle = 0.$$

Hence, $(N \circ X)_u, (N \circ X)_v \in T_pS$. Therefore, $dN_p : T_pS \to T_pS$. □

Remark: From now on, to simplify, we write $N(u, v)$ instead of $(N \circ X)(u, v)$.
Example:

- $S = P$ is a plane, i.e.: $X_{(u,v)} = x_0 + u\,\underline{w}_1 + v\,\underline{w}_2 \Rightarrow X_u = \underline{w}_1,\ X_v = \underline{w}_2 \Rightarrow N(u, v) = \frac{X_u \times X_v}{\|X_u \times X_v\|} = \frac{\underline{w}_1 \times \underline{w}_2}{\|\underline{w}_1 \times \underline{w}_2\|} =$ constant $\Rightarrow N_u = N_v = 0$. Hence, dN_p is the zero map, i.e.: N no change.
- $S = S^2$: $X_{(u,v)} = \left(u, v, \sqrt{1 - u^2 - v^2}\right) \Rightarrow X_u = \left(1, 0, \frac{-u}{\sqrt{1-u^2-v^2}}\right),\ X_v = \left(0, 1, \frac{-v}{\sqrt{1-u^2-v^2}}\right) \Rightarrow N(u, v) = \frac{X_u \times X_v}{\|X_u \times X_v\|} = \left(u, v, \sqrt{1 - u^2 - v^2}\right) \Rightarrow N_u = X_u,\ N_v = X_v \Rightarrow dN_p = \mathrm{id}_{T_pS}$.
- *Saddle*: $X_{(u,v)} = (u, v, v^2 - u^2) \Rightarrow X_u = (1, 0, -2u),\ X_v = (0, 1, 2v)$.
  $\Rightarrow X_u\Big|_{(0,0,0)} = (1, 0, 0), \quad X_v\Big|_{(0,0,0)} = (0, 1, 0) \quad \text{and} \quad N_{(u,v)} = \left(\frac{2u}{\sqrt{4u^2+4v^2+1}}, \frac{-2v}{\sqrt{4u^2+4v^2+1}}, \frac{1}{\sqrt{4u^2+4v^2+1}}\right) \Rightarrow N_u\Big|_{(0,0,0)} = (2, 0, 0) \quad \text{and} \quad N_v\Big|_{(0,0,0)} = (0, -2, 0) \Rightarrow dN_{(0,0,0)}(X_u) = 2X_u\Big|_{(0,0,0)} \quad \text{and} \quad dN_{(0,0,0)}(X_v) = -2X_v\Big|_{(0,0,0)}$.

2.2.3 Classification of Points of Surface

Proposition 2.2.4. *The linear operator $dN_p : T_pS \to T_pS$ is a self-adjoint operator w.r.t. the inner product $\langle , \rangle : T_pS \times T_pS \to \mathbb{R}$, i.e.: $\langle dN_p(x), y \rangle = \langle x, dN_p(y) \rangle$.*

Proof. $\forall w_1, w_2 \in T_pS$, $\exists\ a, b, c, d \in \mathbb{R}$, s.t.: $w_1 = aX_u + bX_v$ and $w_2 = cX_u + dX_v$. Then

$$\langle dN_p(w_1), w_2 \rangle = \langle aN_u + bN_v, cX_u + dX_v \rangle \quad \text{and} \quad \langle w_1, dN_p(w_2) \rangle = \langle aX_u + bX_v, cN_u + dN_v \rangle.$$

Since by the symmetry of $\langle , \rangle$, we have: $\langle N_u, X_u \rangle = \langle X_u, N_u \rangle$ and $\langle N_v, X_v \rangle = \langle X_v, N_v \rangle$, then it suffices to show that $\langle N_u, X_v \rangle = \langle X_u, N_v \rangle$ and $\langle N_v, X_u \rangle = \langle X_v, N_u \rangle$. Note that

$$\frac{\partial}{\partial u}\langle N, X_v \rangle = \frac{\partial}{\partial u} 0 = 0 \Rightarrow \langle N_u, X_v \rangle + \langle N, X_{uv} \rangle = 0 \Rightarrow \langle N_u, X_v \rangle = -\langle N, X_{uv} \rangle$$

and similarly that

$$\frac{\partial}{\partial v}\langle N, X_u \rangle = 0 \Rightarrow \langle N_v, X_u \rangle = -\langle N, X_{uv} \rangle$$

Since $X_{uv} = X_{vu}$, then $\langle N_v, X_u \rangle = \langle N_u, X_v \rangle = \langle X_v, N_u \rangle$ and $\langle N_u, X_v \rangle = \langle N_v, X_u \rangle = \langle X_u, N_v \rangle$. □

Definition 2.2.5. Weingarten Operator
The *self-adjoint* operator $-dN_p : T_pS \to T_pS$ is called the *Weingarten operator* or *shape operator*.
Remark: Let $A : v \mapsto v$ be a self-adjoint operator. Then $\forall i, j$,

$$a_{ij} = \langle Ae_j, e_i \rangle = \langle e_j, Ae_i \rangle = a_{ji}$$

where e_i, e_j are standard basis elements. Hence, in real vector space, every self-adjoint operator is symmetric. Since every symmetric matrix is diagonalizable, then every Weingarten operator $-dN_p$ has an orthonormal basis of real eigenvectors.

Definition 2.2.6. Curvature

- *Principal curvature*: the real eigenvalues $\lambda_1 \leq \lambda_2$ of $-dN_p$ are called the *principal curvatures* of $p \in S$. Furthermore, if $\lambda_1 = \lambda_2 = \lambda$, i.e.: $dN_p = \lambda I$, then all directions in T_pS are called *principal directions* of $p \in S$. Otherwise, if $\lambda_1 < \lambda_2$, then only the unit eigenvectors $e_1, e_2 \in T_pS$ corresponding to λ_1 respectively λ_2 are called the *principal directions* of $p \in S$.
- *Mean curvature*: The *mean curvature* of $p \in S$ is defined as $H := \frac{1}{2}(\lambda_1 + \lambda_2) = \frac{1}{2}\text{tr}(-dN_p)$
- *Gaussian curvature / scalar curvature*: $K := \lambda_1\lambda_2 = \det(-dN_p)$.

Remark: Note that principal curvatures and mean curvature may differ by a sign by using different $N : S \to S^2$, but principal direction and Gaussian curvature do *not* change.
Example:

- $S = sphere$ of radius $r \Rightarrow dN_p = \begin{pmatrix} -\frac{1}{r} & 0 \\ 0 & -\frac{1}{r} \end{pmatrix} \Rightarrow$ principal directions are all directions on T_pS and $H = -\frac{1}{r}$, $K = \frac{1}{r^2}$.
- $S = cylinder$: $X_{(u,v)} = (a\cos u, a\sin u, v) \Rightarrow X_u = (-a\sin u, a\cos u, 0)$, $X_v = (0,0,1) \Rightarrow N_{(u,v)} = (\cos u, \sin u, 0) \Rightarrow N_u = (-\sin u, \cos u, 0)$, $N_v = 0$. Hence, principal directions are: X_u, X_v, $\lambda_1 = -\frac{1}{a}$, $\lambda_2 = 0 \Rightarrow H = -\frac{1}{2a}$, $K = 0$.
- $Saddle$: $X_{(u,v)} = (u, v, v^2 - u^2)$, $p = (0,0,0) \Rightarrow N_u = 2X_u$, $N_v = -2X_v$, so $\lambda_1 = -2$, $\lambda_2 = 2 \Rightarrow H = 0$, $K = -4$.

Definition 2.2.7. Classification of Points of Surface
A point $p \in S$ is called:

- *elliptic* if $\det(-dN_p) > 0$, e.g.: sphere
- *hyperbolic* if $\det(-dN_p) < 0$, e.g.: saddle
- *parabolic* if $\det(-dN_p) = 0$ but $dN_p \neq 0$, e.g.: cylinder
- *planar* if $dN_p = 0$, e.g.: plane

2.2.4 Second Fundamental Form

Definition 2.2.8. Second Fundamental Form

We define a bilinear form $\langle\langle , \rangle\rangle : T_pS \times T_pS \to \mathbb{R}$ by

$$\langle\langle v, w\rangle\rangle := \langle -dN_p(v), w\rangle.$$

The second fundamental form (s.f.f.) $\mathbb{I}_p : T_pS \to \mathbb{R}$ is defined as:

$$\mathbb{I}_p(w) := \langle\langle w, w\rangle\rangle = \langle -dN_p(w), w\rangle.$$

Remark:

- *Check*: $\langle\langle v, w\rangle\rangle = \langle\langle w, v\rangle\rangle$ since $-dN_p$ is self-adjoint and $\langle , \rangle$ is linear.
- WARNING: $\langle\langle v, v\rangle\rangle$ can be negative for $v \neq 0$. Hence, $\langle\langle , \rangle\rangle$ is not an inner product.
- How to understand $\mathbb{I}_p$? Suppose $\alpha : (-\varepsilon, \varepsilon) \to S$ is a p.a.l. with $\alpha(0) = p$, $\alpha'(0) = w$. Then we have $T'_\alpha(0) = \alpha''(0) = K_\alpha(0)\, N_\alpha(0)$. (WARNING : $N_\alpha(0)$ is the normal on α, while $N\big(\alpha(0)\big)$ denotes the normal on S, at p. In general, they are not the same.) Since $\langle T'_\alpha(0), T_\alpha(0)\rangle = 0$, then
$$K_\alpha(0)\, N_\alpha(0) = T'_\alpha(0) \in \mathrm{Span}\big\{N\big(\alpha(0)\big),\ N\big(\alpha(0)\big) \times T_\alpha(0) =: T\big\}$$
i.e.: $K_\alpha N_\alpha = K_n N + K_g T$ for some scalar K_n, K_g and hence we can compute:
$$K_n = \big\langle K_\alpha N_\alpha, N\big\rangle = \big\langle T'_\alpha(0), N\big\rangle = \big\langle \alpha''(0), N\big\rangle$$
$$K_g = \big\langle K_\alpha N_\alpha, T\big\rangle = \big\langle T'_\alpha(0), T\big\rangle = \big\langle \alpha''(0), T\big\rangle$$

Definition 2.2.9. Normal Curvature and Geodesic Curvature
K_n is called the *normal curvature* of α on S and K_g is called the *geodesic curvature* of α on S.
Remark: K_n measures how α bend normal S and K_g measures how α bend within S.
Example: Consider a unit vector $w \in T_pS$. Define the *normal section* of w as the plane spanned by $N(p)$ and w. Then its intersection with S gives a regular curve β p.a.l. with $\beta(0) = p$, $\beta'(0) = w$. This curve β is called the *sectional curve* of w.

$$\beta''(0) = T_\beta(0) = K_\beta(0)\, N_\beta(0) = K_n N + K_g T.$$

Since β is a plane curve, then its normal N_β must lie in that plane, say, the normal section of w. Hence, $K_g = 0$. Therefore,

$$K_n = \begin{cases} K_\beta & \text{, if } N_\beta \text{ is along the direction of } N \\ -K_\beta & \text{, if } N_\beta \text{ is against the direction of } N. \end{cases}$$

Theorem 2.2.10. *Let $w \in T_pS$ be of unit length, and let $\alpha : (-\varepsilon, \varepsilon) \to S$ p.a.l. with $\alpha(0) = p$, and $\alpha'(0) = w$. Then the normal curvature of α on S is equal to $\mathbb{I}_p(w)$.*

Proof.

$$\begin{aligned}\mathbb{I}_p(w) &:= \langle -dN_p(w), w\rangle = \langle -dN_p(\alpha'(0)), \alpha'(0)\rangle = \Big\langle -(N\circ\alpha)'(0), \alpha'(0)\Big\rangle \\ &= \Big\langle (N\circ\alpha)(0), \alpha''(0)\Big\rangle = \Big\langle N(p), K_\alpha(0)\, N_\alpha(0)\Big\rangle := K_n\end{aligned}$$

□

Remark: This Thm gives the geometric interpretation of $\mathbb{I}_p(w)$.

- The normal curvature of any $\alpha : (-\varepsilon, \varepsilon) \to S$ with $\alpha'(0) = w$ are the same $\mathbb{I}_p(w)$.
- $\mathbb{I}_p(w)$ measures the normal curvature of the *sectional curve* of w.

Corollary 2.2.11. *Let $w \in T_pS$ be of unit length. Then the curvature of the sectional curve of w must lie within λ_1 and λ_2, i.e.:*

$$\lambda_1 \le \mathbb{I}_p(w) \le \lambda_2 \quad \forall w \in T_pS \text{ with } ||w|| = 1.$$

Proof. Since $||w|| = 1$, then write $w = \cos\theta\, e_1 + \sin\theta\, e_2$ where e_1 and e_2 are the principal directions of $-dN_p$ and are orthonormal basis of T_pS. Then

$$\mathbb{I}_p(w) := \langle -dN_p(w), w\rangle = \Big\langle -dN_p(\cos\theta\, e_1 + \sin\theta\, e_2), \cos\theta\, e_1 + \sin\theta\, e_2\Big\rangle = \lambda_1\cos^2\theta + \lambda_2\sin^2\theta$$

Hence, $\lambda_1 \le \mathbb{I}_p(w) \le \lambda_2$ and furthermore, the maximum or minimum are obtained when: $\theta = 0$, π respectively $\pi/2$, $3\pi/2$. □

Remark: We can then distinguish the points by:

- *Elliptic points*: $\lambda_1 \le \mathbb{I}_p \le \lambda_2$, $\lambda_1\lambda_2 > 0$.
- *Hyperbolic points*: $\lambda_1 \le \mathbb{I}_p \le \lambda_2$, $\lambda_1\lambda_2 < 0$.
- *Parabolic points*: $\lambda_1 \le \mathbb{I}_p \le \lambda_2$, $\lambda_1\lambda_2 = 0$, $\mathbb{I}_p \ne 0$.
- *Planar points*: $\mathbb{I}_p = 0$.

Definition 2.2.12. Let $N : U \to S^2$ be the normal map w.r.t. $X : U \to V \cap S$. Then we define:

$$e := -\langle N_u, X_u\rangle, \quad f := -\langle N_u, X_v\rangle, \quad \text{and} \quad g := -\langle N_v, X_v\rangle.$$

Proposition 2.2.13. *If $w_1 = aX_u + bX_v$, $w_2 = cX_u + dX_v$, then*

$$\langle\langle w_1, w_2\rangle\rangle = \begin{pmatrix} a & b\end{pmatrix}\begin{pmatrix} e & f\\ f & g\end{pmatrix}\begin{pmatrix} c\\ d\end{pmatrix}.$$

Proposition 2.2.14.

$$\mathbb{I}_p(w) = \begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} e & f \\ f & g \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} \quad \textit{where } w = aX_u + bX_v.$$

Remark:

- As before, we call $\begin{pmatrix} e & f \\ f & g \end{pmatrix}$ the *Second Fundamental Form* of $p \in S$.
- But sometimes N_u and N_v are hard to compute, e.g.: Consider a parametrization $X(u,v) = (v\cos u, v\sin u, u)$, $(u,v) \in \mathbb{R} \times (0,\infty)$, of *helicoid*. Then $X_u = (-v\sin u, v\cos u, 1)$ and $X_v = (\cos u, \sin u, 0)$, so

$$N = \frac{X_u \times X_v}{\|X_u \times X_v\|} = \frac{1}{\sqrt{1+v^2}} \begin{pmatrix} -\sin u \\ \cos u \\ -v \end{pmatrix}$$

  Hence, we see that N_u, N_v are hard to compute. But we have:

$$\langle N, X_u \rangle = 0 \Rightarrow \partial_u \langle N, X_u \rangle = \langle N_u, X_u \rangle + \langle N, X_{uu} \rangle = -e + \langle N, X_{uu} \rangle = 0,$$

  similarly, $0 = -f + \langle N, X_{uv} \rangle$ and $0 = -g + \langle N, X_{vv} \rangle$. Then we can compute $e = \langle N, X_{uu} \rangle$, $f = \langle N, X_{uv} \rangle$ and $g = \langle N, X_{vv} \rangle$. We still use the helicoid as the example to see the powers of these formulas:

$$X_{uu} = (-v\cos u, -v\sin u, 0), \quad X_{uv} = (-\sin u, \cos u, 0), \quad X_{vv} = (0,0,0).$$

  Then $e = 0$, $f = \frac{1}{\sqrt{1+v^2}}$ and $g = 0$

$$\Rightarrow \mathbb{I}_p = \begin{pmatrix} 0 & \frac{1}{\sqrt{1+v^2}} \\ \frac{1}{\sqrt{1+v^2}} & 0 \end{pmatrix} \Rightarrow \mathbb{I}_p(w) = \frac{2ab}{\sqrt{1+v^2}}$$

  for $w = aX_u + bX_v \in T_pS$.

2.2.5 Weingarten Equation

Theorem 2.2.15. *Weingarten equation. Let $\mathcal{B} = \{X_u, X_v\}$ be a basis for T_pS. Then the Weingarten operator $-dN_p : T_pS \to T_pS$ can be computed by*

$$(-dN_p)_{\mathcal{B}} = I_p^{-1}\, \mathbb{I}_p.$$

This formula is called the Weingarten equation.

Proof. Let θ denote the angle between X_u and X_v. Since X_u and X_v are linearly independent, then

$|\cos\theta| < 1$. Hence,

$$\begin{aligned} EG - F^2 &= \|X_u \wedge X_v\|^2 - \langle X_u, X_v\rangle^2 = \|X_u\|^2\|X_v\|^2 - \|X_u\|^2\|X_v\|^2\cos^2\theta \\ &= \|X_u\|^2\|X_v\|^2(1-\cos^2\theta) > 0 \end{aligned}$$

i.e.: I_p is always definite positive, so we have the well-defined-ness of I_p^{-1}. Now, for proving the formula, suppose $(-dN_p)_{\mathcal{B}} = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$, i.e.:

$$-N_u = -dN_p(X_u) = aX_u + bX_v \quad \text{and} \quad -N_v = -dN_p(X_v) = cX_u + dX_v.$$

Hence,

$$\begin{cases} e := \langle -N_u, X_u\rangle = aE + bF \\ f := \langle -N_u, X_v\rangle = aF + bG \quad \text{or} \quad f := \langle -N_v, X_u\rangle = cE + dF \\ g := \langle -N_v, X_v\rangle = cF + dG. \end{cases}$$

Therefore,

$$\mathbb{I}_p = \begin{pmatrix} e & f \\ f & g \end{pmatrix} = \begin{pmatrix} E & F \\ F & G \end{pmatrix}\begin{pmatrix} a & c \\ b & d \end{pmatrix} = I_p(-dN_p)_{\mathcal{B}}$$

□

Corollary 2.2.16. *The Gaussian curvature* $K := \det(-dN_p) = \det(\mathbb{I}_p)/\det(I_p) = \dfrac{eg-f^2}{EG-F^2}$ *and the mean curvature* $H := \frac{1}{2}\mathrm{tr}(-dN_p) = \frac{1}{2}\cdot\dfrac{eG-2fF+gE}{EG-F^2}$.

Example:

- *Helicoid*: $X(u,v) = (v\cos u, v\sin u, u)$. Then

$$I_p = \begin{pmatrix} 1+v^2 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \mathbb{I}_p = \begin{pmatrix} 0 & \frac{1}{\sqrt{1+v^2}} \\ \frac{1}{\sqrt{1+v^2}} & 0 \end{pmatrix}$$

  Hence, by the Weingarten equation,

$$(-dN_p)_{\mathcal{B}} = I_p^{-1}\mathbb{I}_p = \begin{pmatrix} \frac{1}{1+v^2} & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & \frac{1}{\sqrt{1+v^2}} \\ \frac{1}{\sqrt{1+v^2}} & 0 \end{pmatrix} = \begin{pmatrix} 0 & (1+v^2)^{-\frac{3}{2}} \\ (1+v^2)^{-\frac{1}{2}} & 0 \end{pmatrix}$$

  and the mean curvature $H := \frac{1}{2}\mathrm{tr}(-dN_p) = 0$ and the Gaussian curvature $K = \det(-dN_p) = -(1+v^2)^{-2}$.

- $X(u,v) = (\cosh v\cos u, \cosh v\sin u, v)$ (the *catenoid*). Then $X_u = (-\cosh v\sin u, \cosh v\cos u, 0)$

and $X_v = (\sinh v \cos u, \sinh v \sin u, 1)$, so

$$E = \cosh^2 v, \quad F = 0, \quad \text{and} \quad G = \sinh^2 v + 1 = \cosh^2 v \Rightarrow I_p = \begin{pmatrix} \cosh^2 v & 0 \\ 0 & \cosh^2 v \end{pmatrix}$$

Furthermore, $X_{uu} = (-\cosh v \cos u, -\cosh v \sin u, 0)$, $X_{uv} = (-\sinh v \sin u, \sinh v \cos u, 0)$, $X_{vv} = (\cosh v \cos u, \cosh v \sin u, 0)$ and $N = \frac{1}{\cosh v} \begin{pmatrix} \cos u \\ \sin u \\ -\sinh v \end{pmatrix}$, so

$$e = -1, \quad f = 0, \quad g = 1 \Rightarrow \mathbb{I}_p = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

Hence, by the Weingarten equation,

$$\big(-dN_p\big)_{\mathcal{B}} = I_p^{-1}\mathbb{I}_p = \frac{1}{\cosh^2 v}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{\cosh^2 v} & 0 \\ 0 & \frac{1}{\cosh^2 v} \end{pmatrix}$$

and the mean curvature $H = 0$ and the Gaussian curvature $K = -\frac{1}{\cosh^4 v}$.

- Consider a parametrization $X(u, v) = \big((a + r\cos u)\cos v, (a + r\cos u)\sin v, r\sin u\big)$ of torus. Then $X_u = (-r\sin u\cos v, -r\sin u\sin v, r\cos u)$ and $X_v = \big(-(a+r\cos u)\sin v, (a+r\cos u)\cos v, 0\big)$, so

$$E = r^2, \quad F = 0, \quad \text{and} \quad G = (a + r\cos u)^2 \Rightarrow I_p = \begin{pmatrix} r^2 & 0 \\ 0 & (a + r\cos u)^2 \end{pmatrix}$$

Furthermore, $X_{uu} = (-r\cos u\cos v, -r\cos u\sin v, -r\sin u)$, $X_{uv} = (r\sin u\sin v, -r\sin u\cos v, 0)$, $X_{vv} = \big(-(a + r\cos u)\cos v, -(a + r\cos u)\sin v, 0\big)$, and $N = (\cos u\cos v, \cos u\sin v, \sin u)$, so

$$e = r, \quad f = 0, \quad \text{and} \quad g = (a + r\cos u)\cos u \Rightarrow \mathbb{I}_p = \begin{pmatrix} r & 0 \\ 0 & (a + r\cos u)\cos u \end{pmatrix}$$

Hence, the Gaussian curvature

$$K = \frac{\cos u}{r\,(a + r\cos u)}. \quad \text{Thus,} \quad K \begin{cases} > 0 & \text{, when } -\frac{\pi}{2} < u < \frac{\pi}{2} \\ = 0 & \text{, when } u = \frac{\pi}{2} \text{ or } \frac{3\pi}{2} \\ < 0 & \text{, when } \frac{\pi}{2} < u < \frac{3\pi}{2} \end{cases}$$

Moreover,

$$\int_{\left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]^2} K\, dA = \int_{-\frac{\pi}{2}}^{\frac{3\pi}{2}} \int_{-\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{\cos u}{r\,(a + r\cos u)} \cdot r\,(a + r\cos u)\, du\, dv = \int_{-\frac{\pi}{2}}^{\frac{3\pi}{2}} \int_{-\frac{\pi}{2}}^{\frac{3\pi}{2}} \cos u\, du\, dv = 0.$$

Then we may ask: Does every surface satisfy this condition, i.e.: $\int K\, dA = 0$? The answer is

No: For any ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$,

$$\int K\,dA = 4\pi.$$

2.2.6 Umbilical Points

Motivation: Consider $\lambda^2 - 2H\lambda + K = 0$, whose roots are the principal curvatures λ_1, λ_2, since $\frac{1}{2}(\lambda_1+\lambda_2) =: H$ and $\lambda_1\lambda_2 =: K$. Hence, $\lambda = H \pm \sqrt{H^2-K}$, which is a differentiable function w.r.t. $p \in S$, except possible at $H^2 = K$.

Definition 2.2.17. Umbilical Point
A point $p \in S$ is called *umbilical* if $H^2 = K$ at p, or equivalently, if $\lambda_1 = \lambda_2$ at p.

Theorem 2.2.18. *Let S be a connected regular surface. If S is totally umbilical, i.e.: every point on S is umbilical, then S is contained in a sphere or a plane.*

Proof. Let $p \in S$ be any point. Then $-dN_p = \begin{pmatrix}\lambda & 0\\ 0 & \lambda\end{pmatrix}_\beta$ for some $\lambda = \lambda(p)$. Thus, $-N_u = -dN_p(X_u) = \lambda X_u$ and $-N_v = -dN_p(X_v) = \lambda X_v$. Hence,

$$-N_{uv} = \lambda_v X_u + \lambda X_{uv} \quad \text{and} \quad -N_{vu} = \lambda_u X_v + \lambda X_{vu}$$

By the smoothness of N and X, we have $N_{uv} = N_{vu}$ and $X_{uv} = X_{vu}$, so $\lambda_v X_u = \lambda_u X_v$. By the linear independence of $\{X_u, X_v\}$, we have $\lambda_u = \lambda_v = 0$. Since S is connected, then λ is constant.
Case 1: $\lambda = 0 \Rightarrow -dN_p = 0 \Rightarrow S$ is contained in a plane.
Case 2: $\lambda \neq 0$. Consider $d\Big(X + \frac{1}{\lambda}N\Big)_p = dX_p + \frac{1}{\lambda}dN_p = \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}_\beta - \frac{1}{\lambda}\begin{pmatrix}\lambda & 0\\ 0 & \lambda\end{pmatrix}_\beta = 0$. Hence, $X + \frac{1}{\lambda}N = \text{constant } c \Rightarrow \|X - c\| = \frac{1}{|\lambda|} \Rightarrow X$ is contained in a sphere centered at c of radius $\frac{1}{|\lambda|}$. □

2.3 Third Fundamental Form

Definition 2.3.1. Third Fundamental Form $\mathbb{I}I_p$
$\forall w_1, w_2 \in T_pS$,

$$\mathbb{I}I_p(w_1, w_2) := \Big\langle (-dN_p)^2 w_1,\ w_2 \Big\rangle = \langle -dN_p w_1,\ -dN_p w_2\rangle.$$

Theorem 2.3.2. $\mathbb{I}I_p = 2H\mathbb{I}_p - KI_p$.

Proof. By the motivation of umbilical point, we know that $\lambda^2 - 2H\lambda + K = 0$, where λ is a principal curvature, i.e.: $-dN_p w = \lambda w$. Hence,

$$\begin{aligned}\mathbb{I}I_p(w) &:= \Big\langle -dN_p^2 w,\ w\Big\rangle = \langle \lambda^2 w,\ w\rangle = \langle (2H\lambda - K)w,\ w\rangle \\ &= 2H\langle \lambda w,\ w\rangle - K\langle w, w\rangle = (2H\mathbb{I}_p - KI_p)(w).\end{aligned}$$

□

Definition 2.3.3. N-th Fundamental Form $\mathbb{N}_p$
$\forall w_1, w_2 \in T_pS$,

$$\mathbb{N}_p(w_1, w_2) := \Big\langle (-dN_p)^{N-1} w_1,\ w_2 \Big\rangle$$

2.4 Minimal Surface

2.4.1 Definition

Idea: Given a coordinate patch $X : U \to V \cap S$, take a closed disc $\overline{D} \subseteq U$. If we "perturb" the surface $X(\overline{D})$ a little bit, up or down, then we may ask: Will the area go up or down?

Proposition 2.4.1. *Let $\varphi : \overline{D} \times (-\varepsilon, \varepsilon) \to \mathbb{R}^3$ be given by $(u, v, t) \mapsto X(u, v) + th(u, v)N(u, v)$, where N is the normal to S and $h : \overline{D} \to \mathbb{R}$ with $h\big|_{\partial\overline{D}} = 0$ is the "perturbing function", and let $X^t(u, v) := \varphi(u, v, t)$. Then*

$$\left.\frac{dA}{dt}\right|_{t=0} = -\int_D 2hH\sqrt{EG - F^2}\, du\, dv.$$

Proof. $X^t_u = X_u + th_uN + thN_u$, $X^t_v = X_v + th_vN + thN_v$. Then

$$\begin{aligned} E^t &= E + 2th_u\langle X_u, N\rangle + 2th\langle X_u, N_u\rangle + t^2h_u^2\langle N, N\rangle + 2t^2h_uh\langle N, N_u\rangle + t^2h^2\langle N_u, N_u\rangle \\ &= E + 2th_u \cdot 0 + 2th \cdot (-e) + t^2h_u^2 \cdot 1 + 2t^2h_uh \cdot 0 + t^2h^2\langle N_u, N_u\rangle \\ &= E - 2the + O(t^2) \end{aligned}$$

and similarly, $F^t = F - 2thf + O(t^2)$ and $G^t = G - 2thg + O(t^2)$. Hence,

$$\begin{aligned} E^tG^t - (F^t)^2 &= EG - F^2 - 2th(Eg - 2fF + Ge) + O(t^2) \\ &= EG - F^2 - 4th(EG - F^2)H + O(t^2) \\ &= (EG - F^2)(1 - 4thH) + O(t^2) \\ &= (EG - F^2)(1 - 4thH + O(t^2)) \end{aligned}$$

Therefore, $A(t) = \iint_{\overline{D}} \sqrt{E^tG^t - (F^t)^2}\, du\, dv = \iint_{\overline{D}} \sqrt{EG - F^2}\sqrt{1 - 4thH + O(t^2)}\, du\, dv$, so

$$\frac{dA}{dt} = \iint_{\overline{D}} \sqrt{EG - F^2}\, \frac{-2hH + O(t)}{\sqrt{1 - 4thH + O(t^2)}}\, du\, dv \Rightarrow \left.\frac{dA}{dt}\right|_{t=0} = -\int_D 2hH\sqrt{EG - F^2}\, du\, dv.$$

□

Corollary 2.4.2. *$H = 0 \iff A'(0) = 0\ \forall \overline{D} \subseteq U$ and $\forall$ smooth h.*

Proof. ($\Rightarrow$) is obvious. Conversely, suppose on the contrary that $H \neq 0$ at some $p \in \overline{D}$. Then one can choose h such that $h(p) = H(p) \neq 0$. Hence, $A'(0) < 0$, contradicting the assumption. □

Definition 2.4.3. Minimal Surface
A regular parametrized surface is a *minimal surface* if $H = 0$ at every point.

Theorem 2.4.4. *Let S be a minimal surface with parametrization $X(u, v) = (u, v, f(u, v))$. Then f satisfies second order quasi-linear PDE*

$$(1 + f_u^2)f_{vv} - 2f_u f_v f_{uv} + (1 + f_v^2)f_{uu} = 0.$$

When $B^2 - 4AC < 0$, this is an elliptic PDE.

Proof. Let $h : \overline{D} \to \mathbb{R}$ be smooth s.t.: $h\big|_{\partial\overline{D}} = 0$. Consider a family $\{F^t := f + th\}_{t\in(-\varepsilon,\varepsilon)}$ of graphs and then a family $\{X^t = (u, v, F^t)\}_{t\in(-\varepsilon,\varepsilon)}$ of parametrizations. All have the same boundary. Then compute

$$0 = \left.\frac{dA^t}{dt}\right|_{t=0} = \iint_{\overline{D}} \left.\frac{d}{dt}\right|_{t=0} \sqrt{1 + \|\nabla F^t\|^2}\, du\, dv = \iint_{\overline{D}} \frac{\langle \nabla f, \nabla h\rangle}{\sqrt{1 + \|\nabla f\|^2}}\, du\, dv = \iint_{\overline{D}} -h \operatorname{div}\left(\frac{\nabla f}{\sqrt{1 + \|\nabla f\|^2}}\right) du\, dv$$

Since this h is arbitrary, then $-\operatorname{div}\left(\frac{\nabla f}{\sqrt{1+\|\nabla f\|^2}}\right) = 0$. Then we're done since

$$-\operatorname{div}\left(\frac{\nabla f}{\sqrt{1 + \|\nabla f\|^2}}\right) = \frac{(1 + f_u^2)f_{vv} - 2f_u f_v f_{uv} + (1 + f_v^2)f_{uu}}{(1 + \|\nabla f\|^2)^{\frac{3}{2}}}.$$

□

Example: Suppose a solution is of the form $f = \alpha(u) + \beta(v)$. Then by Thm, $(1 + \alpha_u^2)\beta_{vv} + (1 + \beta_v^2)\alpha_{uu} = 0$, so

$$-\frac{1 + \alpha_u^2}{\alpha_{uu}} = \frac{1 + \beta_v^2}{\beta_{vv}}.$$

Hence, there is $\lambda \in \mathbb{R}$, s.t.:

$$-(1 + \alpha_u^2) = \lambda\, \alpha_{uu} \quad \text{and} \quad (1 + \beta_v^2) = \lambda\, \beta_{vv}.$$

Therefore,

$$f(u, v) = \ln\left(\frac{\cos\left(\frac{v}{\lambda}\right)}{\cos\left(\frac{u}{\lambda}\right)}\right)$$

and the minimal surface $(u, v, f(u, v))$ is called the *Scherk's first surface* with $\left(\frac{u}{\lambda}, \frac{v}{\lambda}\right) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \times \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

2.4.2 Isothermal Parametrization

Definition 2.4.5. Isothermal Parametrization

A parametrization $X(u,v)$ of a surface is called *isothermal* if $E = G$ and $F = 0$, or equivalently, since I_p is positive definite, then $\exists\, e^{\varphi(u,v)}$, s.t.: $I_p = e^{\varphi}(du^2 + dv^2)$.

Theorem 2.4.6. *Isothermal parametrization exists on minimal surface. In fact, it exists on any smooth (analytic) surface.*

Proof. We only prove the "minimal part", since the further needs a lot of advanced PDE. Since S is minimal, then $\exists$ smooth $f : \mathbb{R}^2 \to \mathbb{R}$, s.t.: $(u, v, f(u,v))$ is a parametrization of S and the second order PDE

$$(1 + f_u^2)f_{vv} - 2f_u f_v f_{uv} + (1 + f_v^2)f_{uu} = 0$$

is satisfied. If we denote the LHS of the PDE equation by F, then it can be checked that

$$\frac{\partial}{\partial v}\left(\frac{1+f_u^2}{\sqrt{1+f_u^2+f_v^2}}\right) - \frac{\partial}{\partial u}\left(\frac{f_u f_v}{\sqrt{1+f_u^2+f_v^2}}\right) = \frac{-f_v F}{\sqrt{1+f_u^2+f_v^2}} = 0$$

$$\frac{\partial}{\partial u}\left(\frac{1+f_v^2}{\sqrt{1+f_u^2+f_v^2}}\right) - \frac{\partial}{\partial v}\left(\frac{f_u f_v}{\sqrt{1+f_u^2+f_v^2}}\right) = \frac{-f_u F}{\sqrt{1+f_u^2+f_v^2}} = 0$$

Hence, if we define $Q := \frac{1+f_u^2}{a}$, $R := \frac{1+f_v^2}{a}$ and $P = \frac{f_u f_v}{a}$ where $a = \sqrt{1+f_u^2+f_v^2}$, then we get: $\frac{\partial Q}{\partial v} - \frac{\partial P}{\partial u} = 0$ and $\frac{\partial R}{\partial u} - \frac{\partial P}{\partial v} = 0$, so the curl of vector fields $V = (Q, P)$ and $W = (P, R)$ are both zero and hence they are conservative. By Green's Thm,

$$0 = \iint_D \left(\frac{\partial P}{\partial u} - \frac{\partial Q}{\partial v}\right) du\, dv = \int_\gamma Q\, du + P\, dv \quad \text{and} \quad 0 = \iint_D \left(\frac{\partial R}{\partial u} - \frac{\partial P}{\partial v}\right) du\, dv = \int_\gamma P\, du + R\, dv$$

$\forall$ closed loop $\gamma \in D$. Hence, $\exists$ potentials θ, ω, s.t.: $\nabla\theta = V$, $\nabla\omega = W$. Then consider a linear transformation $T : D \to \mathbb{R}^2$ given by $(u,v) \mapsto (u + \theta, v + \omega)$, whose Jacobian is

$$\mathcal{J}(T) = \begin{pmatrix} 1+\theta_u & \theta_v \\ \omega_u & 1+\omega_v \end{pmatrix}$$

Then $\det(\mathcal{J}(T)) = (1+\theta_u)(1+\omega_v) - \theta_v\omega_u = (1+Q)(1+R) - P^2 = \frac{(a+1)^2}{a} > 0$. Hence, we can apply the inverse fct thm: $\exists\, T^{-1}(x,y) = (u,v)$. Then

$$\begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} = \mathcal{J}(T^{-1}) = \mathcal{J}(T)^{-1} = \frac{1}{\det(\mathcal{J}(T))}\begin{pmatrix} 1+\omega_v & -\theta_v \\ -\omega_u & 1+\theta_u \end{pmatrix} = \frac{a}{(1+a)^2}\begin{pmatrix} 1+R & -P \\ -P & 1+Q \end{pmatrix}$$

Now consider a reparametrization $X(x,y) = (u(x,y), v(x,y), f(u(x,y), v(x,y)))$ of S. Then compute:

$$X_x = \left(\frac{a+aR}{(1+a)^2},\ \frac{-aP}{(1+a)^2},\ f_u\frac{a+aR}{(1+a)^2} + f_v\frac{-aP}{(1+a)^2}\right)$$

$$X_y = \left(\frac{-aP}{(1+a)^2},\ \frac{a+aQ}{(1+a)^2},\ f_u\frac{-aP}{(1+a)^2} + f_v\frac{a+aQ}{(1+a)^2}\right)$$

Hence, $E = \langle X_x, X_x\rangle = \frac{a}{(1+a)^2} = G$ and $F = 0$, so $X(x,y)$ is isothermal. □

Theorem 2.4.7. *$\forall\, X(u,v)$ isothermal parametrization of S, we have*

$$\Delta X = 2EHN$$

where N is the normal to S.

Proof. Since X is isothermal, then $\langle X_u, X_u\rangle = \langle X_v, X_v\rangle$ and $\langle X_u, X_v\rangle = 0$. Hence,

$$\frac{\partial}{\partial u}\langle X_u, X_u\rangle = \frac{\partial}{\partial u}\langle X_v, X_v\rangle \Rightarrow \langle X_{uu}, X_u\rangle = \langle X_{vu}, X_v\rangle$$

$$\frac{\partial}{\partial v}\langle X_u, X_u\rangle = \frac{\partial}{\partial v}\langle X_v, X_v\rangle \Rightarrow \langle X_{uv}, X_u\rangle = \langle X_{vv}, X_v\rangle$$

$$\frac{\partial}{\partial u}\langle X_u, X_v\rangle = 0 \Rightarrow \langle X_{uu}, X_v\rangle + \langle X_u, X_{vu}\rangle = 0$$

$$\frac{\partial}{\partial v}\langle X_u, X_v\rangle = 0 \Rightarrow \langle X_{uv}, X_v\rangle + \langle X_u, X_{vv}\rangle = 0$$

Number these from top to bottom by ①, ②, ③ and ④. Then

$$① + ④ \Rightarrow \Big\langle X_{uu} + X_{vv}\ ,\ X_u \Big\rangle = 0$$

$$② + ③ \Rightarrow \Big\langle X_{uu} + X_{vv}\ ,\ X_v \Big\rangle = 0$$

Hence, $\Delta X = X_{uu} + X_{vv}$ is in N direction, so $\Delta X = \lambda N$ for some scalar λ. Since X is isothermal, then $H = \frac{1}{2}\cdot\frac{eG-2fF+gE}{EG-F^2} = \frac{e+g}{2E}$. Therefore,

$$\lambda = \langle \Delta X, N\rangle = \langle X_{uu} + X_{vv}, N\rangle = \langle X_{uu}, N\rangle + \langle X_{vv}, N\rangle = e + g = 2EH.$$

□

Corollary 2.4.8. *S is a minimal surface with isothermal parametrization $X(u,v)$ iff $\Delta X = 0$, i.e.: X is harmonic.*

2.4.3 Perspective of Complex Analysis

Recall:

- $z = u + iv \Rightarrow u = \frac{z+\bar{z}}{2}$, $v = \frac{z-\bar{z}}{2}$. Then we have:

$$\begin{cases} \frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial u} - i\,\frac{\partial}{\partial v}\right) \\ \frac{\partial}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial}{\partial u} + i\,\frac{\partial}{\partial v}\right) \end{cases}$$

- If the complex function φ is holomorphic, i.e.: φ satisfies the Cauchy-Riemann equation $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$, then $\frac{\partial \varphi}{\partial \bar{z}} = 0$.

- We write $X(u,v) = X(z,\bar{z})$.

Definition 2.4.9. $X(z,\bar{z}) = (x^1, x^2, x^3)$. Let $\varphi = (\varphi^1, \varphi^2, \varphi^3)$ where $\varphi^i := \frac{\partial x^j}{\partial z}$. Then we define:

$$(\varphi)^2 := \sum_{j=1}^{3} (\varphi^j)^2 = \sum_{j=1}^{3} \left(\frac{\partial x^j}{\partial z}\right)^2 \quad \text{and} \quad |\varphi|^2 := \sum_{j=1}^{3} |\varphi^j|^2.$$

Proposition 2.4.10. *Let $X\big(x^1(u,v), x^2(u,v), x^3(u,v)\big)$ be a parametrization of a surface S.*

1. *φ^j is holomorphic iff x^j is harmonic.*
2. *X is isothermal iff $(\varphi)^2 = 0$.*
3. *If X is isothermal, then S is regular iff $|\varphi|^2 \neq 0$.*

Proof. 1.

$$\begin{aligned}\frac{\partial \varphi^j}{\partial \bar{z}} &:= \frac{\partial}{\partial \bar{z}}\left(\frac{\partial x^j}{\partial z}\right) = \frac{\partial}{\partial \bar{z}}\left(\frac{1}{2}(x^j_u - i\, x^j_v)\right) = \frac{1}{2}\left(\frac{\partial x^j_u}{\partial \bar{z}} - i\,\frac{\partial x^j_v}{\partial \bar{z}}\right) \\ &= \frac{1}{4}(x^j_{uu} + i\, x^j_{uv}) - \frac{i}{4}(x^j_{vu} + i\, x^j_{vv}) = \frac{1}{4}(x^j_{uu} + x^j_{vv}) = \frac{1}{4}\Delta x^j\end{aligned}$$

2.

$$\begin{aligned}(\varphi)^2 &:= \left(\frac{\partial x^1}{\partial z}\right)^2 + \left(\frac{\partial x^2}{\partial z}\right)^2 + \left(\frac{\partial x^3}{\partial z}\right)^2 \\ &= \frac{1}{4}(x^1_u - i\, x^1_v)^2 + \frac{1}{4}(x^2_u - i\, x^2_v)^2 + \frac{1}{4}(x^3_u - i\, x^3_v)^2 \\ &= \frac{1}{4}\sum_{j=1}^{3}\Big((x^j_u)^2 - 2i\, x^j_u\, x^j_v - (x^j_v)^2\Big) \\ &= \frac{1}{4}\langle X_u, X_u\rangle - \frac{i}{2}\langle X_u, X_v\rangle - \frac{1}{4}\langle X_v, X_v\rangle \\ &= \frac{1}{4}(E - G) - \frac{i}{2}F\end{aligned}$$

3.

$$\begin{aligned}|\varphi|^2 &:= \sum_{j=1}^{3} |\varphi^j|^2 = \sum_{j=1}^{3} \left|\frac{\partial x^j}{\partial z}\right|^2 = \sum_{j=1}^{3} |x^j_u - i\, x^j_v|^2 \\ &= \sum_{j=1}^{3}\Big((x^j_u)^2 + (x^j_v)^2\Big) = \langle X_u, X_u\rangle + \langle X_v, X_v\rangle = 2E\end{aligned}$$

where the last equality is because X is isothermal. Since S is regular $\iff$ $X_u \times X_v \neq 0$ $\iff$ $\|X_u \times X_v\|^2 = EG - F^2 = E^2 > 0$ $\iff$ $|\varphi|^2 = 2E \neq 0$. □

Theorem 2.4.11. *Weierstrass-Enneper integral representation. Let D be a simply-connected domain in $\mathbb{C}$. Suppose a complex function g is meromorphic on D and a complex function f is*

holomorphic on D such that fg^2 is holomorphic on D. The holomorphic map $\varphi = (\varphi^1, \varphi^2, \varphi^3)$ given by

$$\varphi^1 = \frac{1}{2}f(1-g^2), \quad \varphi^2 = \frac{i}{2}f(1+g^2), \quad \textit{and} \quad \varphi^3 = fg$$

determines a minimal surface. Conversely, given a minimal surface S with isothermal parametrization $X(u,v) = (x^1, x^2, x^3)$ with domain containing the origin, there is a holomorphic map $\varphi = (\varphi^1, \varphi^2, \varphi^3)$ such that:

$$x^j(z, \bar{z}) = c^j + 2\,\mathrm{Re}\int_\gamma \varphi^j\, dz$$

where γ is a path from z_0 to z and c^j are constants.

Proof. For the first part, we w.t.s.: $\varphi = \left(\frac{1}{2}f(1-g^2), \frac{i}{2}f(1+g^2), fg\right)$ determines minimal surface: Since fg^2 is holomorphic, then φ is holomorphic. Then compute:

$$(\varphi)^2 := (\varphi^1)^2 + (\varphi^2)^2 + (\varphi^3)^2 = \frac{1}{4}f^2(1-g^2)^2 + \frac{-1}{4}f^2(1+g^2)^2 + f^2g^2$$

$$= \frac{1}{4}f^2\left((1-g^2)^2 - (1+g^2)^2 + 4g^2\right) = 0.$$

Hence, by the previous prop, we're done. For the converse part,

$$dx^j = \frac{\partial x^j}{\partial z}\,dz + \frac{\partial x^j}{\partial \bar{z}}\,d\bar{z} := \varphi^j\,dz + \overline{\varphi}^j\,d\bar{z} = \varphi^j\,dz + \overline{\varphi^j\,dz} = 2\,\mathrm{Re}(\varphi^j\,dz).$$

Therefore,

$$x^j = \int_\gamma dx^j = c^j + \int_\gamma 2\,\mathrm{Re}(\varphi^j\,dz) = c^j + 2\,\mathrm{Re}\int_\gamma \varphi^j\,dz$$

for some constant c^j. □

Example: $f = 1$, $g = z^2 + 2$. Then $X = \mathrm{Re}\begin{pmatrix} \int f(1-g^2)\,dz \\ \int if(1+g^2)\,dz \\ 2\int fg\,dz \end{pmatrix} = \mathrm{Re}\begin{pmatrix} z - \frac{z^3}{3} \\ iz + i\frac{z^3}{3} \\ z^2 \end{pmatrix}$. Replace z by $(u+iv)$ and then we have:

$$X(u,v) = \begin{pmatrix} u - \frac{u^3}{3} + uv^2 \\ -v - u^2v + \frac{v^3}{3} \\ u^2 - v^2 \end{pmatrix}$$

Hence, $X_u = \begin{pmatrix} 1 - u^2 + v^2 \\ -2uv \\ 2u \end{pmatrix}$ and $X_v = \begin{pmatrix} 2uv \\ -1 - u^2 + v^2 \\ -2v \end{pmatrix} \Rightarrow F = 0$, $E = G = (1+u+v)^2$, so X is isothermal. Furthermore, $X_{uu} = \begin{pmatrix} -2u \\ -2v \\ 2 \end{pmatrix}$ and $X_{vv} = \begin{pmatrix} 2u \\ 2v \\ -2 \end{pmatrix}$, so $\Delta X = 0$. Therefore, X is a minimal surface. (This surface has a special name called *Enneper first surface.*)

- *Catenoid family*: Let $f = e^{-i\theta}e^z$ and $g = e^{-z}$. Then

$$\varphi = \Big(\frac{1}{2}f(1-g^2),\ \frac{i}{2}f(1+g^2),\ fg\Big) = (e^{-i\theta}\sinh z,\ ie^{-i\theta}\cosh z,\ e^{-i\theta})$$

  When $\theta = 0$, we can get the usual *catenoid*. When $\theta = \frac{\pi}{2}$, we can get the *helicoid*.

2.5 Gaussian Curvature

Theorem 2.5.1. *The Gaussian curvature of $p \in S$ can be computed by*

$$K = \lim_{n\to\infty} \frac{\operatorname{sgn}\operatorname{area}(N(B_n))}{B_n}$$

where $B_1 \supseteq B_2 \supseteq \cdots \supseteq B_n \supseteq B_{n+1} \supseteq \cdots$ are balls around p with $\lim_{n\to\infty}\operatorname{area}(B_n) = 0$, i.e.: $d(\operatorname{area}\circ N) = K\, d(\operatorname{area})$ where $N : S \to S^2$ Gauss map.

Proof. Let $R_n := X^{-1}(B_n)$. Then $\operatorname{area}(B_n) = \iint_{R_n} \|X_u \times X_v\|\, du\, dv$, so

$$\operatorname{area}(N(B_n)) = \iint_{R_n} \|N_u \times N_v\|\, du\, dv$$

Lemma 2.5.2. $\forall A \in M_3(\mathbb{R})$, $\forall v_1, v_2 \in \mathbb{R}^3$, $\|Av_1 \times Av_2\| = |\det(A)| \cdot \|v_1 \times v_2\|$.

Hence, by Lemma, $\|N_u \times N_v\| = |\det(-dN_p)\| \cdot \|X_u \times X_v\| := |K| \cdot \|X_u \times X_v\|$. Therefore,

$$\lim_{n\to\infty} \frac{\operatorname{area}(N(B_n))}{\operatorname{area}(B_n)} = \lim_{n\to\infty} \frac{\iint_{R_n} |K| \cdot \|X_u \times X_v\|\, du\, dv}{\iint_{R_n} \|X_u \times X_v\|\, du\, dv} = |K(p)|.$$

□

Part II

Modern Differential Geometry

Chapter 3

Smooth Manifolds

3.1 Topological Manifolds

3.1.1 Definitions and Examples

Definition 3.1.1. Topological Manifolds
Let M be a topological space M is called a *topological manifold of dimension* n or a *topological* n*-manifold* if it has the following properties:

1. M is Hausdorff, i.e.: $\forall p \neq q \in M, \exists$ distinct open $U, V \subseteq M$, s.t.: $p \in U$ and $q \in V$.
2. M is second-countable, i.e.: there is a countable basis for $\mathcal{T}_M$.
3. M is locally Euclidean of dimension n, i.e.: every point of M admits an open neighborhood in M that is homeomorphic to open subset of $\mathbb{R}^n$.

Remark:

- In a Hausdorff space, finite subsets are closed and limits of convergent sequences are unique.
- Th second-countablility ensures the existence of partitions of unity which will be introduced later.
- The third property means, intuitively, that M looks locally like $\mathbb{R}^n$. More specifically, $\forall p \in M$, we can find an open subset $U \subseteq M$ containing p, an open subset $\tilde{U} \subseteq \mathbb{R}^n$ and a homeomorphism $\varphi : U \xrightarrow{\cong} \tilde{U}$.

Proposition 3.1.2. *Equivalent Condition-of-the Third Property of Topological Manifolds Proposition*
Let M be a Hausdorff and second-countable topological space. Then M is a topological manifold iff every point of M admits an open neighborhood in M that is homeomorphic to an open ball in $\mathbb{R}^n$.

Proof.
- ($\Leftarrow$): It immediately follows from the def of topological manifolds and from the fact that open balls are open.

- ($\Rightarrow$): We use the notation in the above remark. Consider $x_0 := \varphi(p) \in \tilde{U}$. Since $\tilde{U}$ is open in $\mathbb{R}^n$, then $x_0 \in B_r(x_0) \subseteq \tilde{U}$ for some $r > 0$. Define $W := \varphi^{-1}(B_r(x_0))$. Since φ is a homeomorphism, then φ^{-1} is continuous and hence W is open.

□

Convention: Let M be a topological manifold.

- We write the dimension of M as $\dim M$.
- M^n denotes the topological manifold M of dimension n.

Theorem 3.1.3. *Topological Invariance of Dimension Theorem*
A nonempty n-dimensional topological manifold can not be homeomorphic to an m-dimensional manifold unless $m = n$.

Definition 3.1.4. Chart
Let M be a topological n-manifold. A (*coordinate*) *chart* on M is a pair (U, φ), where U is an open subset of M and $\varphi : U \to \tilde{U}$ is a homeomorphism from U to some open subset $\tilde{U} = \varphi(U) \subseteq \mathbb{R}^n$. Furthermore, given a chart (U, φ) on M:

- The set U is called a (*coordinate*) *domain* or a (*coordinate*) *neighborhood* of each of points in U.
- If $\varphi(U)$ is an open ball in $\mathbb{R}^n$, then U is called a *coordinate ball*. If $\varphi(U)$ is an open cube, then U is called a *coordinate cube*.
- The map φ is called a (*local*) *coordinate map* and the component functions x^i of φ, given by $\varphi(p) = (x^1(p), \cdots, x^2(p))$, are called *local coordinates on U*.

Remark:

- By the def of topological manifolds, each point $p \in M$ i contained in the domain of some chart (U, φ). If $\varphi(p) = 0$, then we say the chart is *centered at p*. If (U, φ) is any chart whose domain contains p, then we can obtain a new chart centered at p by subtracting $\varphi(p)$.
- If (U, φ) is a chart whose domain U contains p, then for shorthand we write that (U, φ) is a chart containing p.
- If we wish to emphasize the coordinate functions $(x^1, \cdots, x^n)$, we somtimes denote the chart by $(U, (x^1, \cdots, x^n))$ or $(U, (x^i))$.

Example (*of topological manifolds*):

- (*Euclidean Spaces*). $\mathbb{R}^n$ is a topological n-manifold:
  1. $\mathbb{R}^n$ is Hausdorff since it is metrizable.
  2. $\mathbb{R}^n$ is second-countable since $\{B_r(p) \mid r \in \mathbb{Q}_+, p \in \mathbb{Q}^n\}$ is a countable basis for $\mathcal{T}_{\text{usual}}$.
  3. Immediately, the third property holds for $\mathbb{R}^n$.

- (*Graphs of Continuous Functions*). Let $U \subseteq \mathbb{R}^n$ be an open subset and let $f : U \to \mathbb{R}^k$ be a continuous function. The *graph of* f is the subset of $\mathbb{R}^n \times \mathbb{R}^k$ defined by

$$\Gamma(f) := \{(x, y) \in \mathbb{R}^n \times \mathbb{R}^k \mid x \in U, y = f(x)\}$$

with the subspace topology. Consider the projection map $\pi_1 : \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ and the restriction $\varphi := \pi_1|_{\Gamma(f)} : \Gamma(f) \to U$ of π_1 to $\Gamma(f)$. By the continuity of π_1, φ is also continuous and by the continuity of f, φ^{-1} is continuous ($\varphi^{-1}(x) = (x, f(x))$). Hence, $\varphi : \Gamma(f) \xrightarrow{\cong} U$ is a homeomorphism, so $\Gamma(f)$ is a topological manifold of dimension n.

Furthermore, in $\Gamma(f)$, $(\Gamma(f), \varphi)$ is a (global) coordinate chart, called *graph coordinates*.

- (*Spheres*). $\forall n \in \mathbb{N} \cup \{0\}$, the n-sphere $\mathbb{S}^n \subseteq \mathbb{R}^{n+1}$ is Hausdorff and second-countable with the subspace topology of $\mathcal{T}_{\text{usual},\mathbb{R}^{n+1}}$.

To show that it is locally Euclidean, $\forall i = 1, \cdots, n+1$, denote by U_i^+ the subset of $\mathbb{R}^{n+1}$ where the i-th coordinate is positive:

$$U_i^+ := \{(x^1, \cdots, x^{n+1}) \in \mathbb{R}^{n+1} \mid x^i > 0\}$$

and similarly for U_i^-. Consider the continuity function $f : \mathbb{B}^n (\subseteq \mathbb{R}^n) \to \mathbb{R}$ given by

$$f(u) := \sqrt{1 - |u|^2}$$

Note that every i-th coordinate x^i of every point on $\mathbb{S}^n$ can be uniquely determined by $x^1, \cdots, \widehat{x^i}, \cdots, x^{n+1}$ by using f: For $x \in U_i^+ \cap \mathbb{S}^n$, $x^i = f(x^1, \cdots, \widehat{x^i}, \cdots, x^{n+1})$ and for $x \in U_i^- \cap \mathbb{S}^n$, $x^i = -f(x^1, \cdots, \widehat{x^i}, \cdots, x^{n+1})$. Hence, we have

$$U_i^+ \cap \mathbb{S}^n = \Gamma(f(x^1, \cdots, \widehat{x^i}, \cdots, x^{n+1})) \text{ and } U_i^- \cap \mathbb{S}^n = \Gamma(-f(x^1, \cdots, \widehat{x^i}, \cdots, x^{n+1}))$$

Therefore, each $U_i^\pm \cap \mathbb{S}^n$ are locally Euclidean of dimension n and hence so is $\mathbb{S}^n$. Furthermore, the maps $\varphi_i^\pm : U_i^\pm \cap \mathbb{S}^n \to \mathbb{S}^n$ given by

$$\varphi_i^\pm(x) := (x^1, \cdots, \widehat{x^i}, \cdots, x^{n+1})$$

are graph coordinates for $U_i^\pm \cap \mathbb{S}^n$.

- (*Projective Spaces*). The *n-dimensional real projective space* $\mathbb{R}P^n$ is defined as the set of all straight lines in $\mathbb{R}^{n+1}$ passing through the origin, or equivalently, the set of all 1-dimensional linear subspaces of $\mathbb{R}^{n+1}$, whose topology is given by the quotient topology determined by

$$\pi : \begin{matrix} \mathbb{R}^{n+1} \setminus \{0\} \to \mathbb{R}P^n \\ x \mapsto [x] \end{matrix}$$

where $[x]$ denotes the subspace spanned by x. Immediately, by the def of $\mathcal{T}_{\mathbb{R}P^n}$, we then can see that $\mathbb{R}P^n$ is Hausdorff and second-countable.

To show that $\mathbb{R}P^n$ is locally Euclidean, $\forall i = 1, \cdots, n+1$, let $\tilde{U}_i \subseteq \mathbb{R}^{n+1} \setminus \{0\}$ be the set of all points in $\mathbb{R}^{n+1} \setminus \{0\}$ with nonzero i-th coordinate, i.e.:

$$\tilde{U}_i := \{x \in \mathbb{R}^{n+1} \setminus \{0\} \mid x^i \neq 0\}$$

and let $U_i := \varphi(\tilde{U}_i) \subseteq \mathbb{R}P^n$. Since $\tilde{U}_i$ is open in $\mathbb{R}^{n+1} \setminus \{0\}$, then U_i is open in $\mathbb{R}P^n$. Note that if $[x] = [y]$, then $x \in \tilde{U}_i \Leftrightarrow y \in \tilde{U}_i$, so $\pi|_{\tilde{U}_i} : \tilde{U}_i \to U_i$ is a quotient map. Define a map $\varphi_i : U_i \to \mathbb{R}^n$ by

$$\varphi_i([x^1, \cdots, x^{n+1}]) := (\frac{x^1}{x^i}, \cdots, \frac{x^{i-1}}{x^i}, \frac{x^{i+1}}{x^i}, \cdots, \frac{x^{n+1}}{x^i})$$

which is well-defined by the def of $\tilde{U}_i$. Note that $\varphi_i \circ \pi|_{\tilde{U}_i}$ is continuous. Since $\pi|_{\tilde{U}_i}$ is quotient, then φ_i is continuous. Since the inverse of φ_i

$$\varphi_i^{-1}(u^1, \cdots, u^n) = [u^1, \cdots, u^{i-1}, 1, u^{i+1}, \cdots, u^n]$$

is continuous, then φ_i is a homeomorphism. Therefore, $\mathbb{R}P^n$ is locally Euclidean of dimension n.

- (*Product Manifolds*). Suppose $M_1^{n_1}, \cdots, M_k^{n_k}$ are topological manifolds. Then the product space $M_1 \times \cdots \times M_k$ is a topological manifold of dimension $n_1 + \cdots + n_k$ as follows. By the def of product topology, it is clearly Hausdorff and second-countable, so only the locally Euclidean property need to be checked:
  Given any point $(p_1, \cdots, p_k) \in M_1 \times \cdots \times M_k$, take a coordinate chart (U_i, φ_i) for each $p_i \in M_i$ on M_i. Then the product map

  $$\varphi_1 \times \cdots \times \varphi_k : U_1 \times \cdots \times U_k \to \varphi(U_1) \times \cdots \times \varphi(U_k) \subseteq \mathbb{R}^{n_1 + \cdots + n_k}$$

  is a homeomorphism.
- (*Tori*). $\forall n \in \mathbb{N}_i, \mathbb{T}^n := \overbrace{\mathbb{S}^1 \times \cdots \times \mathbb{S}^1}^{n\text{-copies}}$ is a topological n-manifold.

3.1.2 Topological Properties

Lemma 3.1.5. *Well-Behaved Basis-for-Topological Manifolds Lemma*
Every topological manifold M^n has a countable basis consisting of precompact coordinate balls.

Proof. First, we consider the special case in which M can be covered by a single chart $(M, \varphi : M \to \tilde{U} \subseteq \mathbb{R}^n)$. Then we define

$$\mathcal{B} := \{B_r(x) \subseteq \mathbb{R}^n \mid x^i \in \mathbb{Q}, r \in \mathbb{Q}_+, B_{r'}(x) \subseteq \tilde{U} \text{ for some } r' > r\}$$

which is countable and each such ball in $\mathcal{B}$ is precompact in $\tilde{U}$. Note that $\mathcal{B}$ forms a basis for the

topology of $\tilde{U}$. Hence, it gives a countable basis for $\mathcal{T}_M$ via the homeomorphism φ.
Now let M be an arbitrary n-manifold. By def, each point of M is in the domain of some chart. Since every open cover of a second-countable space has a countable subcover, then M is covered by countably many charts $\{(U_i, \varphi_i)\}$. By the argument in the first step, each domain U_i admits a countable basis of coordinate balls that are precompact in U_i (hence in M). Then the union of all those countable bases is a countable basis for the topology of M consisting of precompact coordinate balls. □

Connectivity

Proposition 3.1.6. *Connectivity Properties-of-Topological Manifolds Proposition*

1. *M is locally path-connected, i.e.: M has a basis of path-connected open subsets.*
2. *M is connected iff M is path-connected.*
3. *The component of M are the same as its path-component.*
4. *M has countably many components, each of which is an open subsets of M and a connected topological manifold.*

Proof. 1 follows from the Well-Behaved Basis-for-Topological Manifolds Lemma and hence 2 and 3 immediately hold. For 4, by 1, every component of M is open, so the collection of components is an open cover of M and hence, by the second-countability of M, this open cover has a countably many subcover. Since the components are open, they are connected topological manifolds in the subspace topology. □

Compactness

Proposition 3.1.7. *Local Compactness-of-Topological Manifolds PRoposition*
Every topological manifold M is locally compact.

Proof. By the Well-Behaved Basis-for-Topological Manifolds Lemma, let $\mathcal{B}$ be the basis for M consisting of precompact coordinate balls. Hence, $\forall p \in M, \exists B \in \mathcal{B}$, s.t.: $p \in B \subseteq \overline{B}$ where $\overline{B}$ is compact and B is open. □

Definition 3.1.8. Paracompactness
Let X be a topological space.

- A collection $\mathcal{X}$ of subsets of X is said to be *locally finite* if each point in X has an open neighborhood that intersects at most finitely many of the sets in $\mathcal{X}$.
- Given a cover $\mathcal{U}$ of X, another cover $\mathcal{V}$ of X is called a *refinement of* $\mathcal{U}$ if $\forall V \in \mathcal{V}, \exists U \in \mathcal{U}$, s.t.: $V \subseteq U$.
- We say X is *paracompact* if every cover of X admits an open, locally finite refinement.

Lemma 3.1.9. *Closures-and-Unions-of-Local Finite Families Lemma*
Let $\mathcal{X}$ be a locally finite collection of subsets of a topological space X.

1. *The collection* $\overline{\mathcal{X}} := \{\overline{U} \mid U \in \mathcal{X}\}$ *is also locally finite.*
2. $\overline{\bigcup_{U\in\mathcal{X}} U} = \bigcup_{U\in\mathcal{X}} \overline{U}$.

Proof. 1. $\forall p \in X, \exists$ open $p \in V \subseteq X$, s.t.: V only intersects $U_1, \cdots, U_n \in \mathcal{X}$, i.e.: $V \cap U_i \neq \emptyset, \forall i = 1, \cdots, n$, and $V \cap U = \emptyset, \forall U \in \mathcal{X} \setminus \{U_i\}_{i=1}^n$. Since V is open, then $V \cap \overline{U}_i \neq \emptyset, \forall i = 1, \cdots, n$, and $V \cap \overline{U} = \emptyset, \forall \overline{U} \in \overline{\mathcal{X}} \setminus \{\overline{U}_i\}_{i=1}^n$.

2. It suffices to show that $\overline{\bigcup_{U\in\mathcal{X}} U} \subseteq \bigcup_{U\in\mathcal{X}} \overline{U}$: $\forall p \in \overline{\bigcup_{U\in\mathcal{X}} U}$, by 1, $\exists$ open $p \in V \subseteq X$, s.t.: $V \cap \overline{U}_i \neq \emptyset, \forall i = 1, \cdots, n$, and $V \cap \overline{U} = \emptyset, \forall \overline{U} \in \overline{\mathcal{X}} \setminus \{\overline{U}_i\}_{i=1}^n$ for some $U_i \in \mathcal{X}$. Hence, $V \cap U = \emptyset, \forall U \in \mathcal{X} \setminus \{U_i\}_{i=1}^n$, so by the def of closure,

$$p \in V \cap (\bigcup_{U\in\mathcal{X}} U) = \bigcup_{U\in\mathcal{X}} (V \cap U) = \bigcup_{i=1}^{n} (V \cap U_i) = V \cap (\bigcup_{i=1}^{n} U_i)$$

Therefore, $p \in \bigcup_{i=1}^n U_i \subseteq \bigcup_{i=1}^n \overline{U}_i \subseteq \bigcup_{U\in\mathcal{X}} \overline{U}$.

□

Theorem 3.1.10. *Paracompactness-of-Topological Manifolds Theorem*
Every topological manifold M is paracompact. In fact, given any open cover $\mathcal{X}$ of M and any basis $\mathcal{B}$ for $\mathcal{T}_M$, there exists a countable, locally finite open refinement of $\mathcal{X}$ consisting of elements of $\mathcal{B}$.

Proof. By the Local Compactness-of-Topological Manifold Prop, M is locally compact and then by def, M is Hausdorff and second-countable, so there exists an exhaustion $\{K_j\}_{j\in\mathbb{N}_+}$ of M by compact sets. $\forall j$, let $V_j := K_{j+1} \setminus \text{int } K_j$ and $W_j := (\text{int } K_{j+2}) \setminus K_{j-1}$ where $K_0 = \emptyset$. Then V_j is compact and contained in the open set W_j.

$\forall x \in V_j, \exists X_x \in \mathcal{X}$ containing x. Since $\mathcal{B}$ is a basis, then $\exists B_x \in \mathcal{B}$, s.t.: $x \in B_x \subseteq X_x \cap W_j$. Then $\{B_x\}_{x\in V_j}$ is an open cover of V_j. By the compactness of V_j, this open cover has a finite subcover. Hence, the union of all such finite subcovers is a countable open cover of M that refines $\mathcal{X}$.

Since the finite subcover of V_j consists of the sets contained in W_j, and by the def of exhaustion, $W_j \cap W_{j'} = \emptyset$ except when $j - 2 \leq j' \leq j + 2$, then the resulting cover is locally finite. □

Remark: Note that this proof uses a prop in topology that any Hausdorff, second-countable and regionally compact topological space has an exhaustion by compact subsets, which is one of the reason why we assume the topological manifolds are second-countable.

Fundamental Group

Theorem 3.1.11. *Algebraic Constraint-from-Topological Manifolds Theorem*
The fundamental group of a topological manifold M is countable.

Proof. By the Well-Behaved Basis-for-Topological Manifolds Lemma, there is a countable collection $\mathcal{B}$ of coordinate balls covering M. $\forall B, B' \in \mathcal{B}$, by the Connectivity Properties-of-Topological Manifolds Prop, $B \cap B'$ has at most countable components, each of which is path-connected. Let

$\mathcal{X}$ be a countable set containing a point from each component of $B \cap B'$ for each $B, B' \in \mathcal{B}$. $\forall B \in \mathcal{B}, x, x' \in \mathcal{X}$ with $x, x' \in B$, let $h^B_{x,x'}$ be some path from x to x' in B.

Since $\pi_1(M, b_1) \cong \pi_1(M, b_2)$ are isomorphic if b_1, b_2 are in the same component of M, and $\mathcal{X}$ contains at least one point in each component, then we may choose a point $p \in \mathcal{X}$ as based point. Define a *special loop* to be a loop based at p that is a finite product of paths of the form $h^B_{x,x'}$. Since $\mathcal{X}$ is countable, then the set of all special loops are countable. Note that each special loop determines an element of $\pi_1(M, p)$. Hence, it suffices to show that each element of $\pi_1(M, p)$ is represented by a special loop:

Suppose $f : [0, 1] \to M$ is a loop based at p. Consider the collection $\{f^{-1}(B) \mid B \in \mathcal{B}\}$, which is an open cover of $[0, 1]$. By the compactness of $[0, 1]$, this cover has a finite subcover. Hence, there are finitely many numbers $0 = a_0 < a_1 < \cdots < a_k = 1$ such that

$$[a_{i-1}, a_i] \subseteq f^{-1}(B) \text{ for some } B \in \mathcal{B}$$

$\forall i$, let $f_i := f|_{[a_{i-1},a_i]}$ and let $B_i \in \mathcal{B}$ be a coordinate ball containing the image of f_i. Then we have $f(a_i) \in B_i \cap B_{i+1}$. Take $x_i \in \mathcal{X}$ to be in $B_i \cap B_{i+1}$ and take g_i to be a path in $B_i \cap B_{i+1}$ form x_i to $f(a_i)$, where $g_0 = g_k = c_p$ are the constant path based at p since $x_0 = x_k = p$. Denote $\overline{g}_i(t) := g_i(1 - t)$. Then we have:

$$\begin{aligned} f &\simeq f_1 \circ \cdots \circ f_k \\ &\simeq g_0 \circ f_1 \circ \overline{g}_1 \circ g_1 \circ f_2 \circ \overline{g}_2 \circ \cdots \circ \overline{g}_{k-1} \circ g_{k-1} \circ f_k \circ \overline{g}_k \\ &\simeq \tilde{f}_1 \circ \tilde{f}_2 \circ \cdots \circ \tilde{f}_k \end{aligned}$$

where $\tilde{f}_i := g_{i-1} \circ f_i \circ \overline{g}_i$ is a path in B_i from x_{i-1} to x_i. Since B_i is simply connected, then $\tilde{f}_i \simeq h^{B_i}_{x_{i-1},x_i}$ and hence we're done. □

3.2 Smooth Manifolds

3.2.1 Intuition and Construction

Intuition: We may try to make sense of derivatives of dunctions on manifolds. But such derivatives can *not* be invariant under homeomorphisms. For example, the map $\varphi : \mathbb{R}^2 \to \mathbb{R}^2$ given by $(u, v) \mapsto (u^{1/3}, v^{1/3})$ is a homeomorphism, but the differentiable function $f : \mathbb{R}^2 \to \mathbb{R}$ given by $(x, y) \mapsto x$ is such that $f \circ \varphi$ is *not* differentiable at the origin.

Definition 3.2.1. Smooth Maps in Euclidean Spaces
Let U and V be open subsets of Euclidean spaces $\mathbb{R}^n$ respectively $\mathbb{R}^m$. A function $F : U \to V$ is said to be *smooth* or $\mathcal{C}^\infty$ or *infinitely differentiable* if each of its component functions has continuous partial derivatices of all orders.

Furthermore, if F is bijective and has a smooth inverse map, then it is called a *diffeomorphism*.

Motivation: Let M be an arbitrary topological n-manifold. Each point in M is in the domain of a coordinate map $\varphi : U \to \tilde{U} \subseteq \mathbb{R}^n$. A "plausible" def of a smooth function on M intuitively would be to say that $f : M \to \mathbb{R}$ is smooth iff the component function $f \circ \varphi^{-1} : \tilde{U} \to \mathbb{R}$ is smooth in the sense of ordinary calculus. However, this make sense only if this property is independent of the choice of charts. In general, this independence is *not* always true.

Definition 3.2.2. Smooth Atlas
Let M be a topological manifold.

- If $(U, \varphi), (V, \psi)$ are two charts with $U \cap V \neq \emptyset$, then $\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$ is called the *transition map from φ to ψ*.
- Two charts (U, φ) and (V, ψ) are said to be *smoothly compatible* if either $U \cap V = \emptyset$ or the transition map $\psi \circ \varphi^{-1}$ is a diffeomorphism.
- We define an *atlas for M* to be a collection of charts whose domains cover M.
- An atlas $\mathcal{A}$ for M is called a *smooth atlas* if any two charts in $\mathcal{A}$ are smoothly compatible with each other.

Example: On $M = \mathbb{R}^n$,

$$\mathcal{A}_1 = \{(\mathbb{R}^n, \mathrm{id}_{\mathbb{R}^n})\}$$
$$\mathcal{A}_2 = \{(B_1(x), \mathrm{id}_{B_1(x)}) \mid x \in \mathbb{R}^n\}$$

are different smooth atlas. But they define exactly the same batch of smooth functions: a function $f : \mathbb{R}^n \to \mathbb{R}$ is smooth w.r.t. wither atlas iff it is smooth in the sense of ordinary calculus.

Definition 3.2.3. Smooth Structure
Let M be a topological manifold.

- A smooth atlas $\mathcal{A}$ on M is said to be *maximal* if it is not properly contained in any larger other smooth atlas on M.
- A *smooth structure on M* is precisely a maximal smooth atlas.

Remark:

- A given topological manifold may have different smooth structures while it is *not* always possible to find a smooth structure on a given topological manifold.
- In general, it is *not* very convenient to define a smooth structure by explicitly describing a maximal smooth atlas since such an atlas contains very many charts:

Proposition 3.2.4. *Smooth Structure-Determination Proposition*
Let M be a topological manifold.

1. *Every smooth atlas $\mathcal{A}$ for M is contained in a unique maximal smooth atlas, called the smooth structure determined by $\mathcal{A}$.*

2. Two smooth atlas for M determine the same structure iff their union is a smooth atlas.

Proof. 1. Let $\overline{\mathcal{A}}$ denote the set of all charts that are smoothly compatible with every chart in $\mathcal{A}$.

- $\overline{\mathcal{A}}$ *is a smooth atlas:* $\forall (U,\varphi),(V,\psi) \in \overline{\mathcal{A}}$, W.T.S.: $\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$ is smooth. $\forall x := \varphi(p) \in \varphi(U \cap V)$, since the domains of the charts in $\mathcal{A}$ covers M, then $p \in W$ for some chart $(W,\theta) \in \mathcal{A}$. By the def of $\overline{\mathcal{A}}$, both of the maps $\theta \circ \varphi^{-1}$ and $\psi \circ \theta^{-1}$ are smooth. Since $p \in U \cap V \cap W$, then

$$\psi \circ \varphi^{-1} = (\psi \circ \theta^{-1}) \circ (\theta \circ \varphi^{-1})$$

  is smooth on an open neighborhood of x. Hence, $\psi \circ \varphi^{-1}$ is smooth on an open neighborhood of each point in $\varphi(U \cap V)$. Therefore, $\overline{\mathcal{A}}$ is a smooth atlas.
- $\overline{\mathcal{A}}$ *is maximal* immediately by def.
- *Uniqueness:* Let $\mathcal{B}$ be any other maximal smooth atlas containing $\mathcal{A}$. Then by the maximality of $\overline{\mathcal{A}}$ and $\mathcal{B}$, we have $\mathcal{B} \subseteq \overline{\mathcal{A}}$ respectively $\overline{\mathcal{A}} \subseteq \mathcal{B}$.

2. - ($\Rightarrow$): Let $\overline{\mathcal{A}} = \overline{\mathcal{A}}_1 = \overline{\mathcal{A}}_2$. Since $\mathcal{A}_1 \subseteq \overline{\mathcal{A}}$ and $\mathcal{A}_2 \subseteq \overline{\mathcal{A}}$, then

$$\mathcal{A}_1 \cup \mathcal{A}_2 \subseteq \overline{\mathcal{A}}$$

  Hence, any two charts of $\mathcal{A}_1 \cup \mathcal{A}_2$ are smoothly compatible. Since the domains of both of $\mathcal{A}_1$ and $\mathcal{A}_2$ can cover M, then so is $\mathcal{A}_1 \cup \mathcal{A}_2$ and hence $\mathcal{A}_1 \cup \mathcal{A}_2$ is a smooth atlas.
- ($\Leftarrow$): Since $\mathcal{A}_1 \subseteq \mathcal{A}_1 \cup \mathcal{A}_2 \subseteq \overline{\mathcal{A}_1 \cup \mathcal{A}_2}$, then by 1, $\overline{\mathcal{A}}_1 \subseteq \overline{\mathcal{A}_1 \cup \mathcal{A}_2}$ and similarly for $\mathcal{A}_2 \subseteq \overline{\mathcal{A}_1 \cup \mathcal{A}_2}$.

□

Convention: Let $(M,\mathcal{A})$ be a smooth manifold.

- When the smooth structure $\mathcal{A}$ is understood, we usually omit mention of it and just say that M is a smooth manifold.
- Any chart $(U,\varphi) \in \mathcal{A}$ is called a *smooth chart* with
  - U called a *smooth* (*coordinate*) *domain* or *smooth* (*coordinate*) *neighborhood*
  - φ called a *smooth coordinate map.*

  If $\varphi(U)$ is an open ball in Euclidean space, then U is called a *smooth coordinate ball* and similarly a *smooth coordinate cube* is defined.

Definition 3.2.5. Regular Coordinate Balls
Let M be a smooth manifold. A set $B \subseteq M$ is called a *regular coordinate ball* if there is a smooth

chart (B', φ), s.t.: $\overline{B} \subseteq B'$ and that for some positive real numbers $r < r'$,

$$\varphi(B) = B_r(0), \varphi(\overline{B}) = \overline{B}_r(0) \text{ and } \varphi(B') = B_{r'}(0)$$

Remark: Since $\overline{B} \cong \overline{B}_r(0)$, then $\overline{B}$ is compact and hence every regular coordinate ball is precompact in M.

Proposition 3.2.6. *Well-Behaved Basis-for-Smooth Manifolds Proposition*
Every smooth manifold has a countable basis consisting of coordinate balls.

3.2.2 Local Coorodinate Representations

Method: Let M be a smooth manifold. Choose a smooth chart (U, φ) on M, the coordinate map $\varphi : U \to \tilde{U} \subseteq \mathbb{R}^n$ can be regarded as an identification between U and $\tilde{U}$, namely,

$$U =_i \varphi(U) \subseteq \mathbb{R}^n$$

$\forall p \in U$, we can represent it by the coordinate $(x^1, \cdots, x^n) = \varphi(p)$ in $\mathbb{R}^n$. Under the identification, p can be expressed by $(x^1, \cdots, x^n)$ w.r.t. φ, called the *local coordinate representation for p*.
Example (*Polar Coordinates*). In $\mathbb{R}^2$, on $U = \{(x, y) \mid x > 0\}$, we have two related coordinate representations:

$$(r, \theta) = (\sqrt{x^2 + y^2}, \tan^{-1}(y/x)) \text{ and } (x, y) = (r\cos\theta, r\sin\theta)$$

Remark: To ensure that the defined geometric concept, coordinate representation on smooth manifold, are meaningful and valid rather than coordinate "illusions", there are two coping strategies:

- *Coordinate-dependent:* Choose firstly a locally coordinate representation $(x^1, \cdots, x^n)$ and then write down the formula. When taking another locally coordinate representation, prove, by transition maps, that the results after transforming formula are the same.
- *Coordinate-independent / Invariant:* Choose a globally coordinate representation from the first.

3.2.3 Examples of Smooth Manifolds

Example 3.2.7. 0-Dimensional Manifolds
A topological 0-manifold M is just a countable discrete space. $\forall p \in M$, the neighborhood of p that is homeomorphic to an open subset of $\mathbb{R}^0$ is $\{p\}$ itself, and there is exactly one coordinate map $\varphi : \{p\} \to \mathbb{R}^0$. Hence, the set of all charts on M satisfies the smooth compatibility condition. Therefore, each 0-dimensional manifold has a unique smooth structure.

Example 3.2.8. Euclidean Spaces
$\forall n \in \mathbb{N}_+$, $\mathbb{R}^n$ is a smooth n-manifold with the smooth structure determined by the atlas consisting

of single chart $(\mathbb{R}^n, \mathrm{id}_{\mathbb{R}^n})$. We call this the *standard smooth structure on* $\mathbb{R}^n$ and the resulting coordinate map *standard coordinates*.

Example 3.2.9. Another Smooth Structure on $\mathbb{R}$
Consider the homeomorphism $\psi : \mathbb{R} \to \mathbb{R}$ given by $x \mapsto x^3$. Then the atlas consisting of the single chart $(\mathbb{R}, \psi)$ determines a smooth structure on $\mathbb{R}$. Since

$$(\mathrm{id}_{\mathbb{R}} \circ \psi^{-1})(y) = y^{1/3}$$

is *not* smooth at the origin, then the chart $(\mathbb{R}, \psi)$ is *not* smoothly compatible with the standard smooth structure. Hence, the smooth structure on $\mathbb{R}$ determined by ψ is *not* the same as the standard one.

Example 3.2.10. Finite-Dimensional Vectors Spaces
Let V be an n-dimensional real vector space. Given any norm $||\cdot||$ on V, we define a metric $d : V \times V \to \mathbb{R}$ by

$$d(u, v) := ||u - v||$$

and then we can get a topology on V induced d, called the *norm topology* (in fact, it is independent of the choice of norm). With this topology, V is a topological n-manifold, and has a natural smooth structure defined as follows.

Each (ordered) basis $(E_1, \cdots, E_n)$ for V defines a basis isomorphism $E : \mathbb{R}^n \to V$ by

$$E(x) := x^1 E_1 + \cdots + x^n E_n = \sum_{i=1}^{n} x^i E_i$$

It's easy to check that E is a homeomorphism, so (V, E^{-1}) is a chart. The collection of such chart is then an atlas and if we show that it is smooth, then it immediately is a smooth structure on V. Let $(\tilde{E}_1, \cdots, \tilde{E}_n)$ be any other (ordered) basis for V and let $\tilde{E}(x) := \sum_j x^j E_j$ be the corresponding isomorphism. Then there is an invertible matrix (A_{ij}), s.t.:

$$E_i = \sum_j A_{ij} \tilde{E}_j \text{ for all } i \qquad (*)$$

The transition map between the two charts (V, E^{-1}) and $(V, \tilde{E}^{-1})$ is then given by $(\tilde{E}^{-1} \circ E)(x) = \tilde{x} = (\tilde{x}^1, \cdots, \tilde{x}^n)$ where we can compute $\tilde{x}$ explicitly by $E(x) = \tilde{E}(\tilde{x})$:

$$\sum_{j=1}^{n} \tilde{x}^j \tilde{E}_j = \sum_{i=1}^{n} x^i E_i \overset{(*)}{=} \sum_{i,j=1}^{n} x^i A_{ij} \tilde{E}_j \Rightarrow \tilde{x}^j = \sum_i A_{ij} x^i$$

Hence, the map $x \mapsto \tilde{x}$ is an invertible linear map, so it is a diffeomorphism. Therefore, (V, E^{-1}) and $(V, \tilde{E}^{-1})$ are smoothly compatible. Now we get a smooth structure on V and we call this the *standard smooth structure on* V.

Convention (*Einstein Summation Convention*): We often abbreviate such a sum by omitting the summation sign, as in

$$E(x) = x^i E_i \Leftrightarrow E(x) = \sum_{i=1}^{n} x^i E_i$$

We interpret any such expression according to the following rule, called the *Einstein Summation Convention*: If the same index name (i.e.: i above) appears exactly twice in any monomial term, once as an upper index and once as lower index, then that term is understood to be summed over all possible value of that index.

Example 3.2.11. Spaces of Matrices
Let $M(m \times n, \mathbb{R})$ denote the set of $m \times n$ matrices with real entries. Since $M(m \times n, \mathbb{R})$ is a real vector space of dimension mn under matrix addition and scalar multiplication, then it is a smooth mn-manifold. Similarly, $M(m \times n, \mathbb{C})$ is a vector space of dimension $2mn$ over $\mathbb{R}$, and hence a smooth $2mn$-manifold.

Example 3.2.12. Open Submanifolds
Let U be any open subset of $\mathbb{R}^n$. Then U is a topological n-manifold and the single chart (U, id_U) determines a smooth structure on U.

More generally, let M be a smooth n-manifold and let $U \subseteq M$ be any open subset. Define an atlas on U by

$$\mathcal{A}_U := \{\text{smooth charts } (V, \varphi) \text{ for } M \text{ such that } V \subseteq U\}$$

Since $\mathcal{A}_U \subseteq \mathcal{A}_M$ and U is covered by the domains of charts in $\mathcal{A}_U$, then $\mathcal{A}_U$ is a smooth atlas for U. Hence, any open subset of M is itself a smooth n-manifold, namely, endowed with $\overline{\mathcal{A}}_U$, we call any open subset U of M an *open submanifold of* M.

Example 3.2.13. General Linear Group
The *general linear group* $GL(n, \mathbb{R})$ is the invertible $n \times n$ matrices with real entries. Since $GL(n, \mathbb{R})$ is an open submanifold of $M(n \times n, \mathbb{R})$, then $GL(n, \mathbb{R})$ is a smooth n^2-manifold.

Example 3.2.14. Matrices of Full Rank
Suppose $m < n$ and let $M_m(m \times n, \mathbb{R})$ denote the set of all metrices of in $M(m \times n, \mathbb{R})$ of rank m. $\forall A \in M_m(m \times n, \mathbb{R})$, rank $A = m$, which implies that A has some singular $m \times m$ submatrix. By the continuity of the determinant function, this submatrix has nonzero determinant on a neighborhood of A in $M(m \times n, \mathbb{R})$. Hence, A has a neighborhood contained in $M_m(m \times n, \mathbb{R})$. Therefore, $M_m(m \times n, \mathbb{R})$ is an open submanifold of M and hence is itself a smooth mn-manifold. A similar argument shows that $M_n(m \times n, \mathbb{R})$ is a smooth mn-manifold when $n < m$.

Example 3.2.15. Spaces of Linear Transformations
Let V and W be two real vector spaces of dimension $n < \infty$ respectively $m < \infty$ and let $\mathcal{L}(V, W)$ denote the set of linear transformations from V to W. Since $\mathcal{L}(V, W)$ is a mn-dimensional real vector space, then it is a smooth mn-manifold. Furthermore, one way to put global coordinates

on it is to choose bases for V and W, and represent each $T \in \mathcal{L}(V,W)$ by matrix, which yields an isomorphism from $\mathcal{L}(V,W)$ to $M(m \times n, \mathbb{R})$.

Example 3.2.16. Graphs of Smooth Functions
Let $U \subseteq \mathbb{R}^n$ be an open subset and let $f : U \to \mathbb{R}^k$ be a smooth function. We've already seen that the graph of f, $\Gamma(f)$, is a topological n-manifold. Since $\Gamma(f)$ is covered by the single graph coordinate chart $\varphi : \Gamma(f) \to U$ (given in the example of topological manifolds), we can put a canonical smooth structure on $\Gamma(f)$ by declaring the graph coordinate chart $(\Gamma(f), \varphi)$ to be a smooth chart.

Example 3.2.17. Spheres
In the example of topological manifolds, we've seen that $\mathbb{S}^n \subseteq \mathbb{R}^{n+1}$ is a topological n-manifold. Then we put a smooth structure on $\mathbb{S}^n$ as follows.

$\forall i = 1, \cdots, n+1$, let $(U_i^{\pm}, \varphi_i^{\pm})$ denote the graph coordinate charts constructed before. $\forall i \neq j$, the transition map

$$\varphi_i^{\pm} \circ (\varphi_j^{\pm})^{-1}(u^1, \cdots, u^n) = \begin{cases} (u^1, \cdots, \widehat{u^i}, \cdots, \overbrace{\pm\sqrt{1-|u|^2}}^{j\text{-th position}}, \cdots, u^n) & \text{,when } i < j \\ (u^1, \cdots, \underbrace{\pm\sqrt{1-|u|^2}}_{j\text{-th position}}, \cdots, \widehat{u^i}, \cdots, u^n) & \text{,when } j < i \end{cases}$$

is clearly a diffeomorphism. $\forall i$, the transition map $\varphi_i^+ \circ (\varphi_i^-)^{-1} = \varphi_i^- \circ (\varphi_i^+)^{-1} = \mathrm{id}_{\mathbb{B}^n}$ is also obviously a diffeomorphism. Hence, $\{(U_i^{\pm}, \varphi_i^{\pm})\}$ is a smooth atlas, so it determines a smooth structure on $\mathbb{S}^n$, called the *standard smooth structure*.

Example 3.2.18. Level Sets
The preceding example can be generalized as follows. Suppose $U \subseteq \mathbb{R}^n$ is an open subset and $\Phi : U \to \mathbb{R}$ is a smooth function. $\forall c \in \mathbb{R}$, the set $\Phi^{-1}(c)$ is called a *level set of* Φ. Choose some $c \in \mathbb{R}$, let $M := \Phi^{-1}(c)$ and suppose that it happens that the total derivative $D\Phi(a)$ is nonzero $\forall a \in \Phi^{-1}(c)$, i.e.: $\forall a \in \Phi^{-1}(c)$, $\partial\Phi/\partial x^i(a) \neq 0$ for some $i \in \{1, \cdots, n\}$. By the implicit function theorem, there is an open neighborhood U_0 of a, s.t.: $M \cap U_0$ can be expressed as a graph of an equation of the form

$$x^i = f(x^1, \cdots, \widehat{x^i}, \cdots, x^n)$$

for some smooth function $f : \mathbb{R}^{n-1} \to \mathbb{R}$. Therefore, applying the similar argument as in the case of $\mathbb{S}^n$, we can see that M is a topological $(n-1)$-manifold, and has smooth structure.

Example 3.2.19. Projective Spaces
Let (U_i, φ_i) be the coordinate charts of $\mathbb{R}P^n$ constructed in the example of topological manifolds. $\forall i \neq j$, assuming for convenient that $i > j$, the transition map

$$(\varphi_j \circ \varphi_i^{-1})(u^1, \cdots, u^n) = (\frac{u^1}{u^j}, \cdots, \frac{u^{j-1}}{u^j}, \frac{u^{j+1}}{u^j}, \cdots, \frac{u^{i-1}}{u^j}, \frac{1}{u^j}, \frac{u^i}{u^j}, \cdots, \frac{u^n}{u^j})$$

is a diffeomorphic from $\varphi_i(U_i \cap U_j)$ to $\varphi_j(U_i \cap U_j)$. Hence, the charts (U_i, φ_i) are smoothly compatible.

Example 3.2.20. Smooth Product Manifolds
Let $M_1^{n_1}, \cdots, M_k^{n_k}$ be smooth manifolds. We have seen that $M_1 \times \cdots \times M_k$ is a topological $(n_1 + \cdots + n_k)$-manifold in the example of topological manifolds, with the product charts $(U_1 \times \cdots \times U_k, \varphi_1 \times \cdots \times \varphi_k)$. Any two product charts are smoothly compatible since

$$(\psi_1 \times \cdots \times \psi_k) \circ (\varphi_1 \times \cdots \times \varphi_k)^{-1} = (\psi_1 \circ \varphi_1^{-1}) \times \cdots \times (\psi_k \circ \varphi_k^{-1})$$

is smooth. This determines a smooth structure on $M_1 \times \cdots \times M_k$, called the *product smooth manifold structure*. In particular, $\mathbb{T}^n = \underbrace{\mathbb{S}^1 \times \cdots \times \mathbb{S}^1}_{n\text{-copies}}$ has a smooth structure.

Lemma 3.2.21. *Smooth Manifolds-Construction Lemma*
Let M be a set and suppose we are given a collection $\{U_\alpha\}_{\alpha \in \mathcal{I}}$ of subsets of M together with $\varphi_\alpha : U_\alpha \to \mathbb{R}^n$ satisfying the following properties:

1. *$\forall \alpha \in \mathcal{I}, \varphi_\alpha$ is a bijection between U_α and an open subset $\varphi_\alpha(U_\alpha) \subseteq \mathbb{R}^n$.*
2. *$\forall \alpha, \beta \in \mathcal{I}$, the sets $\varphi_\alpha(U_\alpha \cap U_\beta)$ and $\varphi_\beta(U_\alpha \cap U_\beta)$ are open in $\mathbb{R}^n$.*
3. *Whenever $U_\alpha \cap U_\beta \neq \emptyset$, the map $\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$ is smooth.*
4. *Countably many of the sets U_α cover M.*
5. *$\forall p \neq q \in M$, either there exist some U_α containing both p and q or there exist disjoint U_α, U_β with $p \in U_\alpha$ and $q \in U_\beta$.*

Then M has a unique smooth manifold structure such that every $(U_\alpha, \varphi_\alpha)$ is a smooth chart.

Proof. • For defining the topology of M, we claim that the set

$$\mathcal{B} := \{\varphi_\alpha^{-1}(V) \mid V \in \mathcal{T}_{\mathbb{R}^n}, \alpha \in \mathcal{I}$$

is a basis for the topology of M. By 1 and 4, $\mathcal{B}$ covers M, so it suffices to show that $\forall V, W \in \mathcal{T}_{\mathbb{R}^n}, \forall \alpha, \beta \in \mathcal{I}, \forall p \in \varphi_\alpha^{-1}(V) \cap \varphi_\beta^{-1}(W)$, there is a third basis set containing p and contained in the intersection. It suffices to show that

$$\varphi_\alpha^{-1}(V) \cap \varphi_\beta^{-1}(W) \in \mathcal{B}$$

To see this, note that by 3, $(\varphi_\beta \circ \varphi_\alpha^{-1})^{-1}(W)$ is an open subset of $\varphi_\alpha(U_\alpha \cap U_\beta)$ and that by 2, $(\varphi_\beta \circ \varphi_\alpha^{-1})^{-1}(W)$ is also open in $\mathbb{R}^n$. Hence,

$$\varphi_\alpha^{-1}(V) \cap \varphi_\beta^{-1}(W) = \varphi_\alpha^{-1}(V \cap (\varphi_\beta \circ \varphi_\alpha^{-1})^{-1}(W)) \in \mathcal{B}$$

Then we have each map φ_α is a homeomorphism onto its image, so M is locally Euclidean of dimension n. The Hausdorff property follows directly from 5 and the second-countability

follows from 4.

- 3 guarantees that $\{(U_\alpha, \varphi_\alpha)\}$ is a smooth atlas, which determines a smooth structure on M. The uniqueness is omitted.

□

3.3 Manifolds with Boundary

3.3.1 Topological Manifolds with Boundary

Definition 3.3.1. Closed Upper Half-Space
When $n > 0$, the *closed n-dimensional upper half-space* $\mathbb{H}^n \subseteq \mathbb{R}^n$ is defined as

$$\mathbb{H}^n := \{(x^1, \cdots, x^n) \in \mathbb{R}^n \mid x^n \geq 0\}$$

Furthermore, we define the interior and boundary of $\mathbb{H}^n$ respectively by

$$\text{int } \mathbb{H}^n := \{(x^1, \cdots, x^n) \in \mathbb{R}^n \mid x^n > 0\}$$
$$\partial \mathbb{H}^n := \{(x^1, \cdots, x^n) \in \mathbb{R}^n \mid x^n = 0\}$$

for $n > 0$. When $n = 0$, we define $\mathbb{H}^0 := \mathbb{R}^0 = \{0\}, \text{int } \mathbb{H}^n := \mathbb{R}^0$ and $\partial \mathbb{H}^n = \emptyset$.

Definition 3.3.2. Topological Manifolds with Boundary
An n-dimensional topological manifold with boundary is a Hausdorff second-countable space M in which every point has an open neighborhood homeomorphic either to an open subset of $\mathbb{R}^n$ or to a relatively open subset of $\mathbb{H}^n$. Furthermore:

- The pair $(U \subseteq M, \varphi : U \to \mathbb{R}^n)$, where U is open in M, φ is a homeomorphism onto its image and $\varphi(U)$ is an open subset of $\mathbb{R}^n$ or $\mathbb{H}^n$, is called a *chart for M*. Moreover, if $\varphi(U)$ is an open subset of $\mathbb{R}^n$ (which includes the case of an open subset of $\mathbb{H}^n$ that doesn't intersect $\partial \mathbb{H}^n$), then we call (U, φ) an *interior chart*, while if $\varphi(U)$ is a relatively open subset of $\mathbb{H}^n$ such that $\varphi(U) \cap \partial \mathbb{H}^n \neq \emptyset$, then we call (U, φ) an *boundary chart*.
- A boundary chart, whose image is a set of the form $B_r(x) \cap \mathbb{H}^n$ for some $x \in \mathbb{H}^n$ and $r > 0$, is called a *coordinate half-ball*.
- A point $p \in M$ is called an *interior point of M* if it is in the domain of some interior chart, while p is called a *boundary point of M* if it is in the domain of some boundary chart that sends p into $\partial \mathbb{H}^n$. Then we define the *interior of M*, denoted as int M, by the set of all the interior points of M and the *boundary of M*, denoted as ∂M, by the set of all the boundary points of M.

Remark: Each point $p \in M$ is either an interior point or a boundary point: if $p \notin \partial M$, then either it is in the domain of an interior chart or it is in the domain of a boundary chart (U, φ) such that $\varphi(p) \notin \partial \mathbb{H}^n$, in which case the restriction of φ to $U \cap \varphi^{-1}(\text{int } \mathbb{H}^n)$ is an interior chart whose domain

contains p. However, it is *not* obvious that a given point can *not* be simulteneously an interior point w.r.t. one chart and a boundary point w.r.t. another:

Theorem 3.3.3. *Topological Invariance-of-the Boundary Theorem*
Let M be a topological manifold with boundary, then each point of M is either an interior point or a boundary point, but not both, i.e.:

$$M = int\ M \sqcup \partial M$$

Proof. By the above remark, we've shown that $M = \text{int } M \cup \partial M$, so it suffices to show that int $M \cup \partial M = \emptyset$. Suppose on the contrary that there is a point $p \in M$ which is both an interior point and a boundary point, i.e.: $\exists$ charts $(U, \varphi), (V, \psi)$ on M, s.t.: $p \in U \cap V$, $\varphi(U)$ is open in $\mathbb{R}^n$, $\psi(V)$ is open in $\mathbb{H}^n$, $\varphi(U) \cap \partial\mathbb{H}^n = \emptyset$ and $\psi(p) \in \partial\mathbb{H}^n$. Consider $W := U \cap V \neq \emptyset$. Note that $\varphi(W)$ is open in $\mathbb{R}^n$ and that $\psi \circ \varphi^{-1} : W \to \mathbb{R}^n$ is continuous and injective. Hence,

$$(\psi \circ \varphi^{-1})(\varphi(W)) = \psi(W)$$

is also open in $\mathbb{R}^n$, so $\exists x \in \psi(W)$, s.t.: $x^n < 0$ (since $\psi(p) \in \partial\mathbb{H}^n$), contradicting $\psi(W) \subseteq \psi(V) \subseteq \mathbb{H}^n$. □

Remark:

- It's important to distinguish between *topological boundary* and *manifold boundary*:
  - The topological boundary relies on the outer space containing it, say, if we let X be a topological space and let $A \subseteq X$, then

$$\partial A := \{x \in X \mid U \cap A \neq \emptyset, U \cap (X \setminus A) \neq \emptyset, \forall \text{ open } U \ni x\}$$

  - The manifold boundary is an inherent attribute of M itself and has nothing to do with whether M is embedded in a larger space or not.

  For example, the manifold boundary of $\overline{\mathbb{B}}^n$ is always $\mathbb{S}^{n-1}$, while the topological boundary of $\overline{\mathbb{B}}^n$ depends on the specific situations:
  If we regard of $\overline{\mathbb{B}}^n$ as itself, then its topological boundary is $\emptyset$; if we regard of $\overline{\mathbb{B}}^n$ as a subset of $\mathbb{R}^n$, then its topological boundary is $\mathbb{S}^{n-1}$; if we regard of $\overline{\mathbb{B}}^n$ as as a subset of $\mathbb{R}^k$ for $k > n$, then its topological boundary is $\overline{\mathbb{B}}^n$.
- We need to emphasize some traps of terms and nomenclature:
  - Topological manifolds with boundary are *not* generally topological manifolds, since the boundary points do not have locally Euclidean neighborhoods.
  - The boundary of manifolds need *not* always be nonempty. In fact, a topological manifold is a topological manifold with empty boundary. Hence, every topological manifold is a

topological manifold with boundary, but a topological manifold with boundary is a topological manifold iff its boundary is empty.

- (*Convention*):
  - If we only mention the term "manifold" we refer to an manifold without boundary, i.e.: manifold with boundary $\emptyset$.
  - If we wish to emphasize the boundary might or might not be non-empty, then we use the term "manifold with or without boundary".
  - A *closed manifold* is a compact manifold without boundary and an *open manifold* is a non-compact manifold without boundary.

Proposition 3.3.4. *Structure-of-Interior and Boundary of Topological Manifolds Proposition*
Let M be a topological n-manifold with boundary.

1. *$\operatorname{int} M$ is an open subset of M and a topological n-manifold without boundary.*
2. *∂M is a closed subset of M and a topological $(n-1)$-manifold without boundary.*
3. *M is a topological manifold iff $\partial M = \emptyset$.*
4. *If $n = 0$, then $\partial M = \emptyset$ and M is a 0-manifold.*

Proof. 1.
- $\forall p \in \operatorname{int} M, \exists$ a chart (U, φ), s.t.: $\varphi(U) \subseteq \operatorname{int} \mathbb{H}^n$ and hence $U \cap \partial M = \emptyset$, so by the Topological Invariance-of-the Boundary Theorem, $U \subseteq \operatorname{int} M$, which shows that $\operatorname{int} M$ is open in M.
- The Hausdorff-ness and second-countability of $\operatorname{int} M$ inherit immediately from M and the locally Euclidean property is immediately from the first part of 1.

2.
- By the Topological Invariance-of-the Boundary Theorem, $\partial M = M \setminus (\operatorname{int} M)$ is closed in M by 1.
- The Hausdorff-ness and second-countability of ∂M inherit immediately from M. For the locally Euclidean property, let $p \in \partial M$. By def, $\exists$ a chart (U, φ), s.t.: $\varphi(p) \in \partial \mathbb{H}^n$. Consider the map
$$\psi := \varphi|_{\partial M \cap U} : \partial M \cap U \to \varphi(U) \cap \partial \mathbb{H}^n$$
Note that $q \in U \cap \partial M \Leftrightarrow \varphi(q) \in \partial \mathbb{H}^n$. Hence, $\varphi(\partial M \cap U) = \varphi(U) \cap \partial \mathbb{H}^n$, so ψ is a homeomorphism. Since $\varphi(U)$ is open in $\mathbb{H}^n$, then $\varphi(U) \cap \partial \mathbb{H}^n$ is open in $\partial \mathbb{H}^n \cong \mathbb{R}^{n-1}$. Therefore, ∂M is a topological $(n-1)$-manifold without boundary.

3.
- ($\Leftarrow$): $\partial M = \emptyset \Rightarrow M = \operatorname{int} M$, so by 1, we're done.
- ($\Rightarrow$): By def, every point of a topological manifold M is an interior of M and then we're done.

4. Since $\partial \mathbb{H}^0 = \emptyset$, then by def, $\partial M = \emptyset$. Hence, by 3, we're done.

□

Proposition 3.3.5. *Topological Properties-of-Topological Manifolds with Boundary Proposition*
Let M be a topological manifold with boundary.

1. *M has a countable basis consisting of precompact coordinate balls and half-balls.*
2. *M is locally compact.*
3. *M is paracompact.*
4. *M has countably many components, each of which is an open subset of M and a connected topological manifold with boundary.*
5. *The fundamental group of M is countable.*

Proof. Similar as the proof in the topological manifold case. □

3.3.2 Smooth Structure on Topological Manifolds with Boundary

Intuition: Recall that if $U \subseteq \mathbb{H}^n$ is an open subset of $\mathbb{H}^n$, then a map $f : U \to \mathbb{R}^k$ is smooth if $\forall x \in U, \exists$ an open subset $U_x \subseteq \mathbb{R}^n$ containing x, $\exists f_x : U_x \to \mathbb{R}^k$, s.t.: f_x agrees with f on $U_x \cap \mathbb{H}^n$. In fact, if f is smooth, then $f|_{U\cap(\text{int } \mathbb{H}^n)}$ is smooth in the usual sense.

Definition 3.3.6. Smooth Manifolds with Boundary
Let M be a topological manifold with boundary.

- A *smooth structure for M* is defined to be a maximal smooth atlas, as in the case without boundary.
- Together with a smooth structure, M is called a *smooth manifold with boundary.*

Remark:

- Every smooth manifold is automatically a smooth manifold with boundary whose boundary is empty.
- (*Convention*). Let $(M, \mathcal{A})$ be a smooth n-manifold with boundary.
  - When the smooth structure $\mathcal{A}$ is understood, we usually omit mention of it and just say that M is a smooth manifold with boundary.
  - Any chart in $\mathcal{A}$ i called a *smooth chart for M*.
  - *Smooth coordinate balls*, *smooth coordinate half-balls* and *regular coordinate balls in M* are defined as in the obvious ways. Furthermore, a subset $B \subseteq M$ is called a *regular coordinate half-ball* if there is a smooth coordinate half-ball $B' \supseteq \overline{B}$ and a smooth coordinate map $\varphi : B' \to \mathbb{H}^n$ such that for some $r' > r$, we have

$$\varphi(B) = B_r(0) \cap \mathbb{H}^n, \varphi(\overline{B}) = \overline{B}_r(0) \cap \mathbb{H}^n \text{ and } \varphi(B') = B_{r'}(0) \cap \mathbb{H}^n$$

- Note that a product of half-spaces $\mathbb{H}^m \times \mathbb{H}^n$ is *not* itself a half-space. Hence, a finite product of smooth manifold with boundary can *not* generally be considered as a smooth manifold

with boundary:

Proposition 3.3.7. *Boundary Formula-for-Product Smooth Manifolds Proposition*
Let $M_1, \cdots, M_k$ be smooth manifolds and let N be a smooth manifold with boundary. Then the product $M_1 \times \cdots \times M_k \times N$ is a smooth manifold with boundary, whose boundary is

$$\partial(M_1 \times \cdots \times M_k \times N) = M_1 \times \cdots \times M_k \times \partial N$$

Proof. For simplifying, we may consider the case $k = 1$ and the general case can be easily deduced since when $k = t$, $M_1 \times \cdots \times M_t$ is a smooth manifold. Assume M is a smooth manifold.

- *$M \times N$ is a topological manifold with boundary:* The Hausdorff-ness and second-countability of $M \times N$ is directly by the conclusion in topology. For the locally Euclidean property, let the dimensions of M and N be m respectively n. $\forall x \in M, y \in N$, $\exists$ charts $(U \ni x, \varphi), (V \ni y, \psi)$, s.t.: $\varphi(U)$ is open in $\mathbb{R}^m$ and $\psi(V)$ is open in $\mathbb{H}^n$. Consider the map $\varphi \times \psi : U \times V \to \varphi(U) \times \psi(V) \subseteq \mathbb{R}^m \times \mathbb{H}^n$. Note that $\varphi \times \psi$ is a homeomorphism and that $\varphi(U) \times \psi(V) \cong W$ for some open subset W of $\mathbb{H}^{m+n}$. Hence,

  $$(U \times V, \Phi := (\varphi \times \psi) \circ (\varphi(U) \times \psi(V) \xrightarrow{\cong} W) : U \times V \to W \subseteq \mathbb{H}^{m+n})$$

  is a chart containing (x, y), so $M \times N$ is a topological $(m + n)$-manifold with boundary.

- *Smooth Compatiblity:* $\forall$ charts $(U_1 \times V_1, \Phi_1), (U_2 \times V_2, \Phi_2)$ of $M \times N$ with $(U_1 \times V_1) \cap (U_2 \times V_2) = (U_1 \cap U_2) \times (V_1 \cap V_2) \neq \emptyset$, consider the transition map

  $$\Phi_2 \circ \Phi_1^{-1} = (\varphi_2 \times \psi_2) \circ (\varphi_1^{-1} \times \psi_1^{-1}) = (\varphi_2 \circ \varphi_1^{-1}) \times (\psi_2 \circ \psi_1^{-1})$$

  Since $\varphi_2 \circ \varphi^{-1}$ is a diffeomorphism on $U_1 \cap U_2$ and $\psi_2 \circ \psi_1^{-1}$ can be locally extended to a diffeomorphism on $\tilde{V}$ where $\tilde{V}$ is an open subset of $\mathbb{R}^n$ containing $V_1 \cap V_2$, then the transition map $\Phi_2 \circ \Phi_1^{-1}$ can be locally extended to a diffeomorphism on W, where W is an open subset of $\mathbb{R}^m \times \mathbb{R}^n = \mathbb{R}^{m+n}$ containing $(U_1 \times V_1) \cap (U_2 \times V_2)$. Hence, since $\mathcal{A}_M$ covers M and $\mathcal{A}_N$ covers N, then

  $$\mathcal{A} := \{(U \times V, \Phi) \mid (U, \varphi) \in \mathcal{A}_M, (V, \psi) \in \mathcal{A}_N\}$$

  is a smooth atlas on $M \times N$, which determines a smooth structure on $M \times N$.

- *Characterization of boundary:*

  $$\begin{aligned}(x, y) \in \partial(M \times N) &\Leftrightarrow \exists (U \times V, \Phi) \in \mathcal{A}, \text{ s.t.: } \Phi(x, y) \in \partial\mathbb{H}^{m+n} \\ &\Leftrightarrow \exists (U \times V, \Phi) \in \mathcal{A}, \text{ s.t.: } \Phi(x, y) = (\varphi(x), \psi(y)) = (u_1, \cdots, u_m, v_1, \cdots, v_{n-1}, v_n = 0) \\ &\Leftrightarrow \exists (U \times V, \Phi) \in \mathcal{A}, \text{ s.t.: } \psi(y) \in \partial\mathbb{H}^n \\ &\Leftrightarrow (x, y) \in M \times \partial N\end{aligned}$$

  Hence, $\partial(M \times N) = M \times \partial N$.

□

Theorem 3.3.8. *Smooth Invariance-of-the Boundary Theorem*
Let M be a smooth manifold with boundary and let $p \in M$. If there is a smooth chart (U, φ) for M, s.t.: $\varphi(U) \subseteq \mathbb{H}^n$ and $\varphi(p) \in \partial\mathbb{H}^n$, then the same is true for every smooth chart whose domain contains p.

Proof. This Thm is a weaker version of the Topological Invariance-of-the Boundary Thm. □

Chapter 4

Smooth Maps between Manifolds

4.1 Smooth Functions on Manifolds

Convention: From now on, we make a slight distinguish between the terms *functions* and *maps*:

- The term function is for a map whose codomain is $\mathbb{R}$ (a *real-valued function*) or $\mathbb{R}^k$ for some $k > 1$ (a *vector-valued function*).
- The term map (or mapping) means any type of map.

Definition 4.1.1. Smooth Functions on Manifolds
Let M be a smooth n-manifold, let k be a nonnegative integer and let $f : M \to \mathbb{R}^k$ be any function. We say f is a *smooth function* if $\forall p \in M, \exists$ a smooth chart (U, φ) for M, s.t.: $p \in U$ and that the composition $f \circ \varphi^{-1}$ is smooth on $\varphi(U) \subseteq \mathbb{R}^n$.

Remark:

- If M is a smooth n-manifold with boundary, then the def is the same, except that $\varphi(U)$ is an open subset of $\mathbb{R}^n$ or $\mathbb{H}^n$ and that the smoothness of $f \circ \varphi^{-1}$ is in the sense of extension.
- For $k = 1$, the set of all smooth real-valued functions $f : M \to \mathbb{R}$ is denoted by $\mathcal{C}^\infty(M)$. Since the sums and constant multiplies of smooth functions are smooth, then $\mathcal{C}^\infty(M)$ is a vector space over $\mathbb{R}$.

Definition 4.1.2. Coordinate Representations of functions
Let M be a smooth n-manifold with or without boundary, let (U, φ) be a chart for M, let k be a nonnegative and let $f : M \to \mathbb{R}^k$ be any function. The function $\widehat{f} : \varphi^{-1}(U) \to \mathbb{R}^k$ given by

$$\widehat{f}(x) := (f \circ \varphi^{-1})(x)$$

is called the *coordinate representation of* f.

Remark:

- It's clear that f is smooth iff its coordinate representation is smooth in some smooth chart around each point.

- Smooth functions have smooth coordinate representations in every smooth chart.

Example: Consider the real-valued function $f : \mathbb{R}^2 \to \mathbb{R}$ given by $(x, y) \mapsto x^2 + y^2$. In polar coordinates on the set $U = \{(x, y) \in \mathbb{R}^2 \mid x > 0\}$, it has the coordinate representation $\widehat{f}(r, \theta) = r^2$. Since $\widehat{f}$ is smooth, then so is f.

4.2 Smooth Maps between Manifolds

Definition 4.2.1. Smooth Maps between Manifolds
Let M, N be *smooth* manifolds and let $F : M \to N$ be any map. We say that F is a *smooth map* if $\forall p \in M, \exists$ smooth charts (U, φ) containing p and (V, ψ) containing $F(p)$, s.t.: $F(U) \subseteq V$ and that the composition $\psi \circ F \circ \varphi^{-1} : \varphi(U) \to \psi(V)$ is smooth in the ordinary sense.

Remark:

- If M and N are smooth manifolds with boundary, then the smoothness of F is defined in the sense of extension.
- Our previous def of smoothness of functions can be viewed as a special case of this one, by taking $N = \mathbb{R}^k$ and $\psi = \mathrm{id}_{\mathbb{R}^k}$.

Proposition 4.2.2. *Continuity-of-Smooth Maps Proposition*
Every smooth map $F : M \to N$ between smooth manifolds (with or without boundary) is continuous.

Proof. By def, $\forall p \in M, \exists$ smooth charts $(U \ni p, \varphi), (V \ni F(p), \psi)$, s.t.: $F(U) \subseteq V$ and that $\psi \circ F \circ \varphi^{-1} : \varphi(U) \to \psi(V)$ is smooth (hence continuous). Since φ and ψ are homeomorphisms, then φ and ψ^{-1} are continuous, so the restriction

$$F|_U = \psi^{-1} \circ (\psi \circ F \circ \varphi^{-1}) \circ \varphi : U \to V$$

of F into U is continuous. Hence, F is continuous in an open neighborhood of each point in M and therefore, it's continuous on M. □

Proposition 4.2.3. *Equivalent Characterizations-of-Smoothness Proposition*
Let M and N be smooth manifolds with or without boundary and let $F : M \to N$ be a map. Then the followings are equivalent:

1. *F is smooth.*
2. *$\forall p \in M$, $\exists$ smooth charts $(U \ni p, \varphi), (V \ni F(p), \psi)$, s.t.: $U \cap F^{-1}(V)$ is open in M and the composition map $\psi \circ F \circ \varphi^{-1}$ is smooth from $\varphi(U \cap F^{-1}(V))$ to $\psi(V)$.*
3. *F is continuous and there exist smooth atlases $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \mathcal{A}}$ and $\{(V_\beta, \psi_\beta)\}_{\beta \in \mathcal{B}}$ for M respectively N, s.t.: $\forall \alpha, \beta, \psi \circ F \circ \varphi^{-1}$ is a smooth map from $\varphi_\alpha(U_\alpha \cap F^{-1}(V_\beta))$ to $\psi_\beta(V_\beta)$.*

Proof. 1. $\Rightarrow$ 2 : Since $F(U) \subseteq V$, then $U \subseteq F^{-1}(V)$, so $U \cap F^{-1}(V)$, which is open in M, and $\psi \circ F \circ \varphi^{-1} : \varphi(U) = \varphi(U \cap F^{-1}(V)) \to \psi(V)$ is smooth.

2. $\Rightarrow 3$:

   - *F is continuous:* $\forall W \in \mathcal{T}_V, \forall p \in F^{-1}(W)$, by 1, $\exists$ smooth charts $(U \ni p, \varphi), (V \ni F(p), \psi)$, s.t.: $U' := U \cap F^{-1}(V)$ is open in M and the composite map $\widehat{F} := \psi \circ F \circ \varphi^{-1} : \varphi(U') \to \psi(V)$ is smooth (hence continuous). Since $V \cap W$ is open in V and ψ is a homeomorphism, then $\psi(V \cap W)$ is open in $\psi(V)$. Then by the continuity of

   $$\widehat{F}^{-1}(\psi(V \cap W)) = (\varphi \circ F^{-1})(V \cap W) \subseteq \varphi(U')$$

   is open in $\varphi(U')$. Since φ is a homeomorphism, then we have an open neighborhood of p:

   $$\varphi^{-1}(\widehat{F}^{-1}(\psi(V \cap W))) = U' \cap F^{-1}(V \cap W) = U \cap F^{-1}(V \cap W) \subseteq F^{-1}(W)$$

   which is contained in $F^{-1}(W)$. This gives the continuity of F.

   - *Construction of the desired atlas:* Let $\mathcal{A}_M = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \mathcal{A}}$ and $\mathcal{A}_N = \{(V_\beta, \psi_\beta)\}_{\beta \in \mathcal{B}}$ be any maximal smooth atlas of M respectively N and then equip them to M respectively N for smoothness. $\forall \beta \in \mathcal{B}$, since V_β is open in N, then by the continuity of F, $F^{-1}(V_\beta)$ is open in U, so $U_\alpha \cap F^{-1}(V_\beta)$ is open in M, $\forall \alpha \in \mathcal{A}$. Fix α, β and then take an arbitrary point $x_0 \in \varphi_\alpha(U_\alpha \cap F^{-1}(V_\beta))$. Since $p := \varphi_\alpha^{-1}(x_0) \in U_\alpha \cap F^{-1}(V_\beta)$, then $F(p) \in V_\beta$. By 1, $\exists (U_p \ni p, \varphi_p) \in \mathcal{A}_M, (V_p \ni F(p), \psi_p) \in \mathcal{A}_N$, s.t.:

   $$\psi_p \circ F \circ \varphi_p^{-1} : \varphi_p(U_p \cap F^{-1}(V_p)) \to \psi_p(V_p)$$

   is smooth. Hence, in a small enough open neighborhood of x_0,

   $$\psi_\beta \circ F \circ \varphi_\alpha^{-1} = (\psi_\beta \circ \psi_p^{-1}) \circ (\psi_p \circ F \circ \varphi_p^{-1}) \circ (\varphi_p \circ \varphi_\alpha^{-1})$$

   is smooth by the smooth compatiblility of $\mathcal{A}_M$ and $\mathcal{A}_N$. Since the choice of x_0 is arbitrary in $\varphi_\alpha(U_\alpha \cap F^{-1}(V_\beta))$, then $\psi_\beta \circ F \circ \varphi_\alpha^{-1}$ is smooth on $\varphi_\alpha(U_\alpha \cap F^{-1}(V_\beta))$.

3. $\Rightarrow 1 : \forall p \in M$, since atlases $\{(U_\alpha, \varphi_\alpha)\}$ and $\{(V_\beta, \psi_\beta)\}$ cover M respectively N, then $\exists \alpha, \beta$, s.t.: $p \in U_\alpha$ and $F(p) \in V_\beta$. By the continuity of F, $F^{-1}(V_\beta)$ is open in M, so $\widetilde{U} := U_\alpha \cap F^{-1}(V_\beta)$ is an open subset of M containing p which satisfies $F(\widetilde{U}) \subseteq V_\beta$. Set

   $$\widetilde{\varphi} := \varphi_\alpha|_{\widetilde{U}}$$

   Then the charts $(\widetilde{U}, \widetilde{\varphi})$ and (V_β, ψ_β) are desired in 1.

□

Proposition 4.2.4. *Local Behavior-of-Smooth Maps Proposition*
Let M, N be smooth manifolds with or without boundary and let $F : M \to N$ be a map.

1. *If every point $p \in M$ has an open neighborhood U_p such that the restriction $F|_{U_p}$ is smooth, then F is smooth.*

2. Conversely, if F is smooth, then its restriction to every open subset is smooth.

Proof. 1. Since U_p itself is a smooth manifold and $F|_{U_p} : U_p \to N$ is smooth, then by the Equivalent Characterizations-of-Smoothness Prop.1, there are $(U \ni p, \varphi) \in \mathcal{A}_{U_p}, (V \ni (F|_{U_p})(p), \psi) \in \mathcal{A}_N$, s.t.: $U \cap (F|_{U_p})^{-1}(V)$ is open in U_p and $\psi \circ F|_{U_p} \circ \varphi^{-1} : \varphi(U \cap (F|_{U_p})^{-1}(V)) \to \psi(V)$ is smooth. Now we "elevate" the local properties on U_p to the global properties on M:

- Since U is open in U_p and U_p is open in M, then U is open in M. Hence, $(U, \varphi) \in \mathcal{A}_M$.
- Since $\forall x \in U \subseteq U_p$, $(F|_{U_p})(x) = F(x)$, then we have

$$U \cap (F|_{U_p})^{-1}(V) = U \cap \{x \in U_p \mid F(x) \in V\} = U \cap F^{-1}(V)$$

- Since $U \cap (F|_{U_p})^{-1}(V)$ is open in U_p and U_p is open in M, then $U \cap F^{-1}(V)$ is open in M.
- $\psi \circ F \circ \varphi^{-1}$ is then smooth from $U \cap F^{-1}(V)$ to $\psi(V)$.

Therefore, by the Equivalent Characterizations-of-Smoothness Prop.1, F is smooth.

2. $\forall W \in \mathcal{T}_M, W$ is an smooth manifold. $\forall p \in W \subseteq M$, since F is smooth, then by the Equivalent Characterizations-of-Smoothness Prop.1, there are $(U \ni p, \varphi) \in \mathcal{A}_M, (V \ni F(p), \psi) \in \mathcal{A}_N$, s.t.: $U' := U \cap F^{-1}(V)$ is open in M and $\widehat{F} := \psi \circ F \circ \varphi^{-1} : \varphi(U') \to \psi(V)$ is smooth. Now we restrict the global properties on M into the local properties on W:

- Consider $U_W := U \cap W$, containing clearly p. Since U and W are open in M, then U_W is open in M, so U_W is open in W. Define $\varphi_W := \varphi|_{U_W}$. Then $(U_W \ni p, \varphi_W) \in \mathcal{A}_W$.
- Note that

$$(F|_W)^{-1}(V) := \{x \in W \mid F|_W(x) \in V\} = \{x \in W \mid F(x) \in V\} = F^{-1}(V) \cap W \qquad (*)$$

Since U' is open in M and W is open in M, then

$$U_W \cap (F|_W)^{-1}(V) \overset{(*)}{=} (U \cap W) \cap (F^{-1}(V) \cap W) = (U \cap F^{-1}(V)) \cap W = U' \cap W$$

is open in M, so it is also open in W.

- On U_W, consider the map

$$\widehat{F}_W := \psi \circ F|_W \circ \varphi_W^{-1} : \varphi_W(U_W \cap (F|_W)^{-1}(V)) = \varphi_W(U' \cap W) \overset{U' \subseteq U}{=\!=\!=} \varphi(U' \cap W) \to \psi(V)$$

Since φ is homeomorphism, then $\varphi(U' \cap W)$ is open in $\varphi(U')$. Since $\forall y \in \varphi(U' \cap W)$, we have

$$\widehat{F}_W(y) := \psi((F|_W)(\varphi^{-1}(y))) \overset{\varphi^{-1}(y) \in U' \cap W \subseteq W}{=\!=\!=\!=\!=\!=\!=\!=\!=} \psi(F(\varphi^{-1}(y))) := \widehat{F}(y)$$

then $\widehat{F}_W = \widehat{F}|_{\varphi(U' \cap W)}$. Hence, $\widehat{F}_W$ is smooth.

Therefore, by the Equivalent Characterizations-of-Smoothness Prop.1, $F|_W$ is smooth.

□

Corollary 4.2.5. *of Local Behavior-of-Smooth Maps Prop: Gluing Lemma-for-Smooth Maps*
Let M, N be smooth manifolds with or without boundary and let $\{U_\alpha \mid \alpha \in \mathcal{A}\}$ be an open cover of M. Suppose that $\forall \alpha \in \mathcal{A}$, we are given a smooth map $F_\alpha : U_\alpha \to N$ such that the maps agree on overlaps: $\forall \alpha, \beta \in \mathcal{A}$,

$$F_\alpha|_{U_\alpha \cap U_\beta} = F_\beta|_{U_\alpha \cap U_\beta}$$

Then there exists a unique smooth map $F : M \to N$, s.t.: $F|_{U_\alpha} = F_\alpha, \forall \alpha \in \mathcal{A}$.

Definition 4.2.6. Coordinate Representation of Smooth Maps
Let $F : M \to N$ be a smooth map between smooth manifolds with or without boundary and let $(U, \varphi) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_N$. We call $\widehat{F} := \psi \circ F \circ \varphi^{-1}$ the *coordinate representation of F*, which maps $\varphi(U \cap F^{-1}(V))$ to $\psi(V)$.

Proposition 4.2.7. *Basic Properties-of-Smooth Maps Proposition*
Let M, N and P be smooth manifolds with or without boundary.

1. *Every constant map $c : M \to N$ is smooth.*
2. *The identity map of M is smooth.*
3. *If $U \subseteq M$ is an open submanifold with or without boundary, then the inclusion map $\iota : U \hookrightarrow M$ is smooth.*
4. *If $F : M \to N$ and $G : N \to P$ are smooth, then so is $G \circ F : M \to P$.*

Proof. 1 and 2 can easily be checked by the Equivalent Characterizations-of-Smoothness Prop.1. Since $\iota = \mathrm{id}_M|_U$, then 3 is immediately from the Local Behavior-of-Smooth Maps Prop and 2.

For 4, $\forall p \in M$, by the def of smoothness of G, $\exists$ smooth charts $(V \ni F(p), \theta)$ and $(W \ni G(F(p)), \psi)$, s.t.: $G(V) \subseteq W$ and $\psi \circ G \circ \theta^{-1} : \theta(V) \to \psi(W)$ is smooth. By the continuity of F, $F^{-1}(V)$ is an open neighborhood of p in M. Hence, $\exists$ a smooth chart (U, φ) for M, s.t.: $p \in U \subseteq F^{-1}(V)$. Therefore, $\theta \circ F \circ \varphi^{-1} : \varphi(U) \to \theta(V)$ is smooth. Then we have $(G \circ F)(U) \subseteq G(V) \subseteq W$ and

$$\psi \circ (G \circ F) \circ \varphi^{-1} = (\psi \circ G \circ \theta^{-1}) \circ (\theta \circ F \circ \varphi^{-1}) : \varphi(U) \to \psi(W)$$

is smooth since $\varphi(U)$ and $\psi(W)$ are subsets of Euclidean Spaces. □

Proposition 4.2.8. *Component-wise Smoothness Rule*
Let $M_1, \cdots, M_k$ and N be smooth manifolds with or without boundary, such that at most one of $M_1, \cdots, M_k$ has nonempty boundary. $\forall i$, let $\pi_i : M_1 \times \cdots \times M_k \to M_i$ denote the projection onto the M_i factor. A map $F : N \to M_1 \times \cdots \times M_k$ is smooth iff each of the component maps $F_i := \pi_i \circ F : N \to M_i$ is smooth.

Proof. **Claim:** For each i, π_i is smooth.

Proof. of claim: $\forall j, \forall p_j \in M_j$, take any smooth chart $(U_j \ni p_j, \varphi_j)$. Then the chart $(U_1 \times \cdots \times U_k, \varphi_1 \times \cdots \times \varphi_k)$ is a smooth chart for $M_1 \times \cdots \times M_k$ containing $p := (p_1, \cdots, p_k)$. To apply the Equivalent Characterizations-of-Smoothness Prop.1, we check:

- $(U_1 \times \cdots \times U_k) \cap (\varphi_1 \times \cdots \times \varphi_k)^{-1}(U_j) = (U_1 \times \cdots \times U_k) \cap (M_1 \times \cdots \times M_{i-1} \times U_i \times M_{i+1} \times \cdots \times M_k) = U_1 \times \cdots \times U_k$ is open in $M_1 \times \cdots \times M_k$.
- $\forall x := (x_1, \cdots, x_k) \in (\varphi_1 \times \cdots \times \varphi_k)(U_1 \times \cdots \times U_k) = \varphi_1(U_1) \times \cdots \times \varphi_k(U_k)$, we have:

  $$(\varphi_i \circ \pi_i \circ (\varphi_1 \times \cdots \times \varphi_k)^{-1})(x) = x_i$$

  Hence, $\varphi_i \circ \pi_i \circ (\varphi_1 \times \cdots \times \varphi_k)^{-1}$ is the projection map in Euclidean spaces, hence smooth.

□

- ($\Rightarrow$) holds immediately from the Basic Properties-of-Smooth Maps Prop.4.
- ($\Leftarrow$): $\forall q \in N$, let $p = F(q) = (p_1, \cdots, p_k) \in P := M_1 \times \cdots \times M_k$ where $p_j := F_j(q) \in M_j$. Since $F_j : N \to M_j$ is smooth $\forall j$, then by the Equivalent Characterizations-of-Smoothness Prop.1, then $\forall j$, $\exists$ smooth charts $(V_j \ni q, \psi_j)$ and $(U_j \ni p_j, \varphi_j)$, s.t.: $W_j := V_j \cap F_j^{-1}(U_j)$ is open in N and that $\varphi_j \circ F_j \circ \psi_j^{-1} : \psi_j(W_j) \to \varphi_j(U_j)$ is smooth. Since W_J is open in N, then the finite intersection

  $$V := \bigcap_{j=1}^{k} W_j = \bigcap_{j=1}^{k} (V_j \cap F_j^{-1}(U_j))$$

  is open in N, which clearly contains q. Take $\psi := \psi_1|_V$ and then we get a smooth chart (V, ψ) for N containing q. On P, take

  $$(U, \varphi) := (U_1 \times \cdots \times U_k, \varphi_1 \times \cdots \times \varphi_k)$$

  to be a smooth chart containing p. To apply the Equivalent Characterizations-of-Smoothness Prop.1, we check:

  - $V \cap F^{-1}(U) := V \cap \{x \in N \mid F_j(x) \in U_j\} = V \cap (\bigcap_{j=1}^{k} F_j^{-1}(U_j)) \overset{\text{def of } V}{=\!=} V$ is open in N.
  - $\forall y \in \psi(V)$, since ψ is a homeomorphism, then we can consider $x := \psi^{-1}(y) \in V$. Then we have:

    $$(\varphi \circ F \circ \psi^{-1})(y) = (\varphi_1 \times \cdots \times \varphi_k)(F_1(x), \cdots, F_k(x)) = ((\varphi_1 \circ F_1 \circ \psi_1^{-1})(y), \cdots, (\varphi_k \circ F_k \circ \psi_k^{-1})(y))$$

    Since $\varphi_j \circ F_j \circ \psi_j^{-1}$ is smooth from $\psi_j(W_j)$ to $\varphi_j(U_j)$, then $\varphi \circ F \circ \psi^{-1} : \psi(V) \to \varphi(U)$ is smooth.

  Therefore, by the Equivalent Characterizations-of-Smoothness Prop.1, F is smooth.

□

Example (of *Smooth Maps between Manifolds*):

- Any map from a 0-dimensional manifold to smooth manifold with or without boundary is smooth, since the domain of each coordinate representation contains a single point and hence it is constant, hence smooth.
- If the circle $\mathbb{S}^1$ is given its standard smooth structure, then the map $\epsilon : \mathbb{R} \to \mathbb{S}^1$ defined by $\epsilon(t) := \exp(2\pi i t)$ is smooth.
- Th map $\epsilon^n : \mathbb{R}^n \to \mathbb{T}^n$ defined by $\epsilon^n(x^1, \cdots, x^n) := (\exp(2\pi i x^1), \cdots, \exp(2\pi i x^n))$ is smooth by the Component-wise Smoothness Prop.
- Consider the n-sphere $\mathbb{S}^n$ with its smooth structure. The inclusion map $\iota : \mathbb{S}^n \hookrightarrow \mathbb{R}^{n+1}$ is smooth since its coordinate representation w.r.t. any of the graph coordinates is

  $$\widehat{\iota}(u^1, \cdots, u^n) = (u^1, \cdots, u^{i-1}, \pm\sqrt{1 - |u|^2}, u^i, \cdots, u^n)$$

  which is smooth.
- In the def of $\mathbb{R}P^n$, the quotient map $\pi : \mathbb{R}^{n+1} \setminus \{0\} \to \mathbb{R}P^n$ is smooth, since its coordinate representation

  $$\widehat{\pi}(x^1, \cdots, x^n) = (\varphi_i \circ \pi)(x^1, \cdots, x^{n+1}) = \varphi_i[x^1, \cdots, x^{n+1}] = (\frac{x^1}{x^i}, \cdots, \frac{x^{i-1}}{x^i}, \frac{x^{i+1}}{x^i}, \cdots, \frac{x^{n+1}}{x^i})$$

  is smooth.
- $q := \pi|_{\mathbb{S}^n} : \mathbb{S}^n \to \mathbb{R}P^n$ is smooth, since $q = \pi \circ \iota$.
- If $M_1, \cdots, M_k$ are smooth manifolds, then each projection map $\pi_i : M_1 \times \cdots \times M_k \to M_i$ is smooth, since its coordinate representation is a coordinate projection.

4.3 Diffeomorphisms between Manifolds

Definition 4.3.1. Diffeomorphisms between Manifolds
Let M, N be smooth manifolds with or without boundary. A *diffeomorphism from M to N* is a smooth map $F : M \to N$ that is bijective and that has a smooth inverse. Furthermore, if there exists a diffeomorphism between M and N, then we say that M and N are diffeomorphic, denoted as $M \approx N$.

Example:

- Consider the maps $F : \mathbb{B}^n \to \mathbb{R}^n$ and $G : \mathbb{R}^n \to \mathbb{B}^n$ given by

  $$F(x) := \frac{x}{\sqrt{1 - |x|^2}} \text{ and } G(x) := \frac{y}{\sqrt{1 + |y|^2}}$$

  which are smooth and are inverses of each other. Hence, they are both diffeomorphisms. Therefore, $\mathbb{B}^n \approx \mathbb{R}^n$ are diffeomorphic.

- If M is any smooth manifold and (U, φ) is a smooth coordinate chart on M, then $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^n$ is a diffeomorphism.

Proposition 4.3.2. *Properties-of-Diffeomorphisms Proposition*

1. *Every Composition of diffeomorphisms is again a diffeomorphism.*
2. *Every finite product of diffeomorphisms between smooth manifolds is again a diffeomorphism.*
3. *Every diffeomorphism is a homeomorphism and an open map.*
4. *The restriction of a diffeomorphism to an open submanifold with or without boundary is diffeomorphism onto its image.*
5. *$\approx$ is an equivalence relation on the class of all smooth manifolds with or without boundary.*

Proof. Omitted. □

Theorem 4.3.3. *Diffeomorphism Invariance of Dimension Theorem*
A nonempty smooth manifold of dimension m can not be diffeomorphic to an n-dimensional smooth manifold unless $m = n$.

Proof. Let M be a smooth m-manifold, let N be a smooth n-manifold and let $F : M \to N$ be a diffeomorphism. $\forall p \in M$, let $(U \ni p, \varphi)$ and $(V \ni F(p), \psi)$ be smooth charts. Then $\widehat{F} = \psi \circ F \circ \varphi^{-1}$ is a diffeomorphism from an open subset of $\mathbb{R}^m$ to an open subset of $\mathbb{R}^n$. Hence, $m = n$. □

Theorem 4.3.4. *Diffeomorphism Invariance-of-the Boundary Theorem*
Let M, N be smooth manifolds with boundary and let $F : M \to N$ be a diffeomorphism. Then $F(\partial M) = \partial N$ and F restricts to a diffeomorphism from int M to int N.

Proof. For the first statement, it suffices to show that $F(\text{int } M) = \text{int } N$, by the Topological Invariance-of-the Boundary Thm.

- $\forall p \in \text{int M}, \exists$ a smooth chart $(U \ni p, \varphi)$, s.t.: $\varphi(U)$ is open in $\mathbb{R}^n$. Since F is a diffeomorphism, then $F(U)$ is open in N and $F|_U : U \to F(U)$ is a diffeomorphism. Note that $(F(U), \varphi \circ (F|_U)^{-1} : F(U) \to \varphi(U))$ is a smooth chart on N containing $F(p)$. Hence,

$$F(p) \in \text{int } N \Rightarrow F(\text{int } M) \subseteq \text{int } N$$

- Since $F : M \to N$ is a diffeomorphism, then so is its inverse $F^{-1} : N \to M$, so if we do the same argument, we can get $F^{-1}(\text{int } N) \subseteq \text{int } M \Leftrightarrow \text{int } N \subseteq F(\text{int } M)$.

Hence, $F(\text{int } M) = \text{int } N$. For the second statement, note that

$$(F|_{\text{int } M})^{-1} = F^{-1}|_{\text{int } M}$$

is smooth, then $F|_{\text{int } M} : \text{int } M \to \text{int } N$ is a diffeomorphism. □

Chapter 5

Partition of Unity

Intuition: Recall the Gluing Lemma in topology that if two continuous maps agree on the overlapping (open or closed) part, then we can "glue" them together to get a global continuous map. But it's *not* in general true to "glue" together smooth maps defined on *closed* overlapping part and obtain a smooth result. For example, the non-smooth map $f(x) = |x|$ is obtained by "gluing" the smooth maps $f_+ : [0, \infty) \to \mathbb{R}$ and $f_- : (-\infty, 0] \to \mathbb{R}$ given by

$$f_+(x) := x \text{ and } f_-(x) := -x$$

5.1 Cutoff Functions and Smooth Bump Functions

Lemma 5.1.1. *Basic Smooth Function Lemma*
The function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(t) := \begin{cases} e^{-1/t} & , t > 0 \\ 0 & , t \leq 0 \end{cases}$$

is smooth.

Lemma 5.1.2. *Smooth Cutoff Function Lemma*
Given any real numbers $r_1 < r_2$, there exists a smooth function $h : \mathbb{R} \to \mathbb{R}$, s.t.:

$$h(t) = 1 \text{ for } t \leq r_1, 0 < h(t) < 1 \text{ for } r_1 < t < t_2 \text{ and } h(t) = 0 \text{ for } t \geq r_2$$

Proof. Let f be the function of the Basic Smooth Function Lemma, and then set

$$h(t) := \frac{f(r_2 - t)}{f(r_2 - t) + f(t - r_1)}$$

□

Lemma 5.1.3. *Smooth Bump Function Lemma*
Given any positive real numbers $0 < r_1 < r_2$, there is a smooth function $H : \mathbb{R}^n \to \mathbb{R}$, s.t.:

$$H \equiv 1 \text{ on } \overline{B}_{r_1}(0), 0 < H(x) < 1 \text{ for } x \in B_{r_2}(0) \setminus \overline{B}_{r_1}(0) \text{ and } H \equiv 0 \text{ on } \mathbb{R}^n \setminus B_{r_2}(0)$$

Proof. Set $H(x) := h(|x|)$, where h is the smooth cutoff function of the Smooth Cutoff Function Lemma. □

5.2 Definition and Existence of Partition of Unity

Definition 5.2.1. Support
Let $f : X \to \mathbb{R}^k$ be any real-valued or vector-valued function on a topological space X. Then the *support of* f, denoted as supp f, is the closure of the set of points where f is nonzero:

$$\text{supp } f := \overline{\{p \in X \mid f(p) \neq 0\}}$$

Furthermore, we define:

- If supp f is contained in some set $U \subseteq M$, then we say that f is *supported in* U.
- A function f is said to be *completely supported* if supp f is a compact set.

Example: If H is the smooth bump function of the Smooth Bump Function Lemma, then

$$\text{supp } H = \overline{B}_{r_2}(0)$$

Definition 5.2.2. Partition of Unity
Let M be a topological space and let $\mathcal{C} = \{U_\alpha \mid \alpha \in \mathcal{A}\}$ be an arbitrary open cover of M. A *partition of unity subordinate to* $\mathcal{C}$ is a family $\{\psi_\alpha : M \to \mathbb{R} \mid \alpha \in \mathcal{A}\}$ of continuous functions with the following properties:

1. $0 \leq \psi_\alpha(p) \leq 1, \forall \alpha \in \mathcal{A}, p \in M$.
2. supp $\psi_\alpha \subseteq U_\alpha, \forall \alpha \in \mathcal{A}$.
3. (*Local Finiteness Condition*). The family of supports $\{\text{supp } \psi_\alpha \mid \alpha \in \mathcal{A}\}$ is locally finite, i.e.: $\forall p \in M, \exists$ open $U_p \ni p$, s.t.: U_p intersects supp ψ_α for only finitely many values of α.
4. (*Summation Property*). $\forall x \in M, \sum_{\alpha \in \mathcal{A}} \psi_\alpha(p) = 1$.

Furthermore, if M is a smooth manifold with or without boundary, then a family $\{\psi_\alpha : M \to \mathbb{R} \mid \alpha \in \mathcal{A}\}$ of smooth functions with the four properties is called a *smooth partition of unity*.

Remark: Now we get a tool, partition of unity, for gluing smooth functions. There are basically two strategies to do so:

- (*Direct Gluing*). If we can find an open cover $\{U_\alpha\}$ of M satisfying the conditions stated in coro of Local Behavior-of-Smooth Maps Prop: Gluing Lemma-of-Smooth Maps, then we can

just apply it.

- (*Soft Blending with Partition of Unity*). If the local defs are not guaranteed to agree, then we have to resort to a partition of unity: Consider the function

$$f_{\text{global}}(p) := \sum_{\alpha \in \mathcal{A}} \psi_\alpha(p) f_\alpha(p)$$

By the property 4 in the def of partition of unity, this function makes a smooth weighted average in the overlapping part. But we have two troubles:

  - This strategy may change the value of the original functions, so in usage scenarios, we need to check that the initially desired mathematical properties are inherited.
  - We need to guarantee the common use of this strategy, say, the existence of partitions of unity:

Theorem 5.2.3. *Existence-of-Partitions of Unity-on-Smooth Manifolds Theorem*
Let M be a smooth manifold with or without boundary and let $\mathcal{C} = \{U_\alpha \mid \alpha \in \mathcal{A}\}$ be any open cover of M. Then there exists a smooth partition of unity subordinate to $\mathcal{C}$.

Proof. We first deal with the case in which M is a smooth manifold without boundary and the case in which M is a smooth manifold with boundary will be omitted here, sharing the same method and with the core that boundary points are covered by regular coordinate half-balls.

Since each U_α is a smooth manifold, then by the Well-Behaved Basis-for-Smooth Manifolds Prop, it admits a basis $\mathcal{B}_\alpha$ consisting of regular coordinate balls. Then $\mathcal{B} := \bigcup_{\alpha \in \mathcal{A}} \mathcal{B}_\alpha$ is a basis for $\mathcal{T}_M$. By the Paracompactness-of-Topological Manifolds Theorem, $\mathcal{C}$ has a countable, locally finite refinement $\{B_i\} \subseteq \mathcal{B}$. Then by the Closure-and-Unions-of-Locally Finite Families Lemma.1, the cover $\{\overline{B}_i\}$ is also locally finite.

$\forall i$, since B_i is a regular coordinate ball in some U_α, then by the def of regular coordinate balls, $\exists$ a coordinate ball $B_i' \subseteq U_\alpha$, with a smooth coordinate map $\varphi_i : B_i' \to \mathbb{R}^n$, s.t.: $\overline{B}_i \subseteq B_i'$ and that for some $r_i < r_i'$, we have

$$\varphi_i(B_i) = B_{r_i}(0), \varphi_i(\overline{B}_i) = \overline{B}_{r_i}(0) \text{ and } \varphi_i(B_i') = B_{r_i'}(0)$$

For each i, let H_i be the smooth bump function in the Smooth Bump Function Lemma with $r_1 = r_i$ and $r_2 = r_i'$ and then define a function $f_i : M \to \mathbb{R}$ by

$$f_i := \begin{cases} H_i \circ \varphi_i & \text{on } B_i' \\ 0 & \text{on } M \setminus \overline{B}_i \end{cases}$$

which is well-defined (since $f_i|_{B_i' \setminus \overline{B}_i} \equiv 0$ for both two defs) and smooth with support supp $f_i = \overline{B}_i$.

Consider $f : M \to \mathbb{R}$ given by $p \mapsto \sum_i f_i(p)$. By the local finiteness of the cover $\{\overline{B}_i\}$, the sum $\sum_i f_i(p)$ has only finitely many nonzero terms in an open neighborhood of each point $p \in M$ and

hence f is a smooth function. Since each f_i is nonnegative everywhere and positive on B_i, and every point of M is in some B_i, then f is positive everywhere on M. Hence, $\forall i$, we can well-definedly define a function $g_i : M \to \mathbb{R}$ by

$$g_i(x) := \frac{f_i(x)}{f(x)}$$

which is smooth. Clearly by def, we have: $0 \leq g_i(p) \leq 1$ and $\sum_i g_i \equiv 1$.

Now we can define the partition of unity subordinate to $\mathcal{C}$: Since the cover $\{B_i'\}$ is refinement of $\mathcal{C}$, then $\forall i, \exists \alpha_i \in \mathcal{A}$, s.t.: $B_i' \subseteq U_{\alpha_i}$. For each $\alpha \in \mathcal{A}$, define $\psi_\alpha : M \to \mathbb{R}$ by

$$\psi_\alpha := \sum_{i:\alpha_i=\alpha} g_i$$

where if $\{i \mid \alpha_i = \alpha\} = \emptyset$, then we define $\psi_\alpha := 0$. Then we check:

1. By the Closures-and-Unions-of-Local Finite Families Lemma, we have
$$\text{supp } \psi_\alpha = \overline{\bigcup_{i:\alpha_i=\alpha} B_i} \overset{\text{lemma}}{=\!=} \bigcup_{i:\alpha_i=\alpha} \overline{B}_i \subseteq \bigcup_{i:\alpha_i=\alpha} B_i' \subseteq U_\alpha$$
2. $\forall p \in M, 0 \leq \psi_\alpha(p) = \sum_{i:\alpha_i=\alpha} g_i(p) \leq \sum_i g_i(p) = 1$.
3. The local finiteness condition of $\{\text{supp } \psi_\alpha\}$ inherits from $\{f_i\}$.
4. $\sum_\alpha \psi_\alpha = \sum_i g_i \equiv 1$.

Moreover, each ψ_α is smooth, so $\{\psi_\alpha\}_{\alpha \in \mathcal{A}}$ is the desired partition of unity. □

5.3 Applications

5.3.1 Extension of Bump Functions

Definition 5.3.1. Bump Functions
Let X be a topological space, let $A \subseteq X$ be a closed subset and let $U \subseteq X$ be an open subset containing A. A continuous function $\psi : X \to \mathbb{R}$ is called a *bump function for A supported in U* if

$$0 \leq \psi \leq 1 \text{ on } X, \psi|_A \equiv 1 \text{ and supp } \psi \subseteq U$$

Proposition 5.3.2. *Existence-of-Smooth Bump Functions-on-Smooth Manifolds Proposition*
Let M be a smooth manifold with or without boundary. For any closed subset $A \subseteq M$ and any open subset $U \subseteq M$ containing A, there is a smooth bump function for A supported in U.

Proof. Note that $\{U_0 := U, U_1 := M \setminus A\}$ is an open cover of M. Hence, by the Existence-of-Partitions of Unity Theorem, there is a smooth partition of union $\{\psi_0, \psi_1\}$ subordinate to $\{U_0, U_1\}$. Since $\psi_1 \equiv 0$ on A, then

$$\psi_0|_A = \sum_i (\psi_i|_A) \equiv 1$$

Therefore, ψ_0 is the desired smooth bump function. □

5.3.2 Extension of Smooth Functions from Closed Subsets

Definition 5.3.3. Smooth Maps on Subsets of Manifolds
Let M, N be smooth manifolds with or without boundary and let $A \subseteq M$ be an arbitrary subset. We say that a map $F : A \to N$ is *smooth on A* if it has a smooth extension in an open neighborhood of each point, that is, if $\forall p \in A, \exists$ open $W \subseteq M$ containing p, and a smooth map $\widetilde{F} : W \to N$ whose restriction to $W \cap A$ agrees with F.

Lemma 5.3.4. *Exention Lemma for Smooth Functions*
Let M be a smooth manifold with or without boundary, let $A \subseteq M$ be a closed subset and $f : A \to \mathbb{R}^k$ be a smooth function. For any open subset U containing A, there exists a smooth function $\widetilde{f} : M \to \mathbb{R}^k$ such that $\widetilde{f}|_A = f$ and supp $\widetilde{f} \subseteq U$.

Proof. $\forall p \in A$, choose an open neighborhood W_p of p and a smooth function $\widetilde{f}_p : W_p \to \mathbb{R}^k$ that agrees with f on $W_p \cap A$. Since W_p and U are open in M, then so is $W_p \cap U$ and hence we may assume $W_p \subseteq U$. Note that $\{W_p \mid p \in A\} \cup \{M \setminus A\}$ is an open cover of M. Hence, by the Existence-of-Partitions of Unity Thm, there is a smooth partition of unity $\{\psi_p \mid p \in A\} \cup \{\psi_0\}$ subordinate to this cover, with supp $\psi_p \subseteq W_p$ and supp $\psi_0 \subseteq M \setminus A$.

$\forall p \in A$, since W_p is open in M, then the product $\psi_p \widetilde{f}_p$, which is smooth on W_p, has a smooth extension to all of M, which is zero on $M \setminus$ supp ψ_p. Then we can define $\widetilde{f} : M \to \mathbb{R}^k$ by

$$\widetilde{f}(x) := \sum_{p \in A} \psi_p(x) \widetilde{f}_p(x)$$

By the local finiteness of $\{\text{supp } \psi_p\}_{p \in A}$, this sum has only finite number of nonzero terms in an open neighborhood of any point in M. Hence, $\widetilde{f}$ is smooth. $\forall x \in A$, since supp $\psi_0 \subseteq M \setminus A$, then $\psi_0(x) = 0$, so

$$\widetilde{f}(x) := \sum_{p \in A} \psi_p(x) \widetilde{f}_p(x) \xlongequal{\text{def of } \widetilde{f}_p} \sum_{p \in A} \psi_p(x) f(x) = (\psi_0 + \sum_{p \in A} \psi_p(x)) f(x) = f(x)$$

Therefore, $\widetilde{f}$ is an extension of f. For the second statement, it follws from the Closures-and-Unions-od-Local Finite Families Lemma.2 that

$$\text{supp } \widetilde{f} = \overline{\bigcup_{p \in A} \text{supp } \psi_p} = \bigcup_{p \in A} \overline{\text{supp } \psi_p} = \bigcup_{p \in A} \text{supp } \psi_p \subseteq U$$

□

Remark:

- If A is *not* closed, then this lemma may fail: (*Counterexample*). Suppose $M = \mathbb{R}, A = (0, 1)$. Then the function $f : A \to \mathbb{R}$ given by $x \mapsto \frac{1}{x}$ has no extension.

- The assumption in this lemma that the codomain of f is $\mathbb{R}^k$, and *not* some other smooth manifold, is need: (*Counterexample*). The identity map $\mathrm{id}_{\mathbb{S}^1}$ on $\mathbb{S}^1$ is smooth, but does *not* have even a continuous extension to a map from $\mathbb{R}^2$ to $\mathbb{S}^1$.

5.3.3 Exaustion Functions

Definition 5.3.5. Exhaustion Functions
Let X be a topological space. An *exhaustion function for* X is a continuous function $f : X \to \mathbb{R}$ with the property that the set $f^{-1}((-\infty, c])$ is compact $\forall c \in \mathbb{R}$. Here, we call the set $f^{-1}((-\infty, c])$ a *sublevel set of* f.

Example: The functions $f : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{B}^n \to \mathbb{R}$ given by

$$f(x) := |x|^2 \text{ and } g(x) := \frac{1}{1 - |x|^2}$$

are exhaustion functions.
Remark: It's clear that if X is compact, then any continuous real-valued function on X is an exhaustion, so we are interesting such functions only noncompact manifolds.

Proposition 5.3.6. *Existence-of-Smooth Exhaustion Functions Proposition*
Every smooth manifold with or without boundary M admits a smooth positive exhaustion function.

Proof. Let $\{V_j\}_{j=1}^{\infty}$ be any countable open cover of M by precompact open subsets and let $\{\psi_j\}_{j=1}^{\infty}$ be a smooth partition of unity subordinate this cover. Define $f : M \to \mathbb{R}$ by

$$f(p) := \sum_{j=1}^{\infty} j\psi_j(p)$$

which is smooth, by the local finiteness, and positive since $f(p) \geq \sum \psi_j(p) = 1 > 0$.

We claim that f is an exhaustion function $\forall c \in \mathbb{R}$, let $N := [c] + 1 > c$. If $p \notin \bigcup_{j=1}^{N} \overline{V}_j$, then $\psi_j(p) = 0$ for $1 \leq j \leq N$. Hence,

$$f(p) = \sum_{j=N+1}^{\infty} j\psi_j(p) \geq \sum_{j=N+1}^{\infty} N\psi_j(p) = N \sum_{j=1}^{\infty} \psi_j(p) = N > c$$

We restate it equivalently that if $f(p) \leq c$, then $p \in \bigcup_{j=1}^{N} \overline{V}_j$. Therefore, $f^{-1}((-\infty, c]) \subseteq \bigcup_{j=1}^{N} \overline{V}_j$. Note that $\bigcup_{j=1}^{N} \overline{V}_j$ is compact and $f^{-1}((-\infty, c])$ is closed by the continuity of f, so $f^{-1}((-\infty, c])$ is compact. □

5.3.4 Level Sets

Theorem 5.3.7. *Level Sets of Smooth Manifolds Theorem*
Let M be a smooth manifold. If K is any closed subset of M, then there is a smooth nonnegative function $f : M \to \mathbb{R}$, s.t.: $f^{-1}(0) = K$.

Chapter 6

Tangent Vectors

6.1 Geometric Tangent Vectors

Definition 6.1.1. Geometric Tangent Vectors
Given a point $a \in \mathbb{R}^n$, we define the *geometric tangent space to* $\mathbb{R}^n$ *at* a by

$$\mathbb{R}^n_a := \{a\} \times \mathbb{R}^n = \{(a, v) \mid v \in \mathbb{R}^n\}$$

A *geometric tangent vector* in $\mathbb{R}^n$ is an element of $\mathbb{R}^n_a$ for some $a \in \mathbb{R}^n$. Furthermore, we abbreviate a geometric tangent vector (a, v) as v_a (or $v|_a$).

Remark:

- We can think of v_a as the vector v with its initial point at a. Then we can immediately get that $\mathbb{R}^n_a$ is a real vector space, under the operations

  $$v_a + w_a := (v + w)_a \text{ and } c(v_a) := (cv)_a$$

  with the basis $\{(e_i)_a \mid i = 1, \cdots, n\}$. In fact, $\mathbb{R}^n_a$ is, as a vector space, the same as $\mathbb{R}^n$. The reason why we define the notion of geometric tangent space is to regard $\mathbb{R}^n_a$ and $\mathbb{R}^n_b$ as two disjoint sets.

- Any geometric tangent vector $v_a \in \mathbb{R}^n_a$ yields a map $D_v|_a : \mathcal{C}^\infty(\mathbb{R}^n) \to \mathbb{R}$ given by

  $$D_v|_a(f) := D_v f(a) = \frac{d}{dt}|_{t=0} f(a + tv)$$

  which is linear over $\mathbb{R}$ and satisfies the product rule:

  $$D_v|_a(fg) = f(a)D_v|_a(g) + g(a)D_v|_a(f)$$

  Furthermore, by using the representation with respect to the standard basis of $v_a = v^i(e_i)_a$,

we can write $D_v|_a(f)$, by the chain rule, as

$$D_v|_a(f) = v^i \frac{\partial f}{\partial x^i}(a)$$

where we firstly apply the Einstein summation convention and there will be no such reminder anymore at the next time we use this convention.

Definition 6.1.2. Derivations
Given any point $a \in \mathbb{R}^n$, a map $w : \mathcal{C}^\infty(\mathbb{R}^n) \to \mathbb{R}$ is called a derivation at a if it is linear over $\mathbb{R}$ and satisfies the following product rule:

$$w(fg) = f(a)w(g) + g(a)w(f)$$

Furthermore, we denote the set of all derivations of $\mathcal{C}^\infty(\mathbb{R}^n)$ at a as $T_a\mathbb{R}^n$.

Remark: It's clear that $T_a\mathbb{R}^n$ is a vector space over $\mathbb{R}$, under the operations

$$(w_1 + w_2)(f) = w_1(f) + w_2(f) \text{ and } (cw)(f) = c(w(f))$$

Proposition 6.1.3. *Properties-of-Derivations Proposition*
Let $a \in \mathbb{R}^n$, let $w \in T_a\mathbb{R}^n$ and let $f, g \in \mathcal{C}^\infty(\mathbb{R}^n)$.

1. *If f is contant function, then $w(f) = 0$.*
2. *If $f(a) = g(a) = 0$, then $w(fg) = 0$.*

Proof. 1. By the linearity of w, it suffices to show that $w(f_1) = 0$ for the constant function $f_1 \equiv 1$: By the product rule, we have

$$w(f_1) = w(f_1 f_1) = w(f_1) + w(f_1) = 2w(f_1)$$

which implies $w(f_1) = 0$.

2. By the product rule, we have:

$$w(fg) = f(a)w(g) + g(a)w(f) = 0 + 0 = 0$$

□

Theorem 6.1.4. *Directional Derivative Isomorphism Theorem*
Let $a \in \mathbb{R}^n$.

1. *For each geometric tangent vector $v_a \in \mathbb{R}^n_a$, the map $D_v|_a : \mathcal{C}^\infty(\mathbb{R}^n) \to \mathbb{R}$ is a derivation at a.*
2. *The map $v_a \mapsto D_v|_a$ is an isomorphism between $\mathbb{R}^n_a$ and $T_a\mathbb{R}^n$.*

Proof. 1. holds immediately by def.

2. Denote the map $v_a \mapsto D_v|_a$ by $\varphi : \mathbb{R}^n_a \to T_a\mathbb{R}^n$.

   - *φ is linear over $\mathbb{R}$:* $\forall v_a, w_a \in \mathbb{R}^n_a, \forall f \in \mathcal{C}^\infty(\mathbb{R}^n), \forall c \in \mathbb{R}$, we have:

$$\begin{aligned}\varphi(v_a + cw_a)(f) &:= D_{v+cw}|_a(f) := (v^i + cw^i)\frac{\partial f}{\partial x^i}(a) = v^i\frac{\partial f}{\partial x^i}(a) + cw^i\frac{\partial f}{\partial x^i}(a) \\ &= D_v|_a(f) + cD_w|_a(f) := (\varphi(v_a) + c\varphi(w_a))(f)\end{aligned}$$

   - *φ is injective:* $\forall v_a \in \ker\varphi, D_v|_a(f) = 0$ for any $f \in \mathcal{C}^\infty(\mathbb{R}^n)$. Then for each fixed j, take $f := x^j$ to be the jth coordinate function and then we get

$$0 = D_v|_a(f) := v^i\frac{\partial f}{\partial x^i}(a) = v^j$$

     Hence, $v_a = v^i(e_i)_a = 0$.

   - *φ is surjective:* $\forall w \in T_a\mathbb{R}^n$, define $v = v^i e_i \in \mathbb{R}^n$ by $v^i := w(x^i)$. $\forall f \in \mathcal{C}^\infty(\mathbb{R}^n)$, by Taylor's Thm, we write:

$$f(x) = f(a) + \sum_{i=1}^n \frac{\partial f}{\partial x^i}(a)(x^i - a^i) + \sum_{i,j=1}^n (x^i - a^i)(x^j - a^j)\int_0^1 (1-t)\frac{\partial^2 f}{\partial x^i \partial x^j}(a + t(x-a))dt$$

     Note that each term in the last sum above is a product of smooth functions of x that vanish at $x = a$. By the Properties-of-Derivations Prop.2, the derivation w annihilates this entire sum. Hence,

$$\begin{aligned}w(f) &= w(f(a)) + \sum_{i=1}^n w(\frac{\partial f}{\partial x^i}(a)(x^i - a^i)) + 0 \\ &= 0 + \sum_{i=1}^n \frac{\partial f}{\partial x^i}(a)(w(x^i) - w(a^i)) \\ &= \sum_{i=1}^n \frac{\partial f}{\partial x^i}(a)(v^i - 0) = D_v|_a(f)\end{aligned}$$

     Therefore, $w = D_v|_a = \varphi(v_a)$.

□

Corollary 6.1.5. *of Directional Derivative Isomorphism Theorem*
$\forall a \in \mathbb{R}^n$, the n derivations $\partial/\partial x^i|_a$ defined by

$$\frac{\partial}{\partial x^i}|_a(f) := \frac{\partial f}{\partial x^i}(a), \forall i = 1, \cdots, n$$

form a basis for $T_a\mathbb{R}^n$. Therefore, $T_a\mathbb{R}^n$ has dimension n.

Proof. It follows immediately from the Directional Derivative Isomorphism Thm and an observation that $\partial/\partial x^i|_a = D_{e_i}|_a$. □

6.2 Tangent Vectors on Manifolds

Definition 6.2.1. Tangent Vectors on Manifolds
Let M be a smooth manifold with or without boundary and let p be a point of M. A linear map $v : \mathcal{C}^\infty(M) \to \mathbb{R}$ is called a *derivation at* p if it satisfies

$$v(fg) = f(p)v(g) + g(p)v(f), \forall f, g \in \mathcal{C}^\infty(M)$$

The set of all derivations of $\mathcal{C}^\infty(M)$ at p, denoted as T_pM, is a vector space called the *tangent space to* M *at* p. An element of T_pM is called a *tangent vector at* p.

Proposition 6.2.2. *Properties-of-Tangent Vector on Manifolds Proposition*
Let M *be a smooth manifold with or without boundary, let* $p \in M$*, let* $v \in T_pM$ *and let* $f, g \in \mathcal{C}^\infty(M)$.

1. *If* f *is a constant function, then* $v(f) = 0$.
2. *If* $f(p) = g(p) = 0$*, then* $v(fg) = 0$.

Proof. Similar as the proof of the Properties-of-Derivations Prop. □

6.3 Differential of Smooth Maps

Definition 6.3.1. Differential of Smooth Maps
Let M, N be smooth manifolds with or without boundary and let $F : M \to N$ be a smooth map. For each $p \in M$, we define a map

$$dF_p : T_pM \to T_{F(p)}N$$

called the *differential of* F *at* p by

$$dF_p(v)(f) := v(f \circ F), \forall v \in T_pM, f \in \mathcal{C}^\infty(N)$$

Remark (*Well-defined-ness*).

- If $f \in \mathcal{C}^\infty(N)$, then $f \circ F \in \mathcal{C}^\infty(M)$.
- $\forall f, g \in \mathcal{C}^\infty(N)$, we have

$$\begin{aligned} dF_p(v)(fg) &:= v((fg) \circ F) = v((f \circ F)(g \circ F)) \\ &\overset{v \in T_pM}{=\!=} (f \circ F)(p)v(g \circ F) + (g \circ F)(p)v(f \circ F) \\ &:= f(F(p))dF_p(v)(g) + g(F(p))dF_p(v)(f) \end{aligned}$$

  Hence, $dF_p(v) \in T_{F(p)}N$.

Proposition 6.3.2. *Properties-of-Differentials Proposition*
Let M, N *and* P *be smooth manifolds with or without boundary, let* $F : M \to N$ *and* $G : N \to P$

be smooth maps and let $p \in M$.

1. *$dF_p : T_pM \to T_{F(p)}N$ is linear.*
2. *$d(G \circ F)_p = dG_{F(p)} \circ dF_p : T_pM \to T_{(G\circ F)(p)}P$.*
3. *$d(id_M)_p = id_{T_pM} : T_pM \to T_pM$.*
4. *If F is a diffeomorphism, then $dF_p : T_pM \to T_{F(p)}N$ is an isomorphism and $(dF_p)^{-1} = d(F^{-1})_{F(p)}$.*

Proof. 1. $\forall v, w \in T_pM, \forall f \in \mathcal{C}^\infty(N), \forall c \in \mathbb{R}$,

$$\begin{aligned} dF_p(cv + w)(f) &:= (cv + w)(f \circ F) = c(v(f \circ F)) + w(f \circ F) \\ &:= cdF_p(v)(f) + dF_p(w)(f) \end{aligned}$$

2. $\forall v \in T_pM, \forall f \in \mathcal{C}^\infty(P)$,

$$\begin{aligned} d(G \circ F)_p(v)(f) &:= v(f \circ (G \circ F)) = v((f \circ G) \circ F) \\ &:= dF_p(v)(f \circ G) := dG_{F(p)}(dF_p(v))(f) \end{aligned}$$

3. $\forall v \in T_pM, \forall f \in \mathcal{C}^\infty(M)$,

$$d(\mathrm{id}_M)_p(v)(f) := v(f \circ \mathrm{id}_M) = v(f)$$

4. Since F is a diffeomorphism, then $F^{-1} : N \to M$ is smooth with $F^{-1} \circ F = \mathrm{id}_M$. Hence, we can get:
$$\mathrm{id}_{T_pM} \overset{3}{=\!=} d(\mathrm{id}_M)_p = d(F^{-1} \circ F)_p \overset{2}{=\!=} d(F^{-1})_{F(p)} \circ dF_p$$
Therefore, $(dF_p)^{-1} = d(F^{-1})_{F(p)}$, so we're done by 1.

□

6.4 More Properties of Tangent Vectors

6.4.1 Locality

Intuition: Naturally, after defining the previous basic notions, we will be to use coordinate charts to relate the tangent space to a point on a manifold with the Euclidean tangent space. But the def of tangent spaces depends on the whole manifold, while the coordinate charts are in general defined only on open subsets.

Proposition 6.4.1. *Locality-of-Tangent Vectors Proposition*
Let M be a smooth manifold with or without boundary, let $p \in M$ and let $v \in T_pM$. If $f, g \in \mathcal{C}^\infty(M)$ agree on some open neighborhood of p, then $v(f) = v(g)$.

Proof. Consider $h := f - g$, which vanishes in an open neighborhood of p. Let $\varphi \in \mathcal{C}^\infty(M)$ be a smooth bump function that is identically equal to 1 on the support of h and is supported in $M \setminus \{p\}$. Since $\psi \equiv 1$ on which h is nonzero, then $\psi h = h$. Since $h(p) = \psi(p) = 0$, then by the Properties-of-Tangent Vectors on Manifolds Prop, we have

$$v(h) = v(\psi h) = 0$$

Hence, $v(f) = v(g) = 0$. □

6.4.2 Identification

Open Submanifolds

Proposition 6.4.2. *Tangent Spaces-to-Open Submanifolds Proposition*
Let M be a smooth manifold with or without boundary, let $U \subseteq M$ be an open subset and let $\iota : U \hookrightarrow M$ be the inclusion map. $\forall p \in U$, the differential $d\iota_p : T_pU \to T_pM$ is an isomorphism.

Proof.

- *Injectivity:* Let $v \in \ker d\iota_p$ and let B be an open neighborhood of p, s.t.: $\overline{B} \subseteq U$. $\forall f \in \mathcal{C}^\infty(U)$, by the Extension Lemma for Smooth Functions, $\exists \widetilde{f} \in \mathcal{C}^\infty(M)$, s.t.: $\widetilde{f} \equiv f$ on $\overline{B}$. Since f and $\widetilde{f}|_U$ are smooth functions on U that agree in an open neighborhood of p, then by the Locality-of-Tangent Vectors Prop, we have:

$$v(f) = v(\widetilde{f}|_U) = v(\widetilde{f} \circ \iota) := d\iota_p(v)(\widetilde{f}) = 0$$

- *Surjectivity:* $\forall w \in T_pM$, define $v : \mathcal{C}^\infty(U) \to \mathbb{R}$ by $v(f) = w(\widetilde{f})$, where $\widetilde{f}$ is any smooth function on all of M that agrees with f on $\overline{B}$ and which is well-defined by the Locality-of-Tangent Vectors Prop and is a derivation of $\mathcal{C}^\infty(U)$ at p. $\forall g \in \mathcal{C}^\infty(M)$,

$$d\iota_p(v)(g) := v(g \circ \iota) := w(\widetilde{g \circ \iota}) = w(g)$$

where the last two equalities follow from the fact that $g \circ \iota, \widetilde{g \circ \iota}$ and g all agree on B.

□

Vector Spaces

Proposition 6.4.3. *Tangent Sapces-to-Vector Spaces Proposition*
Let V be a finite dimensional vector space with its standard manifold structure. $\forall a \in V$, the map $v \mapsto D_v|_a$ defined by

$$D_v|_a(f) := \frac{d}{dt}|_{t=0} f(a + tv)$$

is a canonical isomorphism from V to T_aV, s.t.: $\forall$ vector space $W, \forall L \in \mathcal{L}(V,W)$, the following diagram commutes:

$$\begin{array}{ccc} V & \xrightarrow{\cong} & T_aV \\ \downarrow{\scriptstyle L} & & \downarrow{\scriptstyle dL_a} \\ W & \xrightarrow{\cong} & T_{L(a)}W \end{array}$$

Proof. Once we choose a basis for V, we may use the same argument as in the proof of the Directional Derivative Isomorphism Thm to show that $D_v|_a$ is indeed a derivation at a and that the map $v \mapsto D_v|_a$ is an isomorphism. As for the commutativity of the diagram, we compute

$$\begin{aligned} dL_a(D_v|_a(f)) &:= D_v|_a(f \circ L) := \frac{d}{dt}|_{t=0} f(L(a+tv)) \\ &= \frac{d}{dt}|_{t=0} f(L(a)+tL(v)) := D_{L(v)}|_{L(a)}(f) \end{aligned}$$

□

Remark: Any vector $v \in V$ can be identified with the directional derivative operator $D_v|_a$.

Product Manifolds

Proposition 6.4.4. *Tangent Spaces-to-Product Manifolds*
Let $M_1, \cdots, M_k$ be smooth manifolds and for each j, let $\pi_j : M_1 \times \cdots \times M_k \to M_j$ be the projection onto the M_j factor. $\forall p = (p_1, \cdots, p_k) \in M_1 \times \cdots \times M_k$, the map $\alpha : T_p(M_1 \times \cdots \times M_k) \to T_{p_1}M_1 \oplus \cdots \oplus T_{p_k}M_k$ given by

$$\alpha(v) := (d(\pi_1)_p(v), \cdots, d(\pi_k)_p(v))$$

is an isomorphism. The same is true if one of M_i is a smooth manifold with boundary.

Proof. Omitted. □

6.4.3 Dimension

Proposition 6.4.5. *Dimension-of-Tangent Spaces-on-Smooth Manifolds*
Let M be an n-dimensional smooth manifold. $\forall p \in M$, the tangent space T_pM is an n-dimensional vector space.

Proof. Let (U, φ) be a smooth coordinate chart containing p. Since φ is a diffeomorphism from U to $\widetilde{U} := \varphi(U) \subseteq \mathbb{R}^n$, then by the Properties-of-Differentials Prop.4, $d\varphi_p$ is an isomorphism between T_pU and $T_{\varphi(p)}\widetilde{U}$. Then by the Tangent Spaces-to-Open Submanifolds Prop, we have

$$T_pM \cong T_pU \text{ and } T_{\varphi(p)}\widetilde{U} \cong T_{\varphi(p)}\mathbb{R}^n$$

Therefore, $T_pM \cong T_{\varphi(p)}\mathbb{R}^n$, which shows that $\dim T_pM = \dim T_{\varphi(p)}\mathbb{R}^n = n$. □

Lemma 6.4.6. *Boundary Tangent Spaces Lemma*
Let $\iota : \mathbb{H}^n \hookrightarrow \mathbb{R}^n$ be the inclusion map. $\forall a \in \partial\mathbb{H}^n$, the differential $d\iota_a : T_a\mathbb{H}^n \to T_a\mathbb{R}^n$ is an isomorphism.

Proof.
- *Injectivity:* Let $v \in T_a\mathbb{H}^n$ be such that $d\iota_a(v) = 0$. $\forall f \in \mathcal{C}^\infty(\mathbb{H}^n)$, let $\widetilde{f}$ be any extension of f to a smooth function on all of $\mathbb{R}^n$. Then $\widetilde{f} \circ \iota = f$. Hence,

$$v(f) = v(\widetilde{f} \circ \iota) := d\iota_a(v)(\widetilde{f}) = 0$$

- *Surjectivity:* $\forall w \in T_a\mathbb{R}^n$, define $v \in T_a\mathbb{H}^n$ by $v(f) := w(\widetilde{f})$ where f is any smooth extension of f. Hence,

$$v(f) = w^i \frac{\partial \widetilde{f}}{\partial x^i}(a)$$

which is independent of the choice of $\widetilde{f}$. Then it's easy to check that v is a derivation at a and that $w = d\iota_a(v)$.

□

Proposition 6.4.7. *Dimension-of-Tangent Spaces-on-Manifolds with Boundary Proposition*
Let M be an n-dimensional smooth manifold with boundary. $\forall p \in M, T_pM$ is an n-dimensional vector space.

Proof. If p is an interior point, then by the Tangent Spaces-to-Open Submanifolds Prop, $T_p(\text{int } M) \cong T_pM$, since int M is an open submanifold of M, and then we're done. Otherwise, if $p \in \partial M$, then let (U, φ) be a smooth boundary chart containing p and let $\widetilde{U} := \varphi(U) \subseteq \partial\mathbb{H}^n$. Note that by the Tangent Spaces-to-Open Submanifolds Prop, $T_pM \cong T_pU$, that since φ is a diffeomorphism, then by the Properties-of-Differentials Prop.4, $T_pU \cong T_{\varphi(p)}\widetilde{U}$, and that by the Boundary Tangent Spaces Lemma, $T_{\varphi(p)}\mathbb{H}^n \cong T_{\varphi(p)}\mathbb{R}^n$. Then we're done. □

6.5 Computations in Coordinates

6.5.1 Coordinate Vectors and Basis

Definition 6.5.1. Coordinate Vectors
Let M be a smooth n-manifold, let $p \in M$ and let (U, φ) be a smooth coordinate chart on M containing p. Then $d\varphi_p : T_pM \to T_{\varphi(p)}\mathbb{R}^n$ is an isomorphism. By the coro of Directional Derivative Isomorphism Thm, the derivations $\partial/\partial x^1|_{\varphi(p)}, \cdots, \partial/\partial x^n|_{\varphi(p)}$ form a basis for $T_{\varphi(p)}\mathbb{R}^n$. Hence, the preimages of these vectors under $d\varphi_p$ form a basis for T_pM:

$$\frac{\partial}{\partial x^i}|_p := (d\varphi_p)^{-1}(\frac{\partial}{\partial x^i}|_{\varphi(p)}) = d(\varphi^{-1})_{\varphi(p)}(\frac{\partial}{\partial x^i}|_{\varphi(p)})$$

called the *coordinate vectors at p* associated with the given coordinate system. Furthermore, the

action of $\partial/\partial x^i|_p$ on a function $f \in \mathcal{C}^\infty(U)$ is then defined by:

$$\frac{\partial}{\partial x^i}|_p(f) = \frac{\partial}{\partial x^i}|_{\varphi(p)}(f \circ \varphi^{-1}) = \frac{\partial \widehat{f}}{\partial x^i}(\widehat{p})$$

where $\widehat{f} = f \circ \varphi^{-1} : \varphi(U) \subseteq \mathbb{R}^n \to \mathbb{R}^n$ is the coordinate representation of f and $\widehat{p} = \varphi(p)$ is the coordinate representation of p.

Remark: For a smooth manifold wth boundary M, if $p \in \text{int } M$ is an interior point, then the def above applies verbatim. For $p \in \partial M$, we need to substitute $\mathbb{H}^n$ for $\mathbb{R}^n$, by using the identification $d\iota_{\varphi(p)} : T_{\varphi(p)}\mathbb{H}^n \to T_{\varphi(p)}\mathbb{R}^n$ in the Boundary Tangent Spaces Lemma and by the interpreting $\partial/\partial x^n|_p$ as a one-sided derivative.

Proposition 6.5.2. *Basis-for-Tangent Spaces Proposition*
Let M be a smooth n-manifold with or without boundary and let $p \in M$. Then T_pM is an n-dimensional vector space and for any smooth chart $(U, (x^i))$ containing p, the coordinate vectors $\partial/\partial x^1|_p, \cdots, \partial/\partial x^n|_p$ form a basis for T_pM.

Remark: By this prop, any tangent vector $v \in T_pM$ can be written uniquely as a linear combination

$$v = v^i \frac{\partial}{\partial x^i}|_p$$

The ordered basis $\{\partial/\partial x^i|_p\}_{i=1}^n$ is called a *coordinate basis for T_pM* and the numbers $(v^1, \cdots, v^n)$ are called the components of v with respect to the coordinate basis. Furthermore, $\forall j$, we can compute v^j by:

$$v(x^j) = (v^i \frac{\partial}{\partial x^i}|_p)(x^j) = v^i \frac{\partial x^j}{\partial x^i}(p) = v^j$$

6.5.2 Differentials in Coordinates

Motivation: Consider a smooth map $F : U \to V$ where $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ are open subsets of Euclidean spaces. $\forall p \in U$, let $dF_p : T_p\mathbb{R}^n \to T_{F(p)}\mathbb{R}^m$ in terms of the standard coordinate bases. If we denote the coordinates in $\mathbb{R}^n$ by $(x^1, \cdots, x^n)$ and the coordinates in $\mathbb{R}^m$ by $(y^1, \cdots, y^m)$, then we use the chain rule to compute the action of dF_p on a typical basis vector as follows:

$$dF_p(\frac{\partial}{\partial x^i}|_p)(f) := \frac{\partial}{\partial x^i}|_p(f \circ F) = \frac{\partial f}{\partial y^j}(F(p))\frac{\partial F^j}{\partial x^i}(p) = (\frac{\partial F^j}{\partial x^i}(p)\frac{\partial}{\partial y^j}|_{F(p)})(f)$$

Hence,

$$dF_p(\frac{\partial}{\partial x^i}|_p) = \frac{\partial F^j}{\partial x^i}(p)\frac{\partial}{\partial y^j}|_{F(p)}$$

In other words, the matrix of dF_p in terms of the coordinate bases is

$$\begin{pmatrix} \partial F^1/\partial x^1(p) & \cdots & \partial F^1/\partial x^n(p) \\ \vdots & & \vdots \\ \partial F^m/\partial x^1(p) & \cdots & \partial F^m/\partial x^n(p) \end{pmatrix}$$

which is the Jacobian matrix of F at p and where the columns of the matrix are the components of the images of the basis vectors. Therefore, in this case, $dF_p : T_p\mathbb{R}^b \to T_{F(p)}\mathbb{R}^m$ corresponds to the total derivative $DF(p) : \mathbb{R}^n \to \mathbb{R}^m$, under our identification of Euclidean spaces with their tangent spaces. The same computation applies when substituting half-spaces for Euclidean spaces.

Method: Now we consider the more general case of a smooth map $F : M \to N$ between smooth manifolds with or without boundary. Choosing smooth coordinate charts $(U \ni p, \varphi)$ for M and $(V \ni F(p), \psi)$ for N, we obtain the coordinate representation

$$\widehat{F} := \psi \circ F \circ \varphi^{-1} : \varphi(U \cap F^{-1}(V)) \to \psi(V)$$

By the computation above, $d\widehat{F}_{\varphi(p)}$ is represented by the Jacobian matrix of $\widehat{F}$ at $\varphi(p)$, with respect to the standard coordinate bases. Note that $F \circ \varphi^{-1} = \psi^{-1} \circ \widehat{F}$. Hence, we compute

$$\begin{aligned} dF_p(\frac{\partial}{\partial x^i}|_p) &:= dF_p(d(\varphi^{-1})_{\varphi(p)}(\frac{\partial}{\partial x^i}|_{\varphi(p)})) = d(\psi^{-1})_{\widehat{F}(\varphi(p))}(d\widehat{F}_{\varphi(p)}(\frac{\partial}{\partial x^i}|_{\varphi(p)})) \\ &= d(\psi^{-1})_{\widehat{F}(\varphi(p))}(\frac{\partial \widehat{F}^j}{\partial x^i}(\varphi(p))\frac{\partial}{\partial y^j}|_{\widehat{F}(\varphi(p))}) \\ &= \frac{\partial \widehat{F}^j}{\partial x^i}(\varphi(p)) d(\psi^{-1})_{\widehat{F}(\varphi(p))}(\frac{\partial}{\partial y^j}|_{\widehat{F}(\varphi(p))}) \\ &:= \frac{\partial \widehat{F}^j}{\partial x^i}(\varphi(p))\frac{\partial}{\partial y^j}|_{F(p)} \end{aligned}$$

Therefore, dF_p is represented in the coordinate bases by the Jacobian matrix of the coordinate representative of F.

6.5.3 Change of Coordinates

Intuition: Let (U, φ) and (V, ψ) be two smooth charts on M with $p \in U \cap V \neq \emptyset$. Denote the coordinate functions of φ by (x^i) and the coordinate functions of ψ by $(\widetilde{x}^i)$. Then any tangent vector at p can be represented with respect to either basis $(\partial/\partial x^i|_p)$ and $(\partial/\partial \widetilde{x}^i|_p)$. We naturally ask what relation are between the two representations.

Proposition 6.5.3. *Change of Coordinate Basis Proposition*
Let M be a smooth manifold with or without boundary, let $p \in M$, let (U, φ) and (V, ψ) be two smooth charts on M both containing p and let $(x^i), (\widetilde{x}^i)$ denote the coordinate functions of φ respectively ψ. Then

$$\frac{\partial}{\partial x^i}|_p = \frac{\partial \widetilde{x}^j}{\partial x^i}(\varphi(p))\frac{\partial}{\partial \widetilde{x}^j}|_p$$

Proof. Since the transition map $\psi \circ \varphi^{-1}$ is a map between open subsets of Euclidean space, then by the analysis in the motivation of Differentials in Coordinates, the differential $d(\psi \circ \varphi^{-1})_{\varphi(p)}$ can be written as

$$d(\psi \circ \varphi^{-1})_{\varphi(p)}(\frac{\partial}{\partial x^i}|_{\varphi(p)}) = \frac{\partial \widetilde{x}^j}{\partial x^i}(\varphi(p))\frac{\partial}{\partial \widetilde{x}^j}|_{\psi(p)} \qquad (*)$$

Hence, we have

$$\frac{\partial}{\partial x^i}|_p := d(\varphi^{-1})_{\varphi(p)}(\frac{\partial}{\partial x^i}|_{\varphi(p)}) = d(\psi^{-1})_{\psi(p)} \circ d(\psi \circ \varphi^{-1})_{\varphi(p)}(\frac{\partial}{\partial x^i}|_{\varphi(p)})$$
$$\overset{(*)}{=\!=} d(\psi^{-1})_{\psi(p)}(\frac{\partial \widetilde{x}^j}{\partial x^i}(\varphi(p))\frac{\partial}{\partial \widetilde{x}^j}|_{\psi(p)}) := \frac{\partial \widetilde{x}^j}{\partial x^i}(\varphi(p))\frac{\partial}{\partial \widetilde{x}^j}|_p$$

□

Remark: This formula is easy to remember, since it looks the same as the chain rule for partial derivatives in $\mathbb{R}^n$.

Corollary 6.5.4. *of Change of Coordinate Bases Proposition: Change of Vector Components* $\forall v \in T_pM, v = v^i\partial/\partial x^i|_p = \widetilde{v}^j\partial/\partial\widetilde{x}^j|_p$. *Then*

$$\widetilde{v}^j = \frac{\partial \widetilde{x}^j}{\partial x^i}(\varphi(p))v^i$$

6.6 Tangent Bundles

Definition 6.6.1. Tangent Bundles
Let M be a smooth manifold with or without boundary. We define the *tangent bundle of* M, denoted as T_M, to be the disjoint union of the tangent spaces at all points of M:

$$TM := \bigsqcup_{p\in M} T_pM$$

Remark:

- We usually write an element of this disjoint union as an ordered pair (p, v) where $p \in M, v \in T_pM$.
- The tangent bundle TM shall admit a natural *projection map* $\pi : TM \to M$ given by $(p, v) \mapsto p$.

Example: For $M = \mathbb{R}^n$,

$$T\mathbb{R}^n := \bigsqcup_{a\in\mathbb{R}^n} T_a\mathbb{R}^n \cong \bigsqcup_{a\in\mathbb{R}^n} \mathbb{R}^n_a := \bigsqcup_{a\in\mathbb{R}^n} \{a\} \times \mathbb{R}^n = \mathbb{R}^n \times \mathbb{R}^n$$

where the isomorphism is the one in the Directional Derivative Isomorphism Thm.2.

Proposition 6.6.2. *Smooth Structure on the Tangent Bundles of Smooth Manifolds Proposition For any smooth* n*-manifold* M*, the tangent bundle* TM *has a natural topology and smooth structure that make it into a* $2n$*-dimensional smooth manifold. With respect to this structure, the projection map* $\pi : TM \to M$ *is smooth.*

Proof. We prove the first statement by applying the Smooth Manifolds-Construction Lemma to TM and $\{(\pi^{-1}(U), \widetilde{\varphi}) \mid (U, \varphi) \in \mathcal{A}_M\}$ where we'll define $\widetilde{\varphi}$ later.

1. Given any smooth chart (U, φ) for M, note that $\pi^{-1}(U) \subseteq TM$ is the set of all tangent vectors to M at all of points of U. Let $(x^1, \cdots, x^n)$ denote the coordinate function of φ and define $\widetilde{\varphi} : \pi^{-1}(U) \to \mathbb{R}^{2n}$ by

$$\widetilde{\varphi}(v^i \frac{\partial}{\partial x^i}|_p) := (x^1(p), \cdots, x^n(p), v^1, \cdots, v^n)$$

whose image is $\varphi(U) \times \mathbb{R}^n$, hence open in $\mathbb{R}^{2n}$. Since φ is a bijection between $U \cong \varphi(U)$, then $\widetilde{\varphi}$ is a bijection onto its image.

2. Now suppose we're given two smooth charts (U, φ) and (V, ψ) for M. Let $(\pi^{-1}(U), \widetilde{\varphi})$ and $(\pi^{-1}(V), \widetilde{\psi})$ be the corresponding charts on TM. The sets

$$\widetilde{\varphi}(\pi^{-1}(U) \cap \pi^{-1}(V)) = \varphi(U \cap V) \times \mathbb{R}^n \text{ and } \widetilde{\psi}(\pi^{-1}(U) \cap \pi^{-1}(V)) = \psi(U \cap V) \times \mathbb{R}^n$$

are open in $\mathbb{R}^{2n}$.

3. If $\pi^{-1}(U) \cap \pi^{-1}(V) \neq \emptyset$, then the transition map $\widetilde{\psi} \circ \widetilde{\varphi}^{-1} : \varphi(U \cap V) \times \mathbb{R}^n \to \psi(U \cap V) \times \mathbb{R}^n$ can be written explicitly, by the coro of Change of Coordinate Bases Prop, as

$$\widetilde{\psi} \circ \widetilde{\varphi}^{-1}(x^1, \cdots, x^n, v^1, \cdots, v^n) = (\widetilde{x}^1(x), \cdots, \widetilde{x}^n(x), \frac{\partial \widetilde{x}^1}{\partial x^i}(x)v^i, \cdots, \frac{\partial \widetilde{x}^n}{\partial x^i}(x)v^i)$$

where (x^i) and $(\widetilde{x}^j)$ denote the coordinate functions of φ respectively ψ. Hence, it's easy to see that $\widetilde{\psi} \circ \widetilde{\varphi}^{-1}$ is smooth.

4. Given any countable cover $\{U_i\}$ of M by smooth coordinate domains, we shall construct a countable cover of TM by $\{\pi^{-1}(U_i)\}$.

5. $\forall (p, v) \neq (q, w) \in TM$, if $p = q$ lie in the same chart (U, φ) for M, then $(p, v), (q, w)$ lie in the same chart $(\pi^{-1}(U), \widetilde{\varphi})$. Otherwise, if $p \neq q$, then by the Hausdorff-ness of M, they lie in two disjoint charts $(U \ni p, \varphi)$ and $(V \ni q, \psi)$, so $(p, v), (q, w)$ lie in disjoint $(\pi^{-1}(U), \widetilde{\varphi})$ respcetively $(\pi^{-1}(V), \widetilde{\psi})$.

To prove the second statement, $\forall$ smooth chart $(U, \varphi = (x^i))$ for M, consider the corresponding chart $(\pi^{-1}(U), \widetilde{\varphi})$ for TM. Note that the coordinate representation of π

$$\varphi \circ \pi \circ \widetilde{\varphi}^{-1}(x^1, \cdots, x^n, v^1, \cdots, v^n) = (x^1, \cdots, x^n)$$

is smooth, so we're done. □

Remark:

- The coordinates $\widetilde{\varphi} = (x^i, v^i)$ given by the def of $\widetilde{\varphi}$ are called the *natural coordinates on* TM. The reason why we say they are "natural" comes from the uniqueness of the smooth structure on TM, by the Smooth Manifolds-Construction Lemma.
- This prop can also be used for smooth manifolds with boundary, with the proof given, on

boundary, by substituting $\mathbb{H}^{2n}$ for $\mathbb{R}^{2n}$ and by rearranging the coordinates in the natural chart $(\pi^{-1}(U), (x^i, v^i))$ into $(\pi^{-1}(U), (v^i, x^i))$ for TM, since $x^n \geq 0$.

Proposition 6.6.3. *Trivial Tangent Bundle Proposition*
Let M be a smooth n-manifold with or without boundary. If M can be covered by a single smooth chart, then TM is diffeomorphic to $M \times \mathbb{R}^n$.

Proof. Let (M, φ) be the global smooth chart for M. Then by the proof of the Smooth Structure on the Tangent Bundles of Smooth Manifolds Prop, the natural coordinate $\widetilde{\varphi}$ is a bijection from TM to $\varphi(M) \times \mathbb{R}^n$, and the smooth structure on TM is defined essentially by declaring $\widetilde{\varphi}$ to be a diffeomorphism. □

Definition 6.6.4. Global Differentials
Let M, N be smooth manifolds with or without boundary and let $F : M \to N$ be a smooth map. By putting together the differentials of F at all of points of M, we obtain a globally defined map between tangent bundles, called the *global differential* or *global tangent map* and denoted by $dF : TM \to TN$.

Remark:

- $dF|_{T_pM} = dF_p, \forall p \in M$.
- When applying the differential of F to a specific vector $v \in T_pM$, we can write either $dF_p(v)$ or $dF(v)$, depending on how much emphasis we wish to give to the point p.

Proposition 6.6.5. *Smoothness of Global Differentials Proposition*
If $F : M \to N$ is a smooth map, then its global differential $dF : TM \to TN$ is a smooth map.

Corollary 6.6.6. *of Smoothness of Global Differentials Proposition: Properties of the Global Differentials*
Let $F : M \to N$ and $G : N \to P$ be smooth maps between smooth manifolds with or without boundary.

1. *$d(G \circ F) = dG \circ dF$.*
2. *$d(id_M) = id_{TM}$.*
3. *If F is a diffeomorphism, then $dF : TM \to TN$ is also a diffeomorphism with $(dF)^{-1} = d(F^{-1})$*

6.7 Velocity of Curves in Manifolds

Motivation: In $\mathbb{R}^n$, the *velocity* of a smooth parametrized curve is the derivatives of each coordinate component with respect to time. But in generally, there is no global coordinate for manifolds, so we define velocity on manifolds by using differentials:

Definition 6.7.1. Velocity
Let M be a topological manifold with or without boundary.

- We define a *curve in* M to be a continuous map $\gamma : J \to M$ where $J \subseteq \mathbb{R}$ is an interval.
- If M is smooth and if the curve γ is smooth, then for any $t_0 \in J$, we define the *velocity of* γ *at* t_0, denoted as $\gamma'(t_0)$, to be the vector

$$\gamma'(t_0) := d\gamma(\frac{d}{dt}|_{t_0}) \in T_{\gamma(t_0)}M$$

  where $d/dt|_{t_0}$ is the standard coordinate basis vector in $T_{t_0}\mathbb{R}$.

Remark: Let $(U \ni \gamma(t_0), (x^i))$ be a smooth chart for M. For t sufficiently close to t_0, we can write the coordinate representation of γ as $\gamma(t) = (\gamma^1(t), \cdots, \gamma^n(t))$. Then as in the section called Differentials in Coordinates, we have:

$$\gamma'(t_0) = \frac{d\gamma^i}{dt}(t_0)\frac{\partial}{\partial x^i}|_{\gamma(t_0)}$$

Proposition 6.7.2. *Realizably of Tangent Vectors by Smooth Curves Proposition*
Let M be a smooth manifold with or without boundary and let $p \in M$. Every $v \in T_pM$ is the velocity of some smooth curve in M.

Proof. We first prove the case when $\partial M \neq \emptyset$: Let $(U, \varphi = (x^i))$ be a smooth chart centered at p and write $v = v^i\partial/\partial x^i|_p$. Take a sufficiently small $\epsilon > 0$ and let $\gamma : (-\epsilon, \epsilon) \to U$ be a curve with coordinate representation

$$\gamma(t) = \varphi^{-1}(tv^1, \cdots, tv^n)$$

which is a smooth curve with $\gamma(0) = p$ and $\gamma'(0) = v$.

Now for the case when $\partial M \neq \emptyset$, if $p \in \text{int } M$, then apply the same argument as above. Otherwise, $\forall p \in \partial M$, let $(U, \varphi = (x^i))$ be a smooth boundary centered at p and write $v = v^i\partial/\partial x^i|_p$.

- If $v^n = 0$, then we define $\gamma : (-\epsilon, \epsilon) \to U$ as before.
- If $v^n > 0$, then we let the domain of γ be $[0, \epsilon)$.
- If $v^n < 0$, then we let the domain of γ be $[-\epsilon, 0)$.

Then we're done. □

Proposition 6.7.3. *Velocity of Composite Curves Proposition*
Let $F : M \to N$ be a smooth map between smooth manifolds with or without boundary and let $\gamma : J \to M$ be a smooth curve. $\forall t_0 \in J$, the velocity at t_0 of the composite curve $F \circ \gamma : J \to N$ is given by

$$(F \circ \gamma)'(t_0) = dF(\gamma'(t_0))$$

Proof.

$$(F \circ \gamma)'(t_0) := d(F \circ \gamma)(\frac{d}{dt}|_{t_0}) = (dF \circ d\gamma)(\frac{d}{dt}|_{t_0}) := dF(\gamma'(t_0))$$

□

Corollary 6.7.4. *of Velocity of Composite Curves Proposition*
Let $F : M \to N$ be a smooth map between smooth manifolds with or without boundary, let $p \in M$ and let $v \in T_pM$. Then $\forall$ smooth curve $\gamma : J \to M$ with $0 \in J, \gamma(0) = p$ and $\gamma'(0) = v$, we have:

$$dF_p(v) = (F \circ \gamma)'(0)$$

Appendix A

Preliminary Theorems from Analysis

Theorem A.0.1. *Chain Rule*
Let $f, g : \mathbb{R} \to \mathbb{R}$ be differentiable at a respectively $f(a)$. Then $g \circ f$ is differentiable at a, and

$$(g \circ f)'(a) = g'(f(a)) \cdot f'(a)$$

Theorem A.0.2. *Inverse Function Theorem (IFT)*
Let $U \subseteq \mathbb{R}^n$ be open and let $f : U \to \mathbb{R}^n$ be $\mathcal{C}^1$. If $Df(a)$ is invertible with $a \in U$, then $\exists$ open $U' \ni a$, $V' \ni f(a)$, s.t.:

- *$f|_{U'} : U' \to V'$ is bijective*
- *$(f|_{U'})^{-1}$ is $\mathcal{C}^1$*
- *$\forall y \in V'$, $D(f|_{U'})^{-1}(y) = [Df(g(y))]^{-1}$*

Theorem A.0.3. *Implicit Function Theorem*
Let $U \subseteq \mathbb{R}^{n+m}$ be open, let $F : U \to \mathbb{R}^m$ be $\mathcal{C}^1$, and let $(a, b) \in U$, where $a \in \mathbb{R}^n$ and $b \in \mathbb{R}^m$. Suppose $F(a, b) = 0$. If $D_y F(a, b)$ is invertible, then $\exists$ open $A \ni a$, $B \ni b$, s.t.: $g(a) = b$ for some $\mathcal{C}^1$ map g and that $F(x, g(x)) = 0$, $\forall x \in A$. Moreover,

$$Dg(x) = -\left[D_y F(x, g(x))\right]^{-1} \circ D_x F(x, g(x)).$$